

Workshop FortiManager 2º Edição



Data: 31/08/2024

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<https://workshopfmq.secinfra.com.br/>

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1.OBJETIVO

Inicialmente, gostaria de agradecer a confiança depositada no meu trabalho, ao se inscrever no nosso primeiro Workshop.

Gostaria de lembrar o grande objetivo do nosso Workshop: Implementarmos um ambiente Hub & Spoke, utilizando das ferramentas SD-WAN e ADVPN da Fortinet, agregando ainda a esse ambiente, a funcionalidade de Self Healing através do nosso roteamento BGP, e implementando e gerenciando esse ambiente através do FortiManager, facilitando a implementação de novas unidades (Spokes) ou mesmo em grande escala.

2.AMBIENTE DE LABORATÓRIO

Nosso ambiente de laboratório foi planejado para ser executado através da ferramenta "PNetLAB", porém, caso o participante tenha domínio de ferramentas similares como EVE-NG e GNS3, poderão ser utilizados também.

No nosso laboratório, iremos utilizar as seguintes imagens:

- (1) Roteador Cisco, versão "LinuxL3-AdvEnterpriseK9-M2_157_3_May_2018";
- (1) FortiManager-VM, versão 7.4.3;
- (3) FortiGate-VM, versão 7.0.15;
- (3) VirtualPC.

As instalações e configurações necessárias, foram disponibilizadas no canal do YouTube "SECINFRA", e estão disponíveis no link a seguir:

<https://www.youtube.com/playlist?list=PLkEWllBrbusq6UfdPQYAVB07H-Kh128Gc>

Caso enfrente problemas ou dificuldades nos passos descritos nos vídeos, sinta-se a vontade para entrar em contato através do WhatsApp (54) 99357-8866.

3.DIA DO WORKSHOP

No dia do Workshop (31/08/2024), é muito importante que todos os participantes tenham pontualidade, a sala será aberta 15 minutos antes do horário previsto de início (10:00 GMT-3), e pontualmente as 10:15 será iniciada a gravação e a apresentação do conteúdo.

Durante o horário das 09:45 às 10:15, caso tenha algum problema com o ambiente do laboratório, poderá utilizar desse tempo para tirar dúvidas com o Instrutor, porém é recomendado que as dúvidas já tenham sido tiradas anteriormente.

O Cronograma planejado é o seguinte:

- 10:00 as 10:15 – Tempo de espera dos participantes e dúvidas do ambiente de laboratório;
- 10:15 as 10:45 – Validação da topologia, o instrutor irá repassar todas as configurações realizadas no Roteador Cisco, FortiGate HUB e FortiManager;
- 10:45 as 11:30 – Nivelamento e configurações iniciais nos FortiGates Spoke, nessa etapa iremos apresentar as principais funcionalidades que serão utilizadas no FortiManager durante o WorkShop, e iremos aproveitar para configurar os FortiGates Spoke utilizando essas funcionalidades.
- 11:30 as 13:00 – SD-WAN Overlay Template, nesta etapa iremos realizar a configuração inicial do SD-WAN Overlay Template, também iremos explicar como ele funciona, e qual a funcionalidade de cada configuração.
- 13:00 as 14:30 – Pausa para almoço.
- 14:30 as 15:30 – Templates – Nesta etapa iremos realizar diversos ajustes nos templates gerados pelo SD-WAN Overlay Template, também será demonstrado mais funcionalidades possíveis de uso de variáveis, inclusive através de Jinja Script.
- 15:30 as 16:30 – Ajustes finais e implementação – Nesta etapa iremos realizar alguns ajustes finais, implementando algumas boas práticas, tanto para melhor funcionamento do ADVPN, como para termos uma convergência mais rápida de VPNs e BGP, e realizaremos a instalação das políticas e templates nos FortiGates.
- 16:30 as 18:00 – Testes finais e dúvidas – Nesta etapa, iremos realizar alguns testes em conjunto, e após isso teremos um tempo reservado para dúvidas e também para pedidos de testes em específico no ambiente.

4. PASSO A PASSO WORKSHOP

Abaixo, iremos deixar todos os passos que iremos executar em conjunto no Workshop, de modo que os participantes possam já ter uma noção prévia, antes da execução em conjunto, também será de grande valia quando desejar estudar novamente o conteúdo do Workshop.

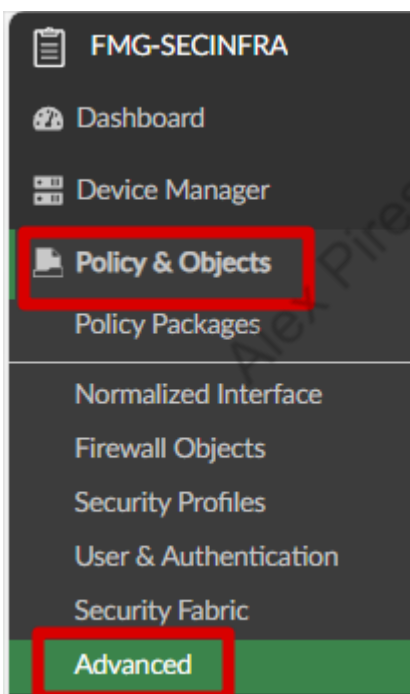
Vale mencionar que esse não é o conteúdo completo do Workshop, e sim apenas um guia básico, a explicação, passo a passo detalhado e comentários do Instrutor, só estará disponível durante o Workshop e na gravação do Workshop, então, por exemplo, na etapa de Nivelamento, não teremos o conteúdo de demonstração das funcionalidades, e sim somente os passos que serão executados nessa etapa, que serão necessários para a conclusão do ambiente do Workshop.

Pedimos a gentileza de não compartilhar esse material, ele é de uso exclusivo do participante do Workshop!

4.1. NIVELAMENTO

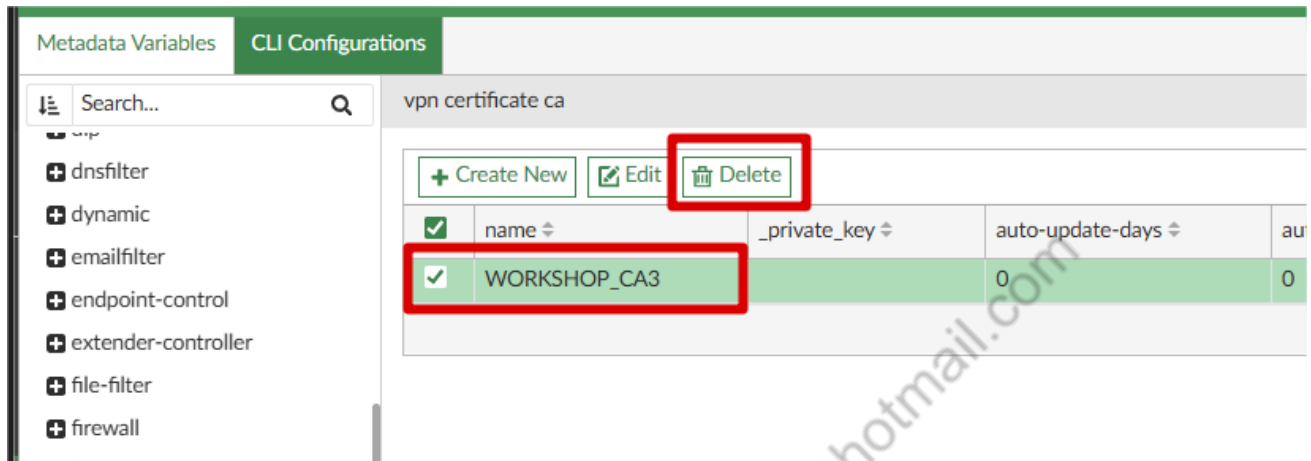
4.1.1. Exclusão do Certificado da ADOM

Nota: Esse passo somente é necessário quando estamos trabalhando com ambiente Trial, ou seja, não licenciado. Em ambientes reais, não é necessário.



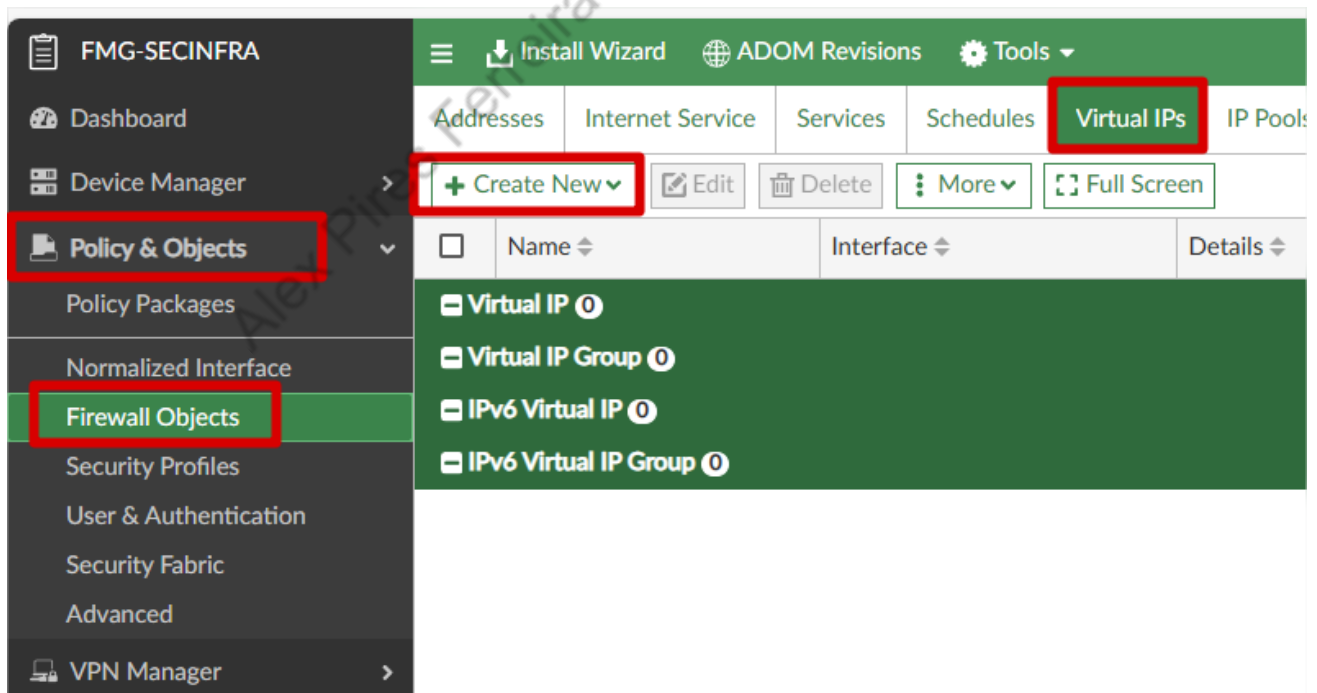


Em cli Configurations, vamos acessar vpn > certificate > ca, localizar o certificado com o nome da ADOM e excluí-lo.



4.1.2.DNAT PARA FORTIMANAGER

Vamos criar os VIPs, um para cada link, conforme os passos abaixo:



Addresses	Internet Service	Services	Schedules	Virtual IPs	IP Pools
+ Create New Edit Delete More Full Screen					
Virtual IP		Interface		Details	
Virtual IP Group					

Dnat para o LINK 1:

VIP Type	IPv4 IPv6
Name	DNAT_FMG_LINK1
Comments	
Color	 Change
Status	☒
Interface	port2
Configure Default Value	☒

Network

Type	Static NAT	
External IP Address/Range	100.100.100.2	100.100.100.2
Mapped IPv4 Address/Range	192.168.31.254	192.168.31.254
Mapped IPv6 Address/Range		
Source Interface Filter	Click to select	

Create New Virtual IP

Type

Static NAT

External IP Address/Range

100.100.100.2

100.100.100.2

Mapped IPv4 Address/Range

192.168.31.254

192.168.31.254

Mapped IPv6 Address/Range

Source Interface Filter

Click to select

Optional Filters

Port Forwarding

Protocol

TCP

Port Mapping Type

One to oneMany to many

External Service Port (1 to 65535)

541

Map to IPv4 Port

541

Map to IPv6 Port

Enable ARP Reply

Add To Groups

Click to select

Advanced Options >

Per-Device Mapping >

Revision

Change Note

OK

Cancel

DNAT Para o link 2:

Create New Virtual IP

VIP Type

IPv4 IPv6

Name

DNAT_FMG_LINK2

Comments

Color

 Change

Status

☒

Interface

port3

Configure Default Value

☒

Network

Type

Static NAT

External IP Address/Range

110.110.110.2

110.110.110.2

Mapped IPv4 Address/Range

192.168.31.254

192.168.31.254

Mapped IPv6 Address/Range

Optional Filters

☐

Port Forwarding

☒

Protocol

TCP

Port Mapping Type

One to one Many to many

External Service Port (1 to 65535)

541

Map to IPv4 Port

541

Map to IPv6 Port

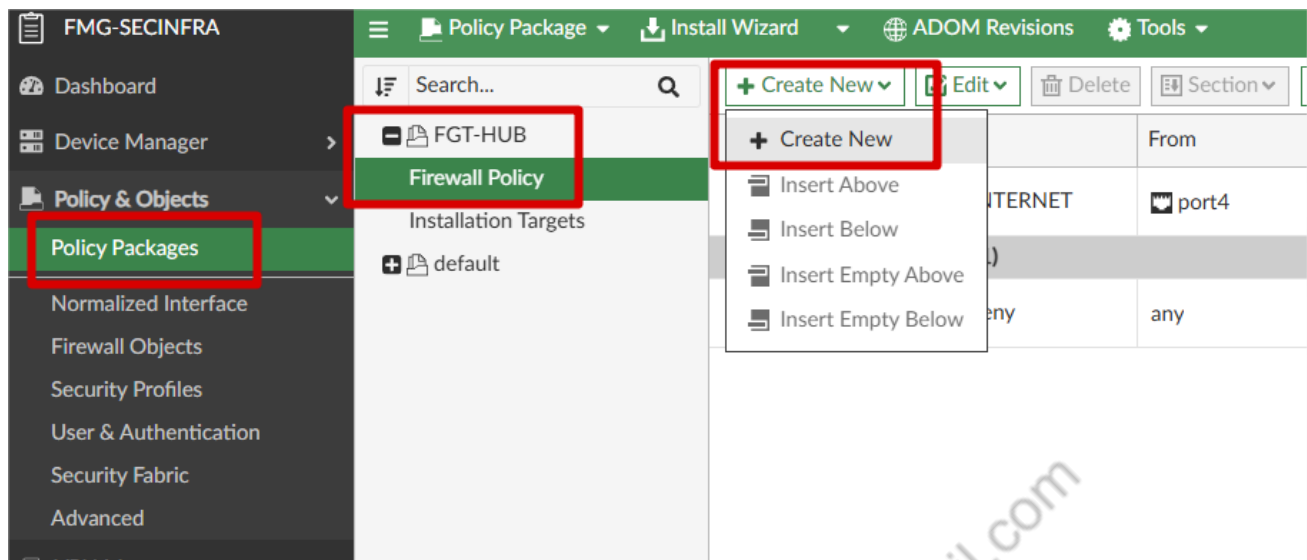
Enable ARP Reply

☒

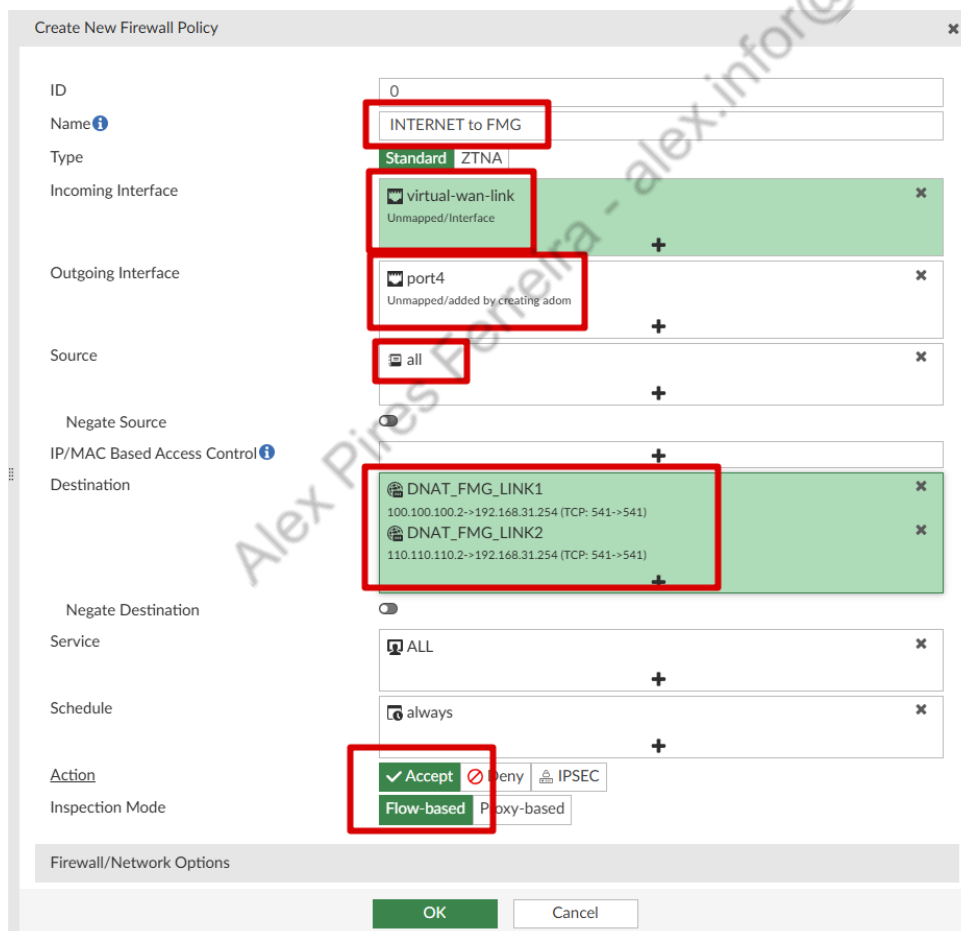
Add To Groups

Click to select

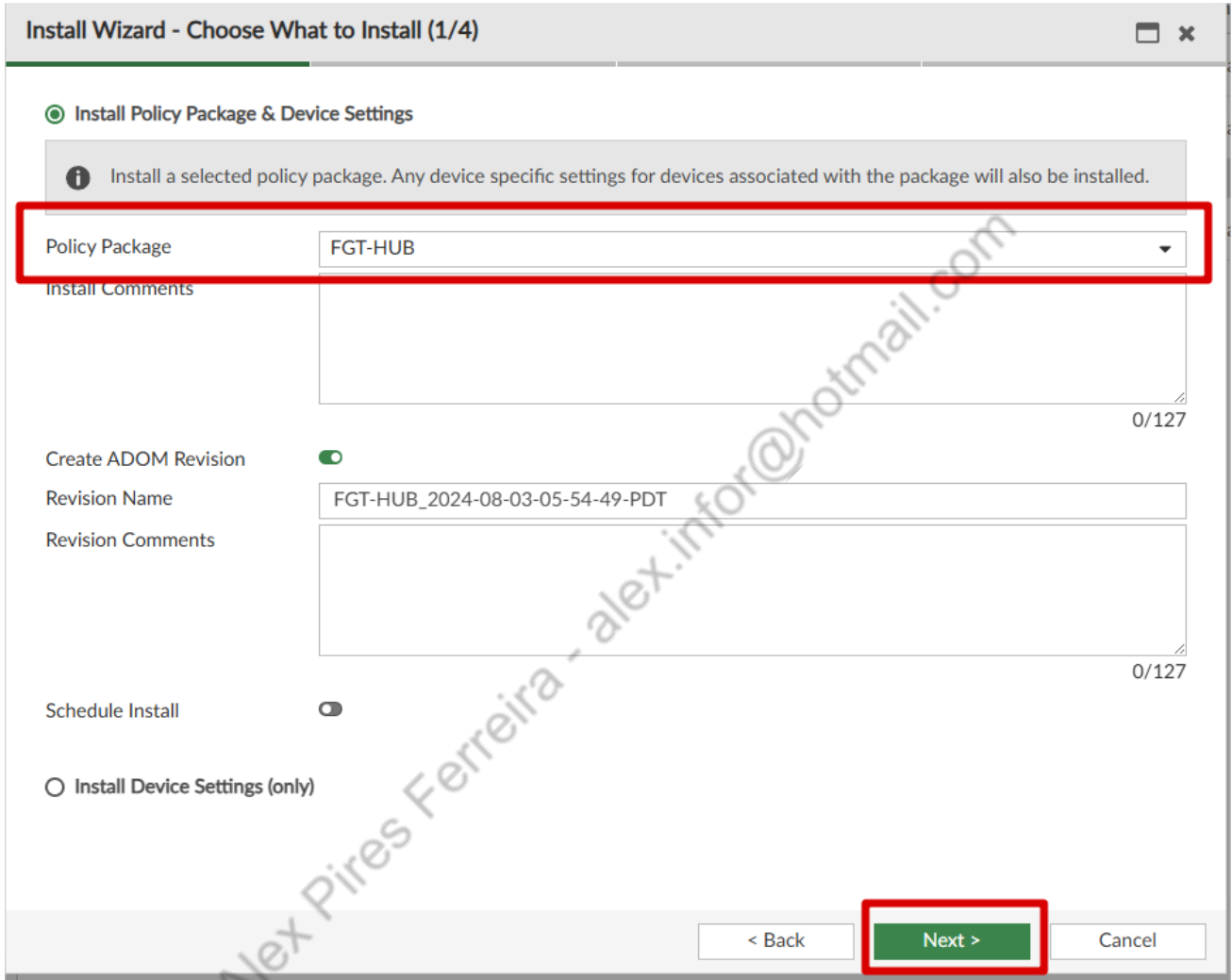
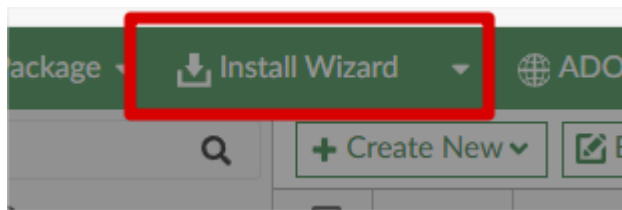
Agora vamos criar a política:



Na próxima tela, caso você tenha criado a zona "DMZ" no FGT-HUB para a port4, você deverá antes criar uma "normalized interface" com o nome de "DMZ" e usar ela na "Outgoing Interface":



Agora podemos instalar o Policy Package:

A screenshot of a 'Install Wizard' dialog box titled 'Install Wizard - Choose What to Install (1/4)'. The dialog has a light gray header and a white body. The first step is 'Install Policy Package & Device Settings', indicated by a green circle. An information icon and text state: 'Install a selected policy package. Any device specific settings for devices associated with the package will also be installed.' Below this, a red box highlights the 'Policy Package' dropdown menu, which is set to 'FGT-HUB'. Underneath is a text area for 'Install Comments' with a character count of '0/127'. The 'Create ADOM Revision' section has a toggle switch turned on. It includes a 'Revision Name' field with the text 'FGT-HUB_2024-08-03-05-54-49-PDT' and a 'Revision Comments' text area with a '0/127' character count. The 'Schedule Install' toggle switch is turned off. At the bottom, there are three buttons: '< Back', 'Next >', and 'Cancel'. The 'Next >' button is highlighted with a red box. A diagonal watermark 'Alex Pires Ferreira - alex.infor@hotmail.com' is visible across the lower half of the dialog.

Install Wizard - Select Devices to Install (FGT-HUB) (2/4)

Please select one or more devices to install (Use checkbox or Ctrl or Shift key for multiple selections)

Search...

☒	Device Name ⇅	IP Address ⇅	Platform ⇅	⚙
☒	FGT-HUB	192.168.31.1	FortiGate-VM64-KVM	

< Back Next > Cancel

Na próxima tela, podes escolher para dar um "Install Preview" e verificar o que será instalado, após isso, pode clicar em close e depois Install.

Install Wizard - Validate Devices (FGT-HUB) (3/4)

Installation Preparation Total: 2/2 ✓ Success: 2, ⚠ Warning: 0, ✖ Error: 0 Show Details

- ✓ Interface Validation
- ✓ Policy and Object Validation
- ✓ Ready to Install

☒ Install Preview ☐ Policy Package Diff

Search...

☒	Device Name ⇅	Status ⇅	⚙
☒	FGT-HUB[root]	Connection Up	

< Back Install Cancel

Após a instalação, só clicar em Finish.

4.1.3.CONFIGURAÇÕES CONSOLE FORTIGATE FGT-FILIAL-1

Vamos configurar um dos links, e a port1 para nossa gerencia, na porta 1, lembre-se de por um IP da sua rede local!

```
FortiGate-VM64-KVM # config sys in
FortiGate-VM64-KVM (interface) # edit port2
FortiGate-VM64-KVM (port2) # set ip 120.120.120.2/30
FortiGate-VM64-KVM (port2) # set allowaccess ping
FortiGate-VM64-KVM (port2) # next
FortiGate-VM64-KVM (interface) # edit port1
FortiGate-VM64-KVM (port1) # set mode static
FortiGate-VM64-KVM (port1) # set ip 192.168.99.13/24
FortiGate-VM64-KVM (port1) # end
```

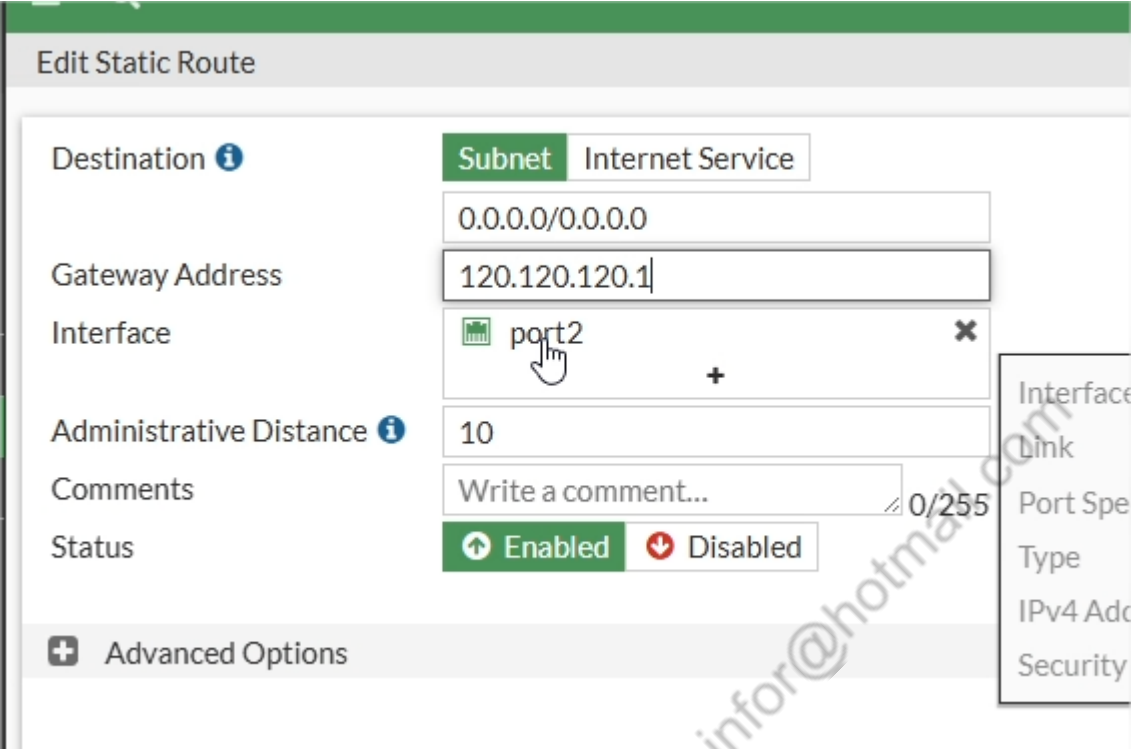
4.1.4.CONFIGURAÇÕES CONSOLE FORTIGATE FGT-FILIAL-2

Vamos configurar um dos links e a port1 para nossa gerencia, na porta1 lembre-se de por um IP da sua rede local!

```
FortiGate-VM64-KVM # config sys in
FortiGate-VM64-KVM (interface) # edit port1
FortiGate-VM64-KVM (port1) # set mode static
FortiGate-VM64-KVM (port1) # set ip 192.168.99.14/24
FortiGate-VM64-KVM (port1) # next
FortiGate-VM64-KVM (interface) # edit port2
FortiGate-VM64-KVM (port2) # set ip 140.140.140.2/30
FortiGate-VM64-KVM (port2) # set allowaccess ping
FortiGate-VM64-KVM (port2) # end
FortiGate-VM64-KVM #
```

4.1.5. CONFIGURAÇÕES INTERFACE GRÁFICA FORTIGATE FGT-FILIAL-1

Criar Static Route:



Edit Static Route

Destination ⓘ **Subnet** Internet Service
 0.0.0.0/0.0.0.0

Gateway Address
 120.120.120.1

Interface
 port2

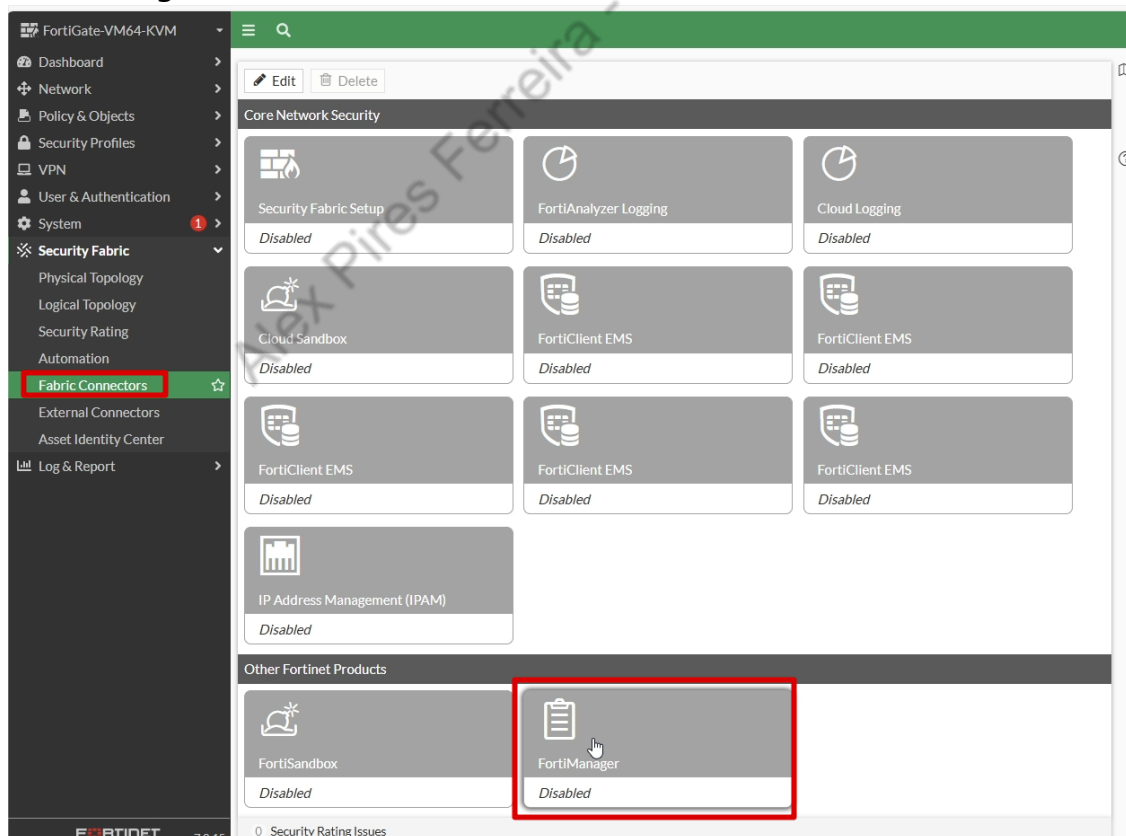
Administrative Distance ⓘ
 10

Comments
 Write a comment... 0/255

Status
 Enabled Disabled

+ Advanced Options

Conectar ao FortiManager, pelo caminho: Security Fabric > Fabric Connectors > FortiManager:



Configuração:

Edit Fabric Connector

Other Fortinet Products


FortiManager

FortiManager Settings

Status	☒ Enabled ☐ Disabled
Type	☒ On-Premises ☐ FortiManager Cloud ☐ FortiGate Cloud
Mode	☒ Normal ☐ Backup
IP/Domain name	100.100.100.2 X
	110.110.110.2 X
	+

Aceitar serial:

Verify FortiManager Serial Number

The FortiManager's access to the FortiGate will be authenticated by the FortiManager certificate. The serial number from the certificate must match the serial number observed on the FortiManager.

 The obtained serial number from the FortiManager certificate is: **FMG-VMTM24010902**

Do you wish to accept the serial number and certificate—verifying that they match the correct FortiManager?

4.1.6.CONFIGURAÇÕES INTERFACE GRÁFICA FORTIGATE FGT-FILIAL-2

Criar Static Route:

New Static Route

Destination

Subnet
Internet Service

0.0.0.0/0.0.0.0

Gateway Address

140.140.140.1

Interface

port2

Administrative Distance

10

Comments

Write a comment...

Status

Enabled
Disabled

+
Advanced Options

Conectar ao FortiManager, pelo caminho: Security Fabric > Fabric Connectors > FortiManager:

FortiGate-VM64-KVM
Dashboard
Network
Policy & Objects
Security Profiles
VPN
User & Authentication
System
Security Fabric
Physical Topology
Logical Topology
Security Rating
Automation
Fabric Connectors
External Connectors
Asset Identity Center
Log & Report

Edit
Delete

Core Network Security

Security Fabric Setup
Disabled

FortiAnalyzer Logging
Disabled

Cloud Logging
Disabled

Cloud Sandbox
Disabled

FortiClient EMS
Disabled

FortiClient EMS
Disabled

FortiClient EMS
Disabled

FortiClient EMS
Disabled

FortiClient EMS
Disabled

IP Address Management (IPAM)
Disabled

Other Fortinet Products

FortiSandbox
Disabled

FortiManager
Disabled

Configuração:

Edit Fabric Connector

Other Fortinet Products

 FortiManager

FortiManager Settings

Status	☒ Enabled ☐ Disabled
Type	☒ On-Premises ☐ FortiManager Cloud ☐ FortiGate Cloud
Mode	☒ Normal ☐ Backup
IP/Domain name	100.100.100.2 x 110.110.110.2 x +

Aceitar serial:

Verify FortiManager Serial Number

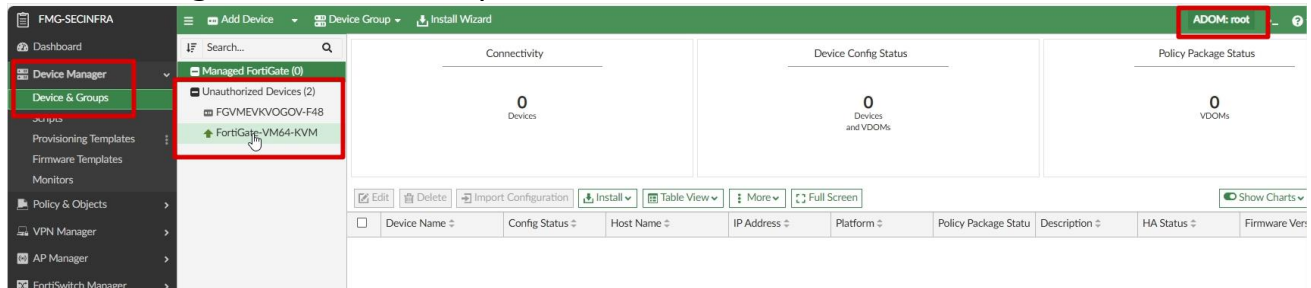
The FortiManager's access to the FortiGate will be authenticated by the FortiManager certificate. The serial number from the certificate must match the serial number observed on the FortiManager.

 The obtained serial number from the FortiManager certificate is: **FMG-VMTM24010902**

Do you wish to accept the serial number and certificate—verifying that they match the correct FortiManager?

4.1.7. AUTORIZAR FORTIGATES FILIAIS NO FORTIMANAGER

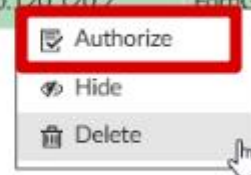
Acessar o FortiManager na adom "root", localizar os FortiGate não autorizados em Device Manager > Device & Groups:



Clicar com o botão direito, e autorizar um por vez, pelo IP Address, poderá saber qual é o FILIAL 1 (120.120.120.2) e qual é o FILIAL 2 (140.140.140.2):

☐	Device Name	Platform	Serial Number	IP Address	Firmware Version	Management Mode
☐	FGVMEVKVOGOV-F48	FortiGate-VM64-...	FGVMEVKVOGOV-F48	140.140.140.2	FortiGate 7.0.15,build0632 (G/	Configuration & Logging
☒	FortiGate-VM64-KVM	FortiGate-VM64-...	FGVMEVDN806-AAI	120.120.120.2	FortiGate 7.0.15,build0632 (G/	Configuration & Logging

☐	FGVMEVKVOGOV-F48	FortiGate-VM64-...	FGVMEVKVOGOV-F48	140.140.140.2	FortiGate 7.0.15,build0632 (G/
☒	FortiGate-VM64-KVM	FortiGate-VM64-...	FGVMEVDN806-AAI	120.120.120.2	FortiGate 7.0.15,build0632 (G/



Adicionar os FortiGate à ADOM WORKSHOP, ajustando o nome:

Authorize Device

Add the following device(s) to ADOM:
WORKSHOP (FortiGate 7.0)

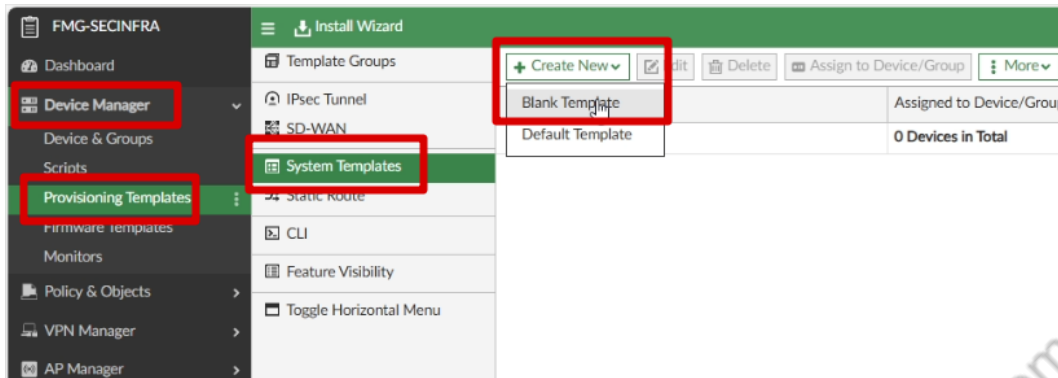
Name	Assign New Device Name	Assign Policy Package	Assign Provisioning Template	Assign Dashboard Config
FortiGate-VM64-...	FGT-FILIAL-1	Click to select	+	Click to select

OK
Cancel

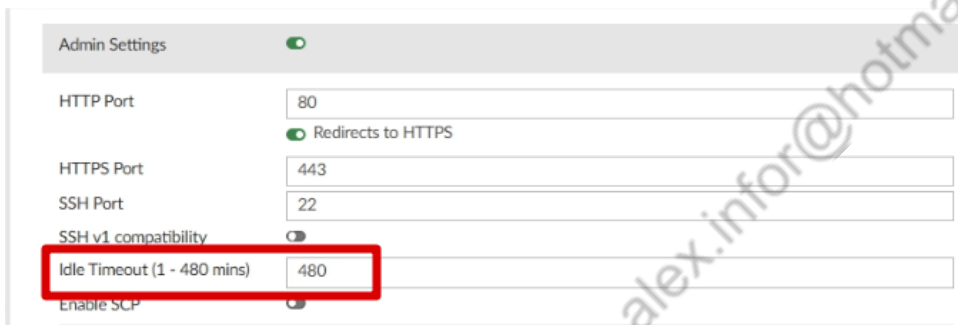
4.1.8.CONFIGURAÇÃO SYSTEM TEMPLATE

De volta à ADOM WORKSHOP, iremos criar os templates.

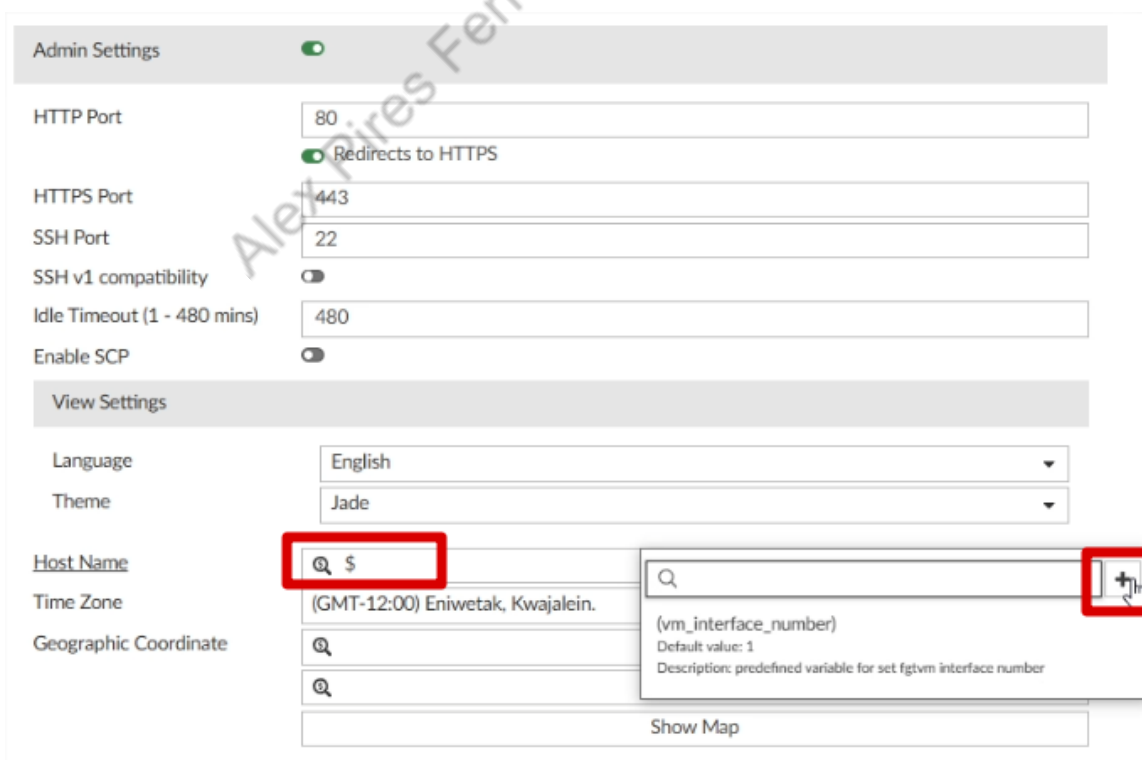
System Templates:



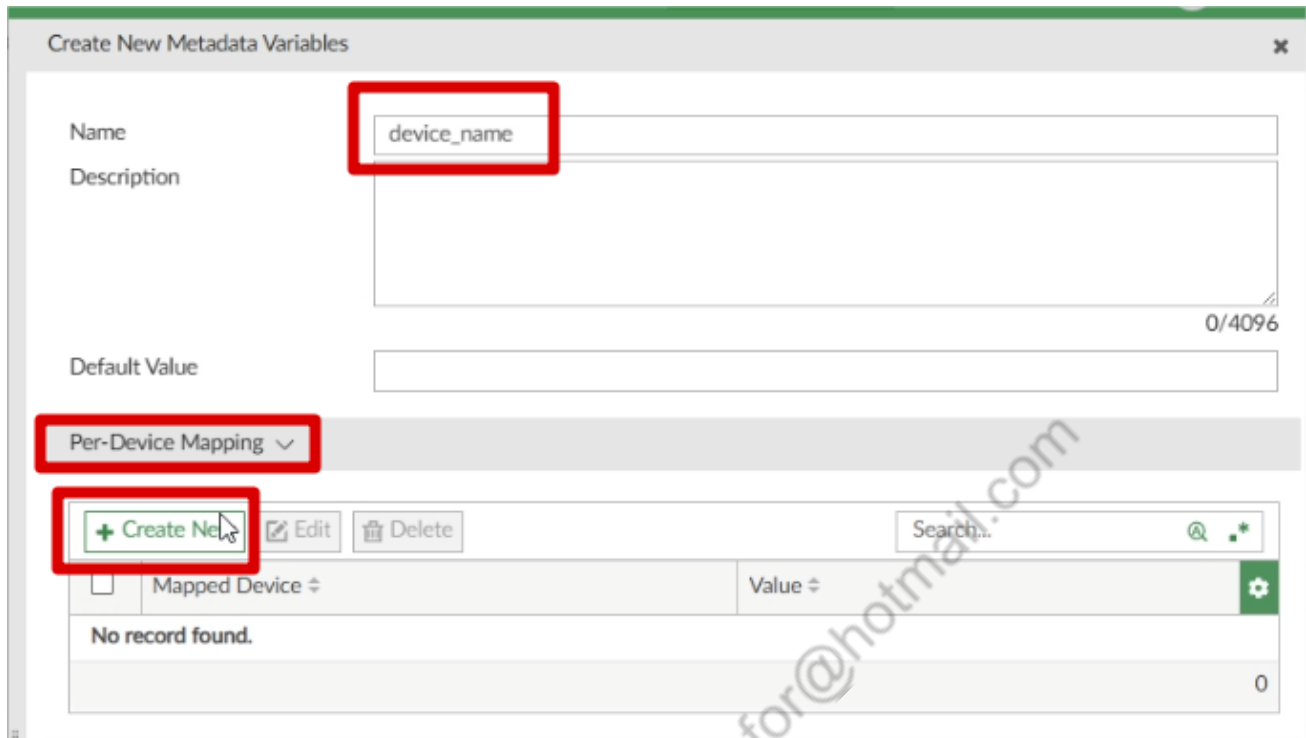
Admin Settings:



Ainda em Admin Settings, em "Host Name", iremos criar uma variável, para isso digite "\$", e um menu à direita irá abrir, clique no "+":



O nome da variável será “device_name”, expanda a guia “Per-Device Mapping” e clique em Create New:



Create New Metadata Variables

Name: device_name

Description:

Default Value:

Per-Device Mapping

+ Create New Edit Delete

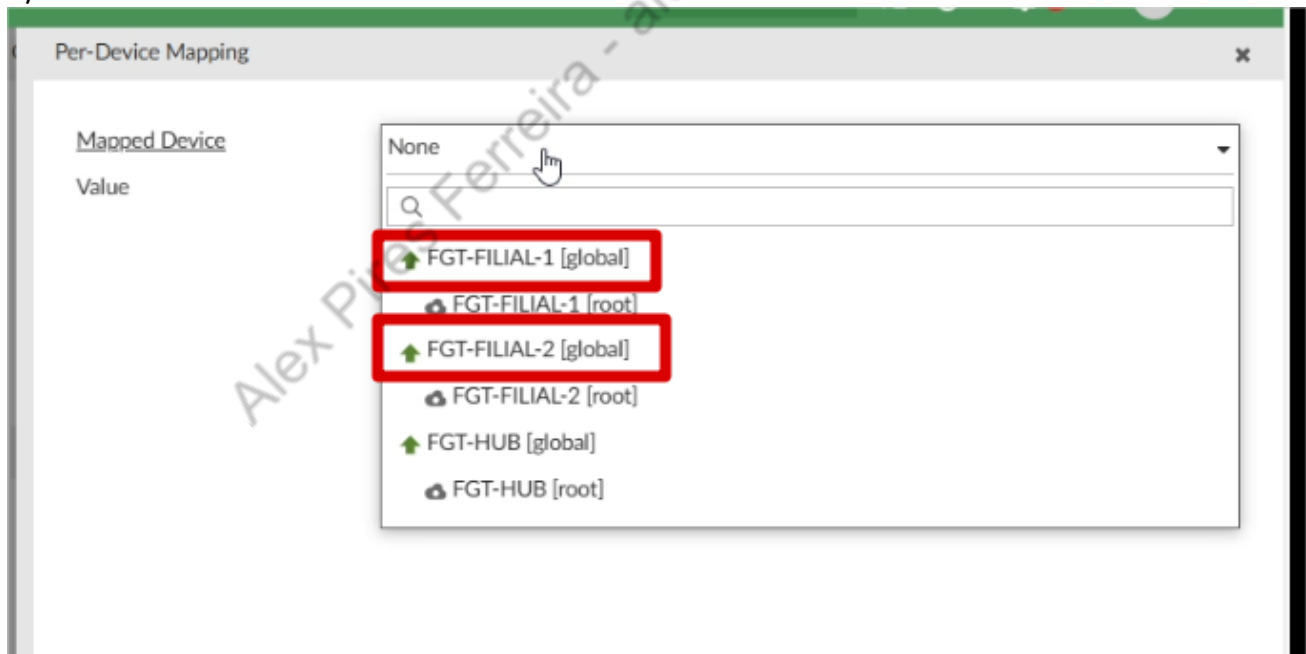
Search...

Mapped Device Value

No record found.

0

Vamos criar um mapeamento para cada FortiGate, com o valor FGT-FILIAL-1 e FGT-FILIAL-2, lembre-se de usar o device “GLOBAL”:



Per-Device Mapping

Mapped Device

Value

None

FGT-FILIAL-1 [global]

FGT-FILIAL-1 [root]

FGT-FILIAL-2 [global]

FGT-FILIAL-2 [root]

FGT-HUB [global]

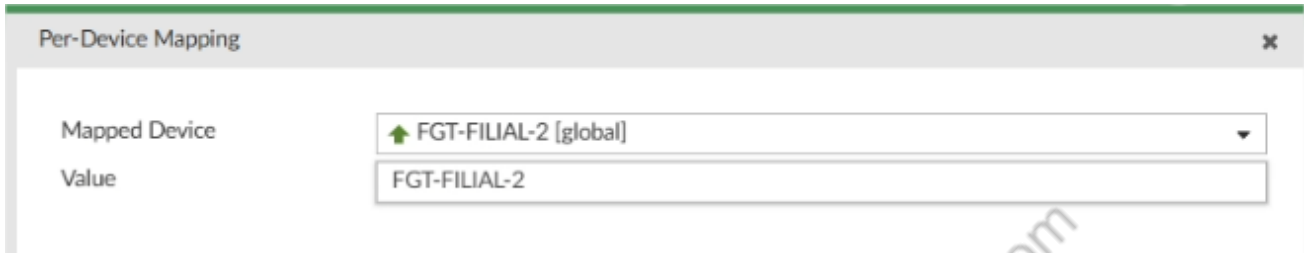
FGT-HUB [root]



Per-Device Mapping

Mapped Device: ↑ FGT-FILIAL-1 [global]

Value: FGT-FILIAL-1

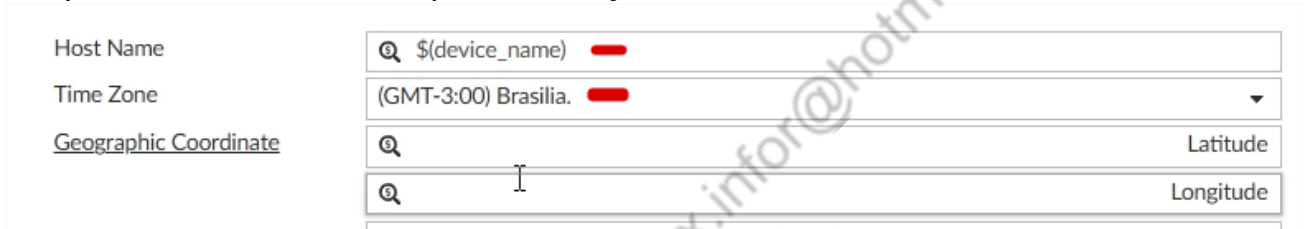


Per-Device Mapping

Mapped Device: ↑ FGT-FILIAL-2 [global]

Value: FGT-FILIAL-2

Após adicionar os 2 mapeamentos, clique em OK, e informe a variável recém criada no campo "Host Name", vamos aproveitar e ajustar o Fuso Horário também:



Host Name: 🔍 \$(device_name) —

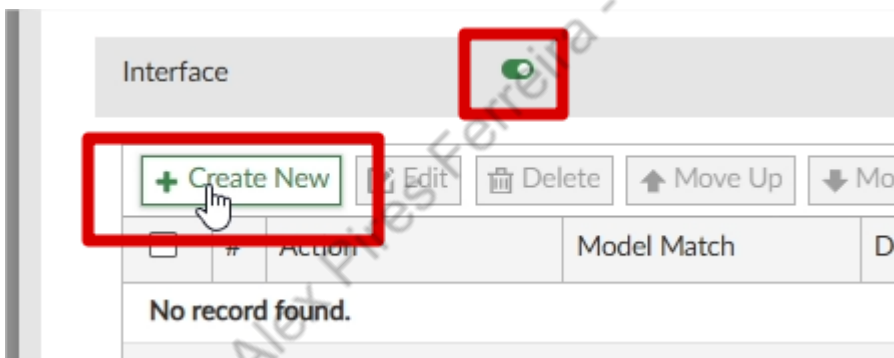
Time Zone: (GMT-3:00) Brasilia. —

Geographic Coordinate

🔍 Latitude

🔍 Longitude

No fim do template, temos a aba "INTERFACE", ative a mesma e clique em Create New:



Interface 🔍

+ Create New Edit Delete Move Up Move Down

	#	Action	Model Match	De
No record found.				

Agora iremos realizar a configuração de todas as interfaces, para isso iremos utilizar variáveis também, abaixo prints de como será configurado, e das variáveis com os seus valores para cada unidade:

Name

port2_prefix

Description

Default Value

Per-Device Mapping

+ Create New

Edit

Delete

Search...

☐	Mapped Device	Value
☐	↑ FGT-FILIAL-1 [global]	120.120.120.
☐	↑ FGT-FILIAL-2 [global]	140.140.140.

Action

Model

Interface Name

IP/Netmask

Administrative Access

Config Interface

all

port2

\$(port2_prefix)2/30

☐ dnp
☐ ftm
☒ ping
☐ snmp
☐ telnet

☐ fabric
☐ http
☐ probe-response
☐ speed-test

☐ fgfm
☐ https
☐ radius-acct
☐ ssh

Name

port3_prefix

Description

Default Value

0/4096

Per-Device Mapping ▾

+ Create New

Edit

Delete

Search...

☐	Mapped Device ▾	Value ▾	
☐	↑ FGT-FILIAL-1 [global]	130.130.130.	
☐	↑ FGT-FILIAL-2 [global]	150.150.150.	
			2

Action

Config Interface ▾

Model

all ▾

Interface Name

port3

IP/Netmask

\$(port3_prefix)2/30

Administrative Access

☐ dnp
☐ ftm
☒ ping
☐ snmp
☐ telnet

☐ fabric
☐ http
☐ probe-response
☐ speed-test

☐ fgfm
☐ https
☐ radius-acct
☐ ssh

Name

Description

Default Value

Per-Device Mapping ▾

☐	Mapped Device ▾	Value ▾
☐	↑ FGT-FILIAL-1 [global]	192.168.101.
☐	↑ FGT-FILIAL-2 [global]	192.168.102.

Action

Model

Interface Name

IP/Netmask

Administrative Access

☐ dnp	☐ fabric	☐ fgfm
☐ ftm	☐ http	☐ https
☒ ping	☐ probe-response	☐ radius-acct
☐ snmp	☐ speed-test	☐ ssh
☐ telnet		

Action	DHCP Server		
Model	all		
Interface Name	port4		
ID	2		
Default Gateway	\$(port4_prefix)1		
Netmask	255.255.255.0		
IP Range	Start IP	End IP	Action
	\$(port4_prefix)10	\$(port4_prefix)200	x +

Action	Add Zone
Model	all
Zone Name	LAN_FILIAL
Interface Members	port4 x +

4.1.9. CONFIGURAÇÃO SDWAN TEMPLATE "FILIAIS"

FMG-SECINFRA
Dashboard
Device Manager
Device & Groups
Scripts
Provisioning Templates
Firmware Templates
Monitors
Policy & Objects

Install Wizard

Template Groups

IPsec Tunnel

SD-WAN

System Templates

Static Route

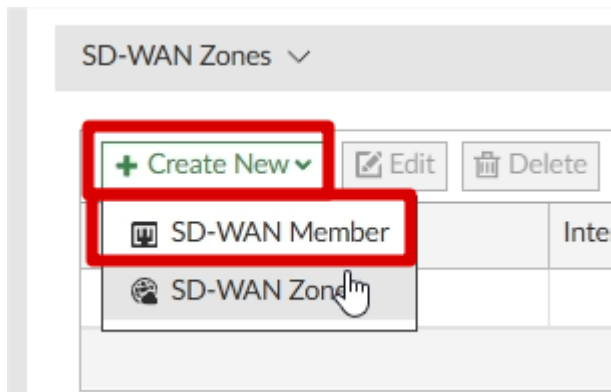
CLI

Feature Visibility

Toggle Horizontal Menu

No record found.

Name	FILIAIS
Description	
SD-WAN Status	☒



Criaremos os 2 membros, usando as variáveis que criamos no System Templates:

Sequence Number
Interface Member
SD-WAN Zone
Gateway IP
Cost
Priority
Status
Installation Target

☒

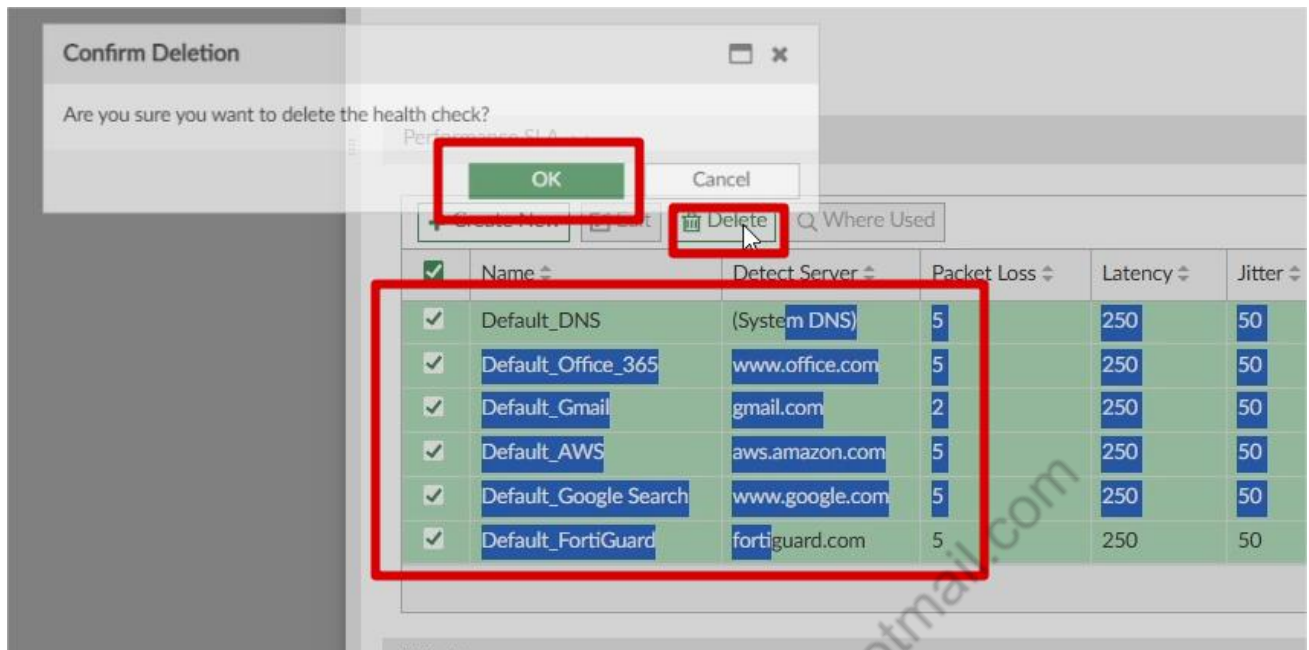
Sequence Number
Interface Member
SD-WAN Zone
Gateway IP
Cost
Priority
Status
Installation Target

☒

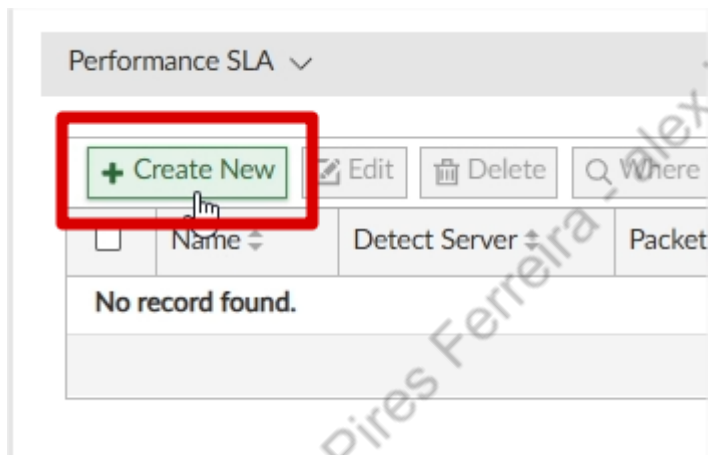
A configuração ficará assim:

+ Create New Edit Delete Where Used Search...							
☐	ID	Interface	Gateway	Cost	Priority	Status	
☐	virtual-wan-link						
☐	1	port2	\${port2_prefix}1	0		Enable	
☐	2	port3	\${port3_prefix}1	0		Enable	
							3

Vamos deletar todos os SLAs padrões:



Vamos criar um novo:



Name
SLA_INTERNET
IP Version
IPv4 IPv6
Probe Mode
Active
Enable Probe Packets
Protocol
Ping
Server
8.8.8.8
Participants
All SD-WAN Members Specify
port2 port3
2 entries selected
Installation Target
Click to select

SLA Target
Latency Threshold Jitter Threshold Packet Loss Threshold Mos Threshold Action
+ Add Target

SLA Target
Latency Threshold Jitter Threshold Packet Loss Threshold Mos Threshold
50 ms 10 ms 2 %
Link Status
Check Interval 500 Milliseconds
Failure Before Inactive 5
Restore Link After 5 Ch
Probe Timeout 500 Milliseconds
Action When Inactive
Update Static Route
Cascade Interfaces
Advanced Options >
OK Cancel

Vamos criar uma SD-WAN Rule simples para a internet:

SD-WAN Rules ▾					
+ Create New Edit Delete More ▾					
☐	ID	Name	Source	Destination	Criteria
☐		sd-wan	All	All	Source IP

Já vamos aproveitar para criar o objeto "RD_LAN":

Name
SAIDA_INTERNET
Status
☒
IP Version
IPv4 IPv6
Installation Target

Click to select

Source

Source Address

Click to select

User Group

Click to select

Destination

Address

Click to select

Route Tag

0

Internet Service

Click to select

Selected 0 (Total: 21)

IP/Netmask: 10.0.0.0/255.0.0.0

☐ RFC1918-172
IP/Netmask: 172.16.0.0/255.240.0.0

☐ RFC1918-192
IP/Netmask: 192.168.0.0/255.255.0.0

☐ SSLVPN_TUNNEL_ADDR1
IP Range: 10.212.134.200-10.212.134.210

☐ wildcard.dropbox.com
FQDN:*.dropbox.com

☐ wildcard.google.com
FQDN:*.google.com

OK Cancel

Category

Name **RD_LAN**

Color  [Change](#)

Type Subnet

IP/Netmask **192.168.30.0/255.255.255.0**

Interface ☐ any

Static Route Configuration ☐

Comments

Add To Groups

Advanced Options >

Per-Device Mapping >

Revision

Change Note

Vamos criar um mapeamento, para cada filial, com sua respectiva rede LAN:

Per-Device Mapping ▾

[+ Create New](#) [Edit](#) [Delete](#)

☐	Mapped Device *	Details *
☐	↑ FGT-FILIAL-1 [root]	IP/Netmask: 192.168.101.0/255.255.255.0
☐	↑ FGT-FILIAL-2 [root]	IP/Netmask: 192.168.102.0/255.255.255.0

Após, clicar em OK e usar o objeto na SDWAN Rule:

Source	
Source Address	 RD_LAN IP/Netmask: 192.168.30.0/255.255.255.0
User Group	Click to select

Como destino, vamos usar o grupo padrão RFC1918-GRP, não se preocupe, após iremos utilizar a opção “DST-NEGATE”:

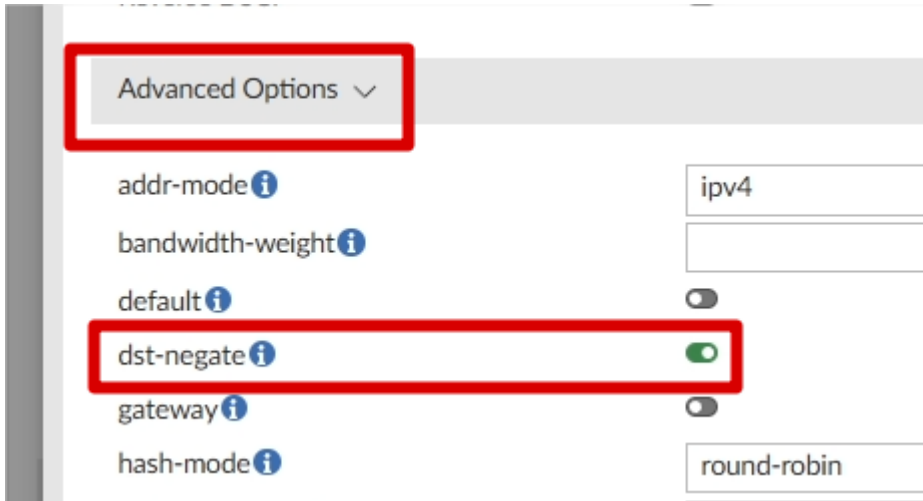
Destination	
Address	 RFC1918-GRP Group Members (3): RFC1918-10, RFC1918-172, RFC1918-192
Route Tag	0

1 entry selected

Vamos definir a estratégia da SDWAN RULE:

Outgoing Interfaces	
Strategy	Lowest Cost (SLA)
Interface Preference	+
	port2 port3
Zone Preference	+
Measured SLA	Click to select
Required SLA Target	+
	SLA_INTERNET#1 Ping: 8.8.8.8 ; Latency: 50ms, Jitter: 10ms, Packet Loss: 2%
Quality Criteria	Latency

Em advanced options, vamos marcar dst-negate:



Advanced Options ▾

addr-mode ⓘ ipv4

bandwidth-weight ⓘ

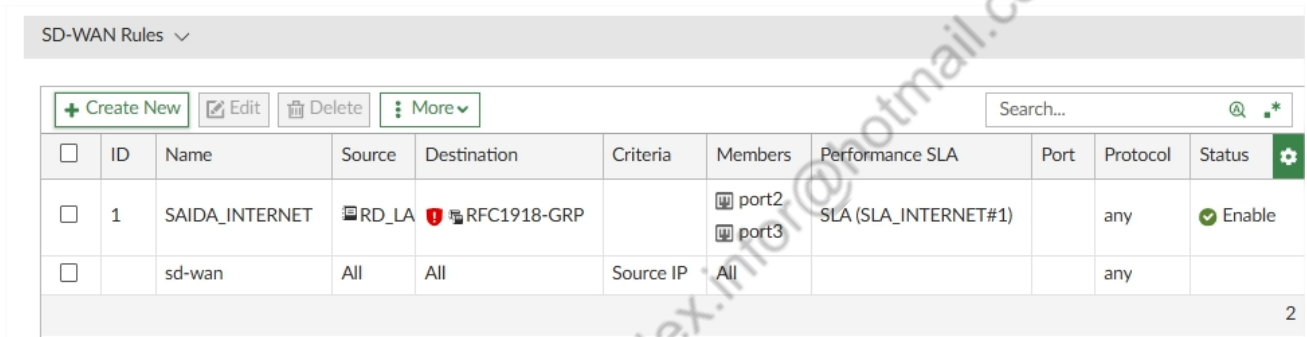
default ⓘ

dst-negate ⓘ **on**

gateway ⓘ

hash-mode ⓘ round-robin

A SDWAN Rule ficará assim:



SD-WAN Rules ▾

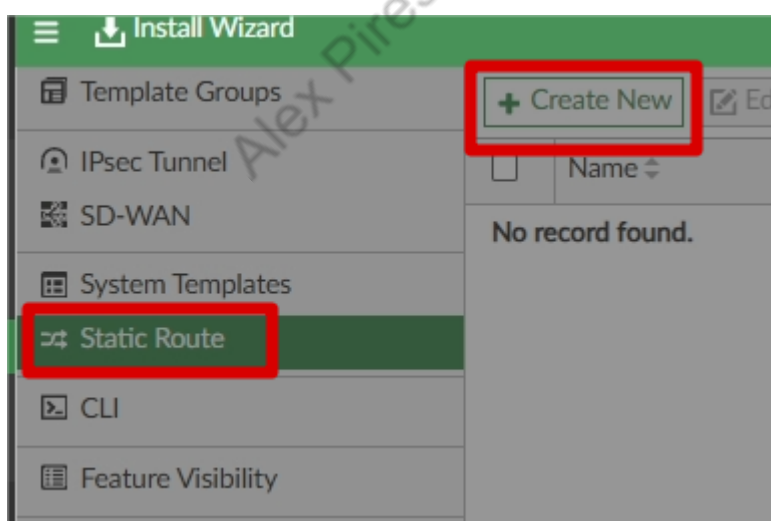
[+ Create New](#) [Edit](#) [Delete](#) [More ▾](#)

☐	ID	Name	Source	Destination	Criteria	Members	Performance SLA	Port	Protocol	Status	
☐	1	SAIDA_INTERNET	RD_LA	RFC1918-GRP		port2 port3	SLA (SLA_INTERNET#1)		any	Enable	
☐		sd-wan	All	All	Source IP	All			any		

2

Após isso podemos clicar em OK, e o template SDWAN já estará terminado.

4.1.10. STATIC ROUTE TEMPLATE “FILIAIS”



Create New Static Route Template

Name

Description

☐ Destination

☐ IPv4

☐ IPv6

IPv4 Static Route

Type IPv4

Destination

Gateway Address

☐ SD-WAN ☐

SD-WAN Zone

Administrative Distance

Description

Após isso, é muito importante, clicar em Advanced Options e alterar o "SEQ-NUM" para 1:

Advanced Options

bfd ☐

blackhole ☐

dst-type

dstaddr

dynamic-gateway ☐

link-monitor-exempt ☐

priority

seq-num

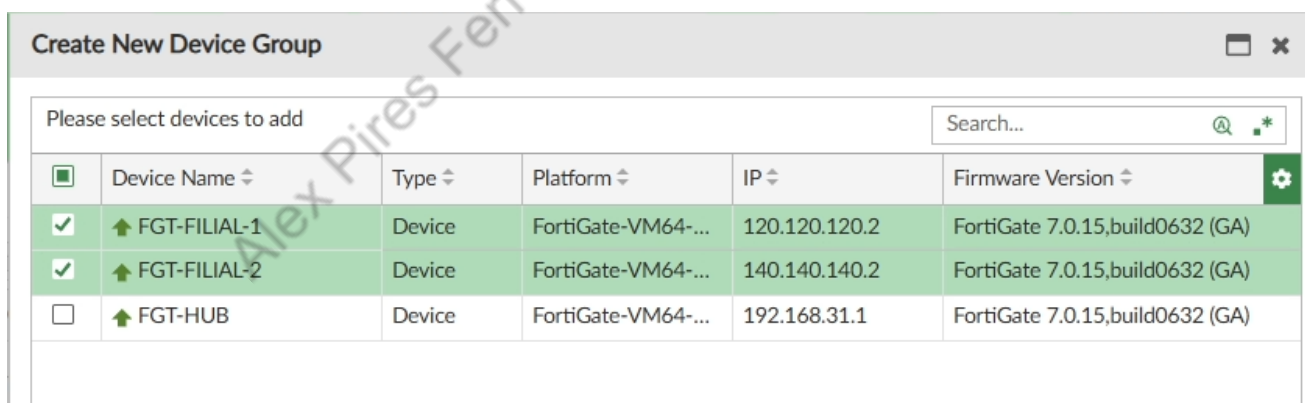
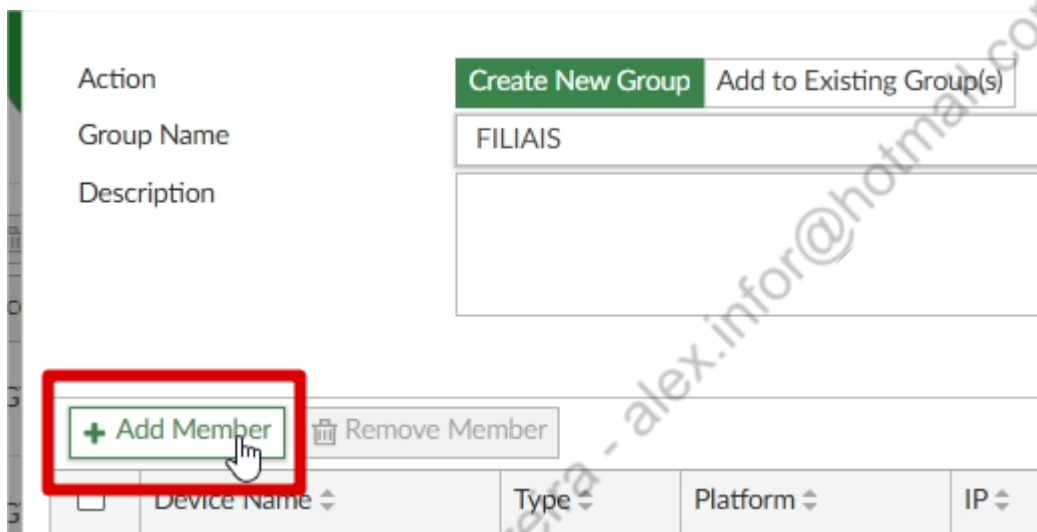
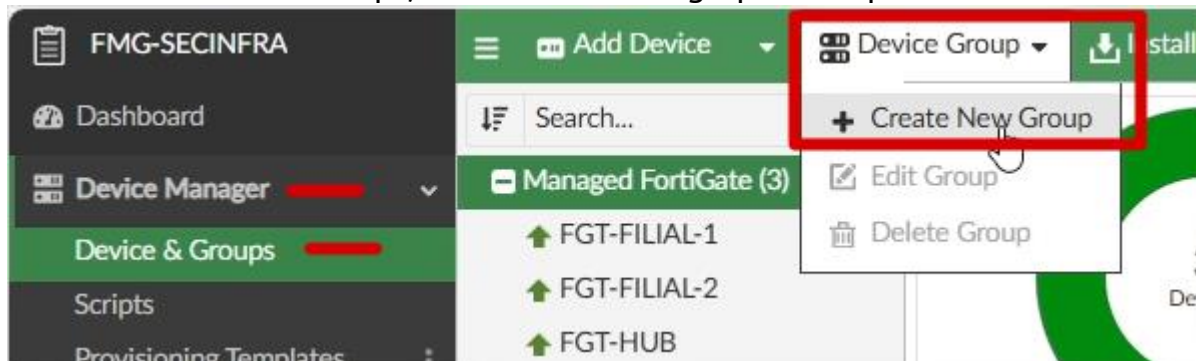
vrf

weight

Após isso, basta clicar em OK, e nosso template estará criado.

4.1.11. DEVICE GROUP E TEMPLATE GROUP

Na tela de Device & Groups, vamos criar nosso grupo de dispositivos:



Create New Device Group

Action

Create New Group

Add to Existing Group(s)

Group Name

FILIAIS

Description

0/128

+ Add Member

Remove Member

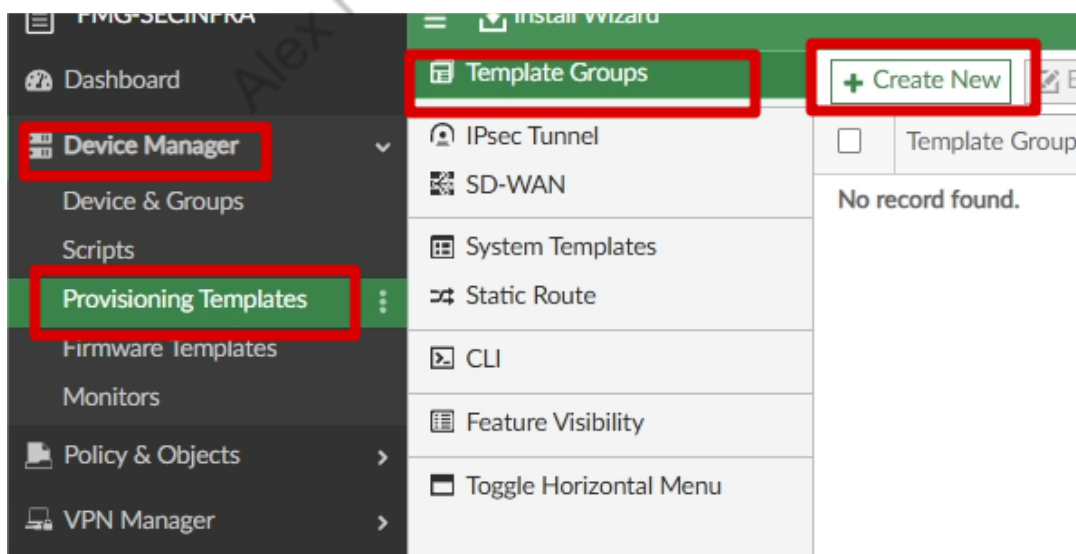
Search...

☐	Device Name	Type	Platform	IP	Firmware Version	
☐	↑ FGT-FILIAL-1	Device	FortiGate-VM64-...	120.120.120.2	FortiGate 7.0.15,build0632 (GA)	
☐	↑ FGT-FILIAL-2	Device	FortiGate-VM64-...	140.140.140.2	FortiGate 7.0.15,build0632 (GA)	

2

Cancel

Agora iremos criar nosso Template Group, e após isso, vincular ao Device Group:



Create New Template Group

Name

Description

Provisioning Templates * Only one template can be selected for each template type.

Selezione o System Template, SDWAN Template e Static Route Template:

Provisioning Templates - FILIAIS

☒ System Template (1/2)

default

☒ FILIAIS

☐ Threat Weight Template (0/0)

☐ IPsec Tunnel Template (0/3)

☐ BGP Template (0/2)

☒ Static Route Template (1/1)

☒ FILIAIS

☐ NSX-T Service Template (0/0)

☒ SD-WAN Template (1/1)

☒ FILIAIS

☐ CLI Template (0/0)

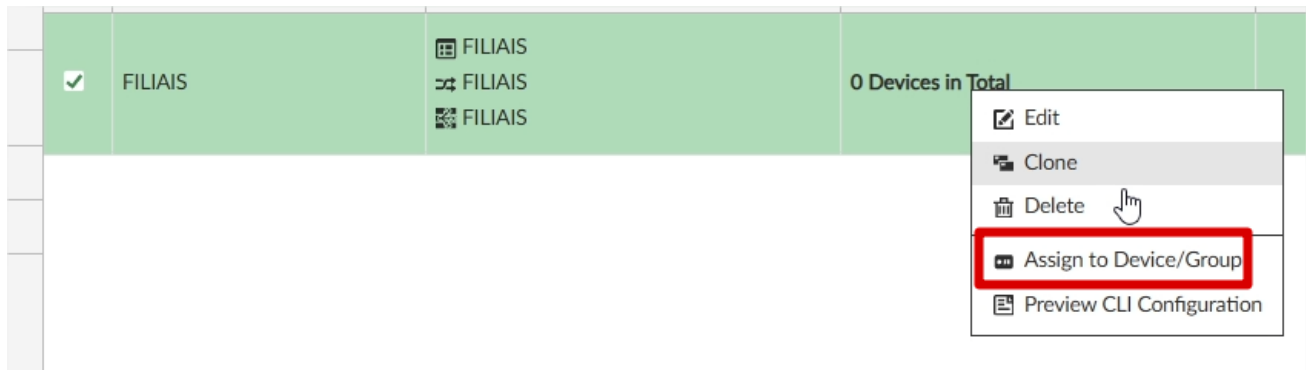
☐ CLI Template Group (0/0)

☐ AP Profile (0/74)

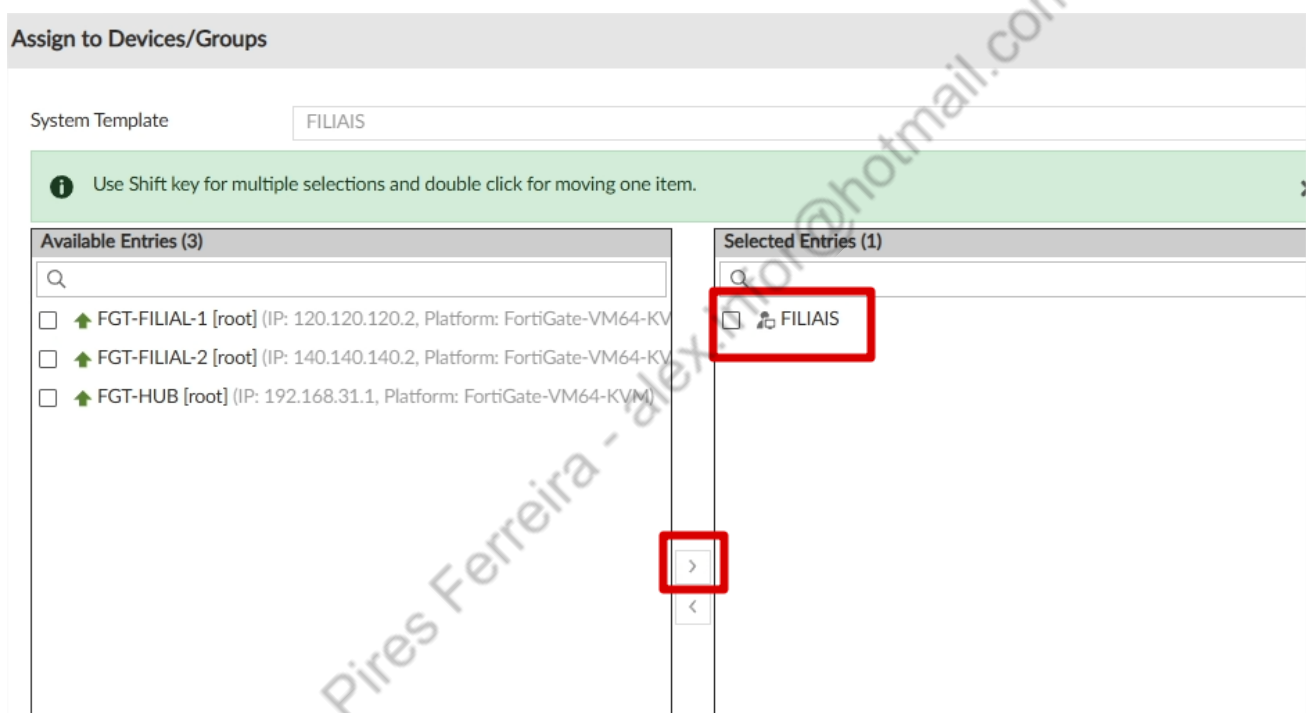
☐ FortiSwitch Template (0/0)

☐ FortiExtender Profile (0/0)

Após clicar em OK, clique com o botão direito no Template Group e clique em Assign Device/Group:

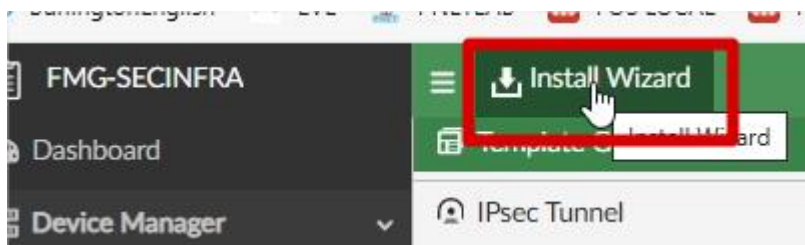


Mova o Grupo “FILIAIS” para a tela da direita:



4.1.12. INSTALAÇÃO DEVICES

Vamos testar nossos templates, mandando instalar nos Devices:



Install Wizard - Choose What to Install (1/4)

☐ Install Policy Package & Device Settings

☒ **Install Device Settings (only)**

i Install only device settings for a select set of devices. Policy and Object changes will not be updated from the last install.

Install Comments

0/127

< Back **Next >** Cancel

Install Wizard - Select Devices to Install (2/4)

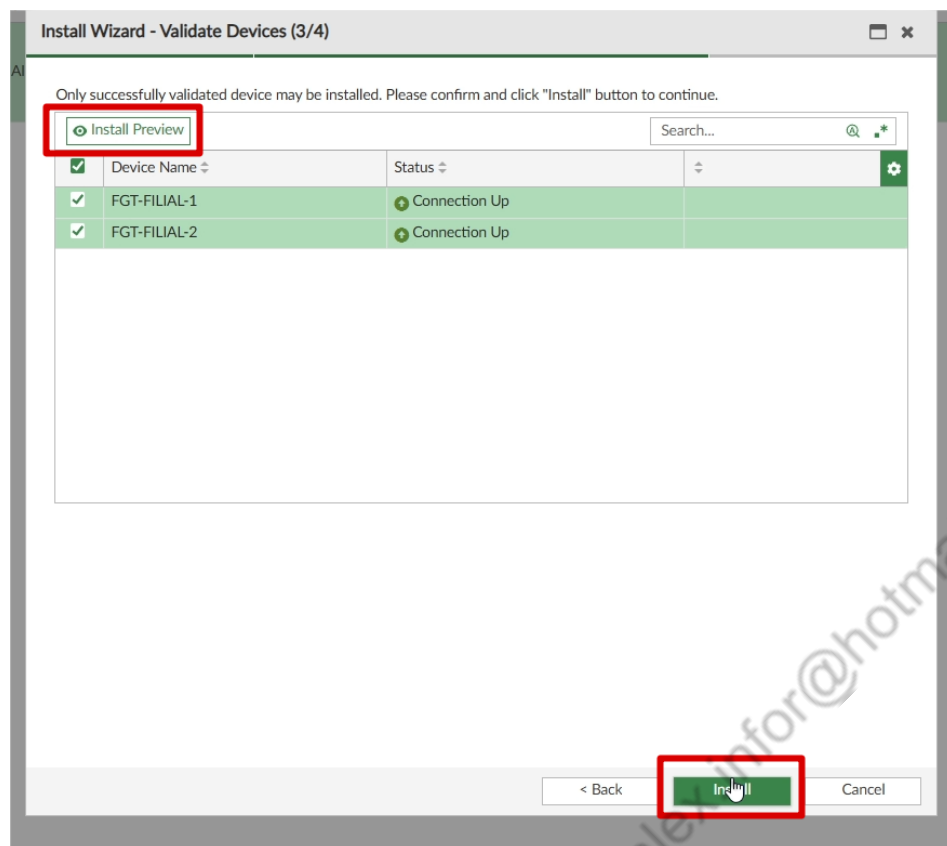
i Please select one or more devices to install (Use checkbox or Ctrl or Shift key for multiple selections)

Search...

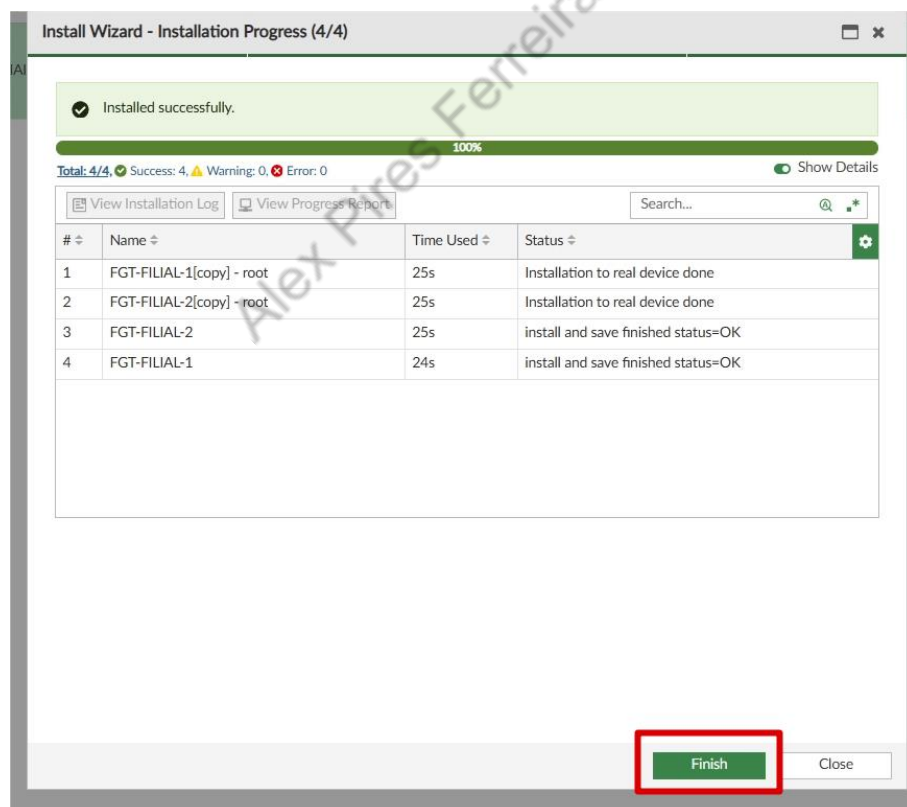
☒	Device Name	IP Address	Platform
☒	FILIAIS		
☒	FGT-FILIAL-1	120.120.120.2	FortiGate-VM64-KVM
☒	FGT-FILIAL-2	140.140.140.2	FortiGate-VM64-KVM

< Back **Next >** Cancel

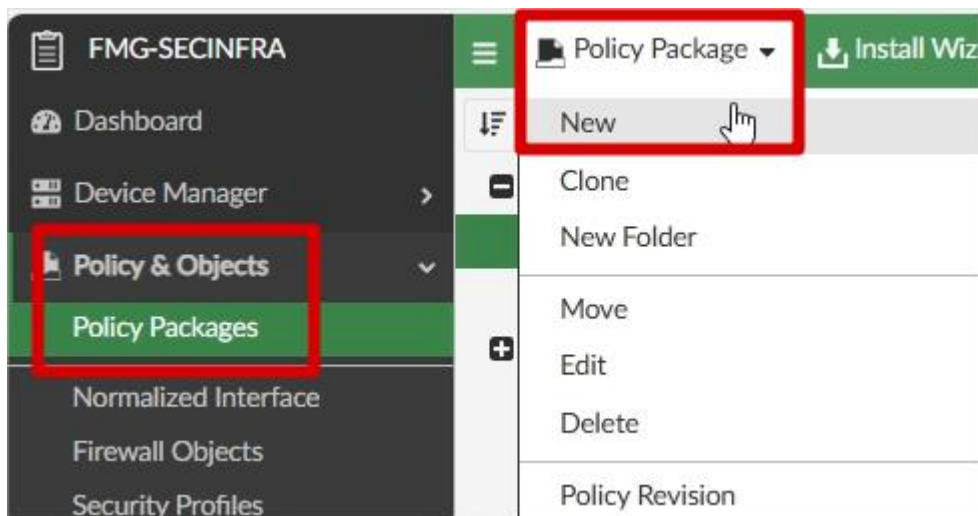
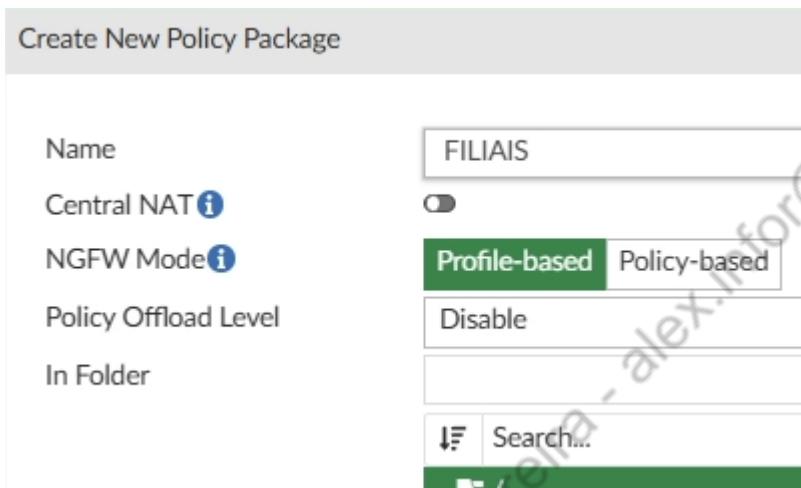
Nesta tela, você pode clicar em Install Preview, para revisar o que será enviado para os FortiGates, e após isso clicar em “Close” e INSTALL:



Aguardar o término, e fechar a tela.

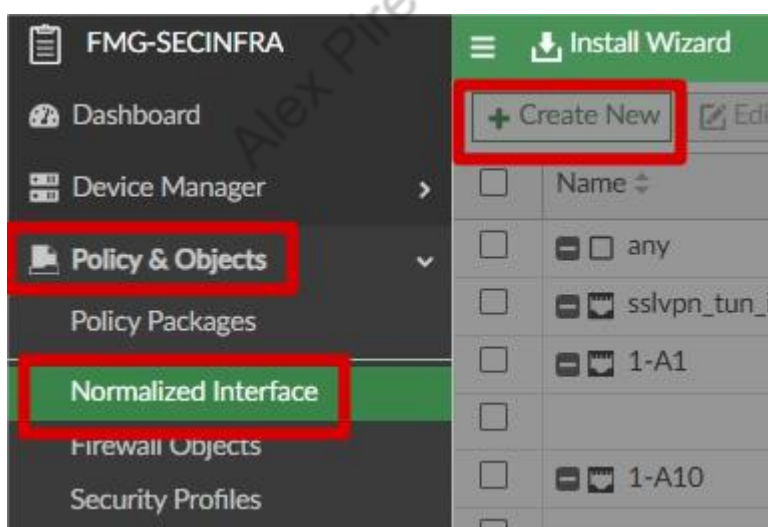


4.1.13. Criação de Policy Package “FILIAIS”

The screenshot shows the 'Create New Policy Package' form. The 'Name' field is filled with 'FILIAIS'. The 'Central NAT' toggle is turned off. The 'NGFW Mode' is set to 'Profile-based'. The 'Policy Offload Level' is set to 'Disable'. The 'In Folder' field is empty. At the bottom, there is a search bar with the text 'Search...'.

Agora que criamos o Policy Package, antes de criarmos a primeira policy, precisamos “Normalizar” a zona “LAN_FILIAL”, para isso, siga os passos abaixo:



Create New Normalized Interface

Name LAN_FILIAL

Description

Color  Change

Wildcard 

Per-Platform Mapping ▾

+ Create New ✎ Edit 🗑 Delete

☐	Name ▾	Device Interface Name ▾	Shapi
☐	all	LAN_FILIAL	

Agora vamos criar a primeira Policy:

Policy Package ▾

Search...

+ Create New ▾ ✎ Edit ▾ 🗑 Delete

 FGT-HUB

Firewall Policy

Installation Targets

 FILIAIS

Firewall Policy

Installation Targets

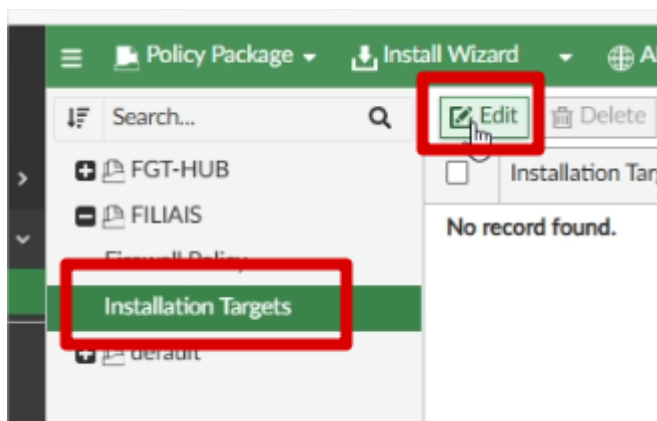
 default

☐	#	Name
☐		 Implicit (1/1 Total:1)
☐	1	Implicit Deny

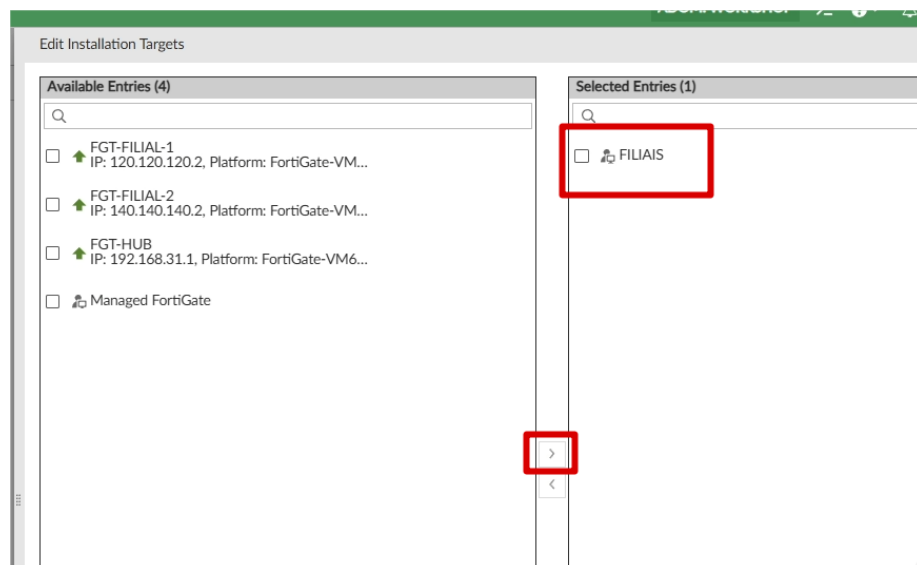
Create New Firewall Policy

ID	0
Name	LAN to INTERNET
Type	Standard ZTNA
Incoming Interface	LAN_FILIAL Unmapped/Interface
Outgoing Interface	virtual-wan-link Unmapped/Interface
Source	RD_LAN IP/Netmask: 192.168.30.0/255.255.255.0
Negate Source	☐
IP/MAC Based Access Control	☐
Destination	RFC1918-GRP Group Members (3): RFC1918-10, RFC1918-172, RFC1918-192
Negate Destination	☐
Service	ALL
Schedule	always
Action	Accept Deny IPSEC
Inspection Mode	Flow-based Proxy-based
Firewall/Network Options	
NAT	☒ NAT NAT46 NAT64
IP Pool Configuration	Use Outgoing Interface Address Use Dynamic IP Pool
Preserve Source Port	☐
Protocol Options	default

Agora vamos em Installation Targets, e adicionar o grupo FILIAL:

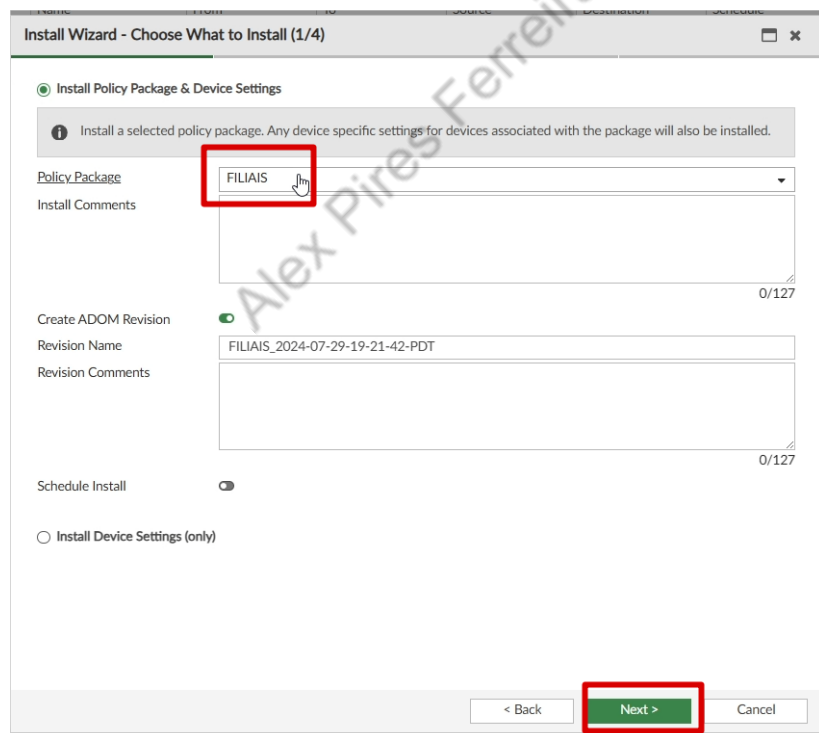
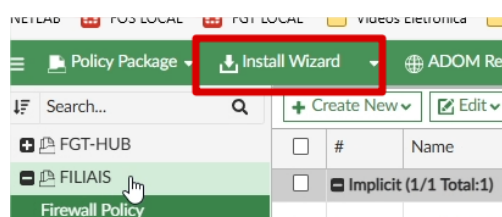


Deixe o grupo FILIAIS na caixa da direita, usando o botão do meio para mover, após selecionar.



Após isso pode apertar OK.

Vamos agora instalar o Policy Package:



Install Wizard - Select Devices to Install (FILIAIS) (2/4)

Please select one or more devices to install (Use checkbox or Ctrl or Shift key for multiple selections)

Search...

☒	Device Name ⇅	IP Address ⇅	Platform ⇅	
☒	FILIAIS			

< Back Next > Cancel

Agora você pode clicar em Install Preview, e ver o que será instalado, após validar, clique em CLOSE e depois em INSTALL:

Install Wizard - Validate Devices (FILIAIS) (3/4)

Installation Preparation Total: 3/3, Success: 3, Warning: 0, Error: 0 Show Details

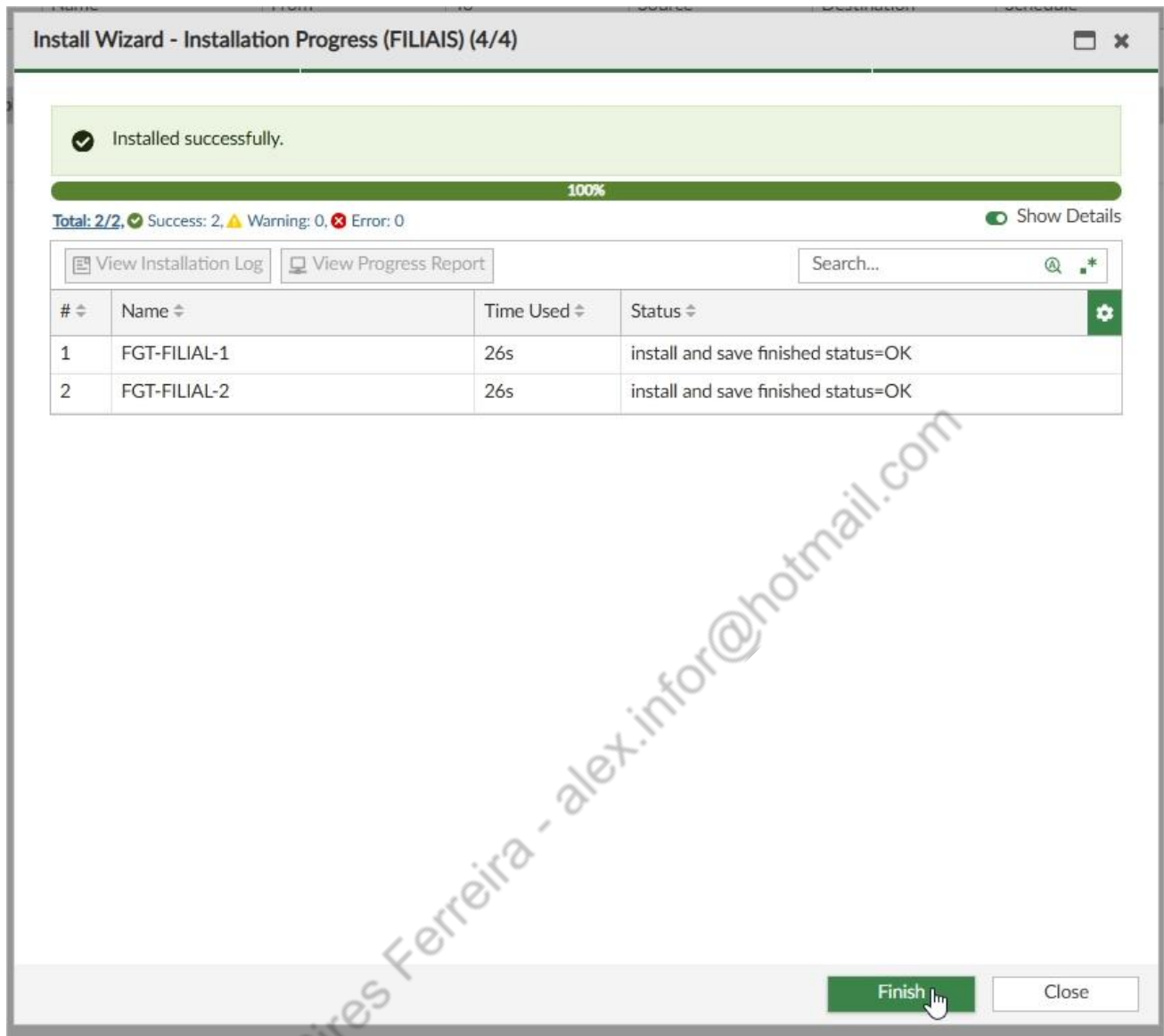
- ✓ Interface Validation
- ✓ Policy and Object Validation
- ✓ Ready to Install

Install Preview Policy Package Diff Search...

☒	Device Name ⇅	Status ⇅	
☒	FGT-FILIAL-1[root]	Connection Up	
☒	FGT-FILIAL-2[root]	Connection Up	

< Back Install Cancel

Após concluir, clicar em FINISH:

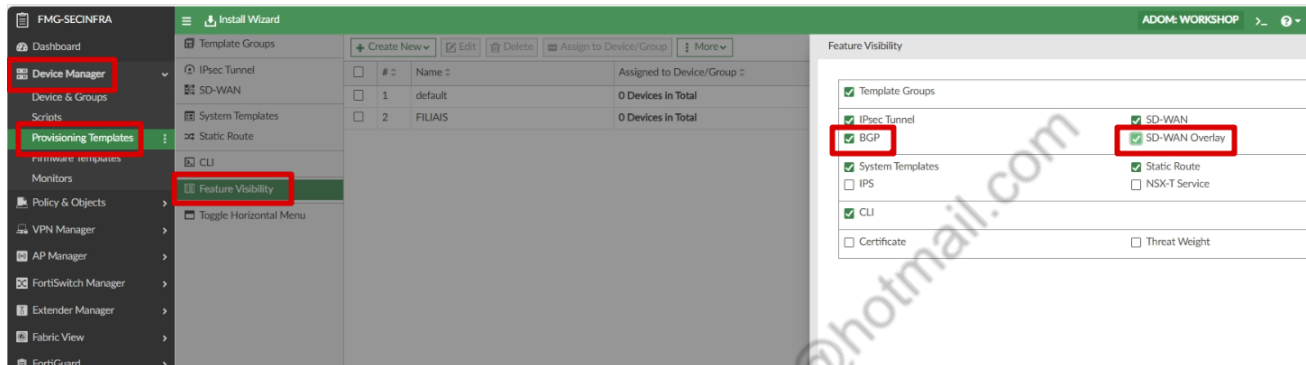


4.2. SDWAN OVERLAY TEMPLATE

Vamos começar a configuração do SDWAN Overlay Template

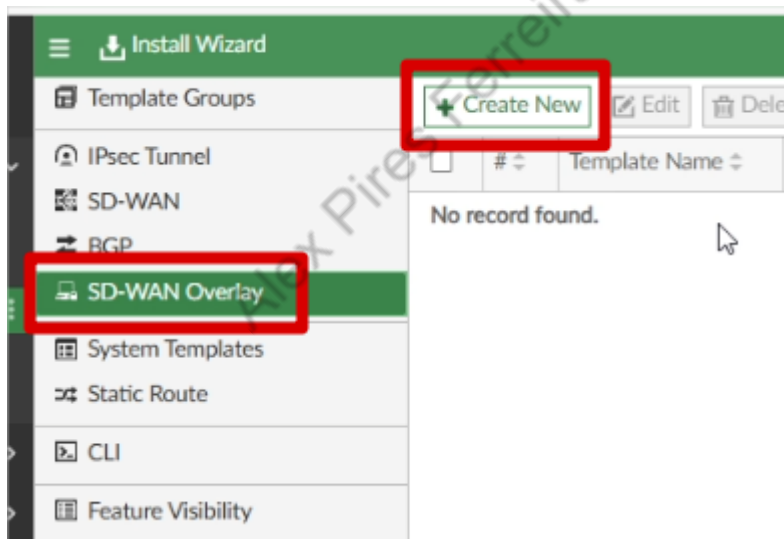
4.2.1.HABILITAR SDWAN OVERLAY TEMPLATE E BGP TEMPLATE

Vamos em Device Manager > Provisioning Templates > Feature Visibility, e marcamos SDWAN Overlay e BGP.



4.2.2.CRIAÇÃO SDWAN OVERLAY TEMPLATE

No menu SDWAN OVERLAY, vamos criar um template:



Vamos ajustar as redes para Loopback IP Address e Overlay Network

Create New SD-WAN Overlay Template - Region Settings (1/5)

Name
Description

HUB_WORKSHOP

0/255

Select New Topology

Single HUB

Dual HUB
(Primary & Secondary)

Dual HUB
(Primary & Primary)

Advanced

Loopback IP Address
Overlay Network
BGP-AS Number
Auto-Discovery VPN

172.31.0.0/255.255.0.0
10.254.0.0/255.255.0.0
65000
☒

Na próxima tela, iremos selecionar o FGT-HUB como Standalone HUB, e o Device Group FILIAIS no Device Group Assignment.

Create New SD-WAN Overlay Template - Role Assignment (2/5)

Name
Topology

HUB_WORKSHOP
Single HUB Dual HUB (Primary & Secondary) Dual HUB (Primary & Primary)

HUB

Standalone HUB

↑ FGT-HUB

Branch

Device Group Assignment
Automatic Branch ID Assignment

FILIAIS
☐

Aqui iremos configurar as portas dos links 1 e 2, e as redes do FGT-HUB:

HUB

 Standalone HUB

Underlay

Network Advertisement

Advanced >

 FGT-HUB

#	Private Link 	Override IP 	Action
WAN Underlay 1	☐ port2	☐	x +
WAN Underlay 2	☐ port3	☐	x +

Connected Static

#	Interface	Action
Network Prefix 1	192.168.30.0/24	x +
Network Prefix 2	192.168.31.0/24	x +

Branch Route Maps

Route map in ☐

Route map out ☐

Aqui iremos configurar as portas dos links 1 e 2 das filiais, assim como a rede da filial, usando nossa variável:

Branch

 Branch Device Group

Underlay

Network Advertisement

Advanced v

 FILIAIS

#	Private Link 	Action
WAN Underlay 1	☐ port2	x +
WAN Underlay 2	☐ port3	x +

Connected Static

#	Interface	Action
Network Prefix 1	\${port4_prefix}0/24	x +

Página 49 | 101

Na penúltima tela, iremos marcar as seguintes opções, na primeira opção, lembrar de marcar o SD-Wan template que criamos anteriormente para as Filiais:

Create New SD-WAN Overlay Template - SD-WAN Template Options (4/5)

Add Overlay Objects to SD-WAN Template

FILIAIS

Add Overlay Interfaces and Zones

Add Health Check Servers for Each HUB as Performance SLA

Normalize Interfaces

Após isso, a última tela é apenas para conferência, revise e confirme!

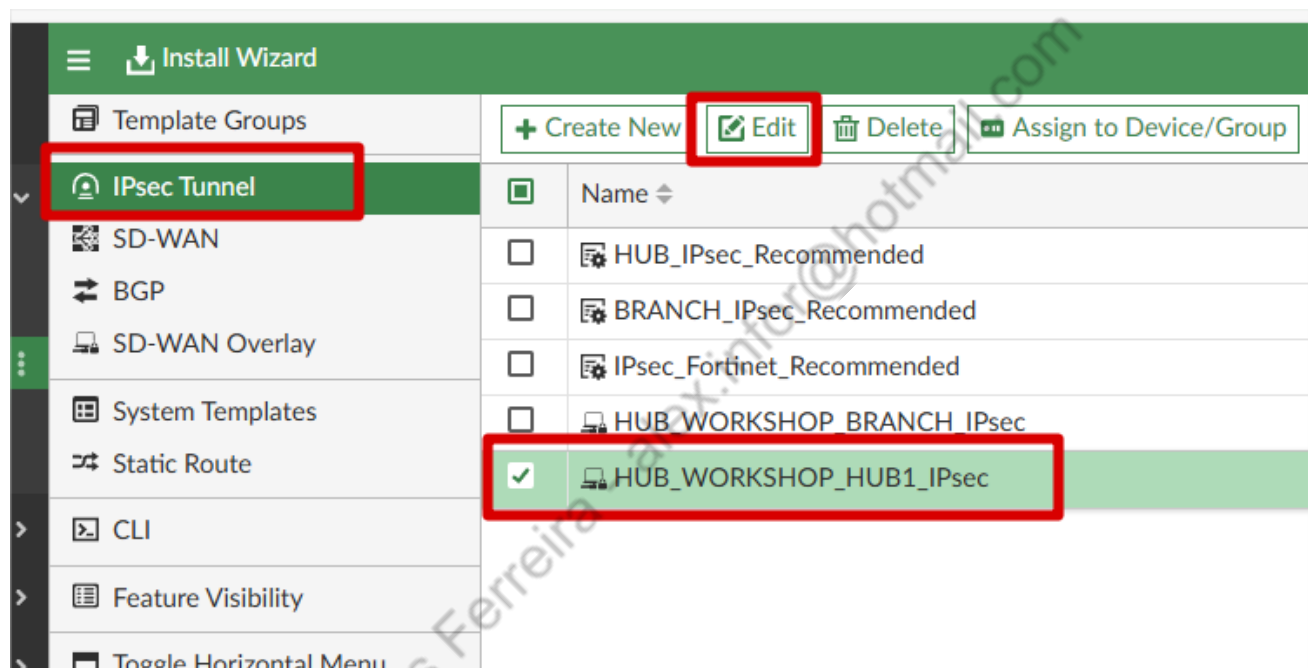
Alex Pires Ferreira - alex.infor@hotmail.com

4.3. TEMPLATES E VARIÁVEIS

Nesta etapa, iremos realizar ajustes nos templates gerados pelo SDWAN Overlay Templates, e também iremos repassar em algumas configurações, mostrando a lógica implementada pelo SD-WAN Overlay Template.

4.3.1.AJUSTES TEMPLATES IPSEC

Vamos ajustar os templates IPSEC, vamos começar pelo template do HUB:

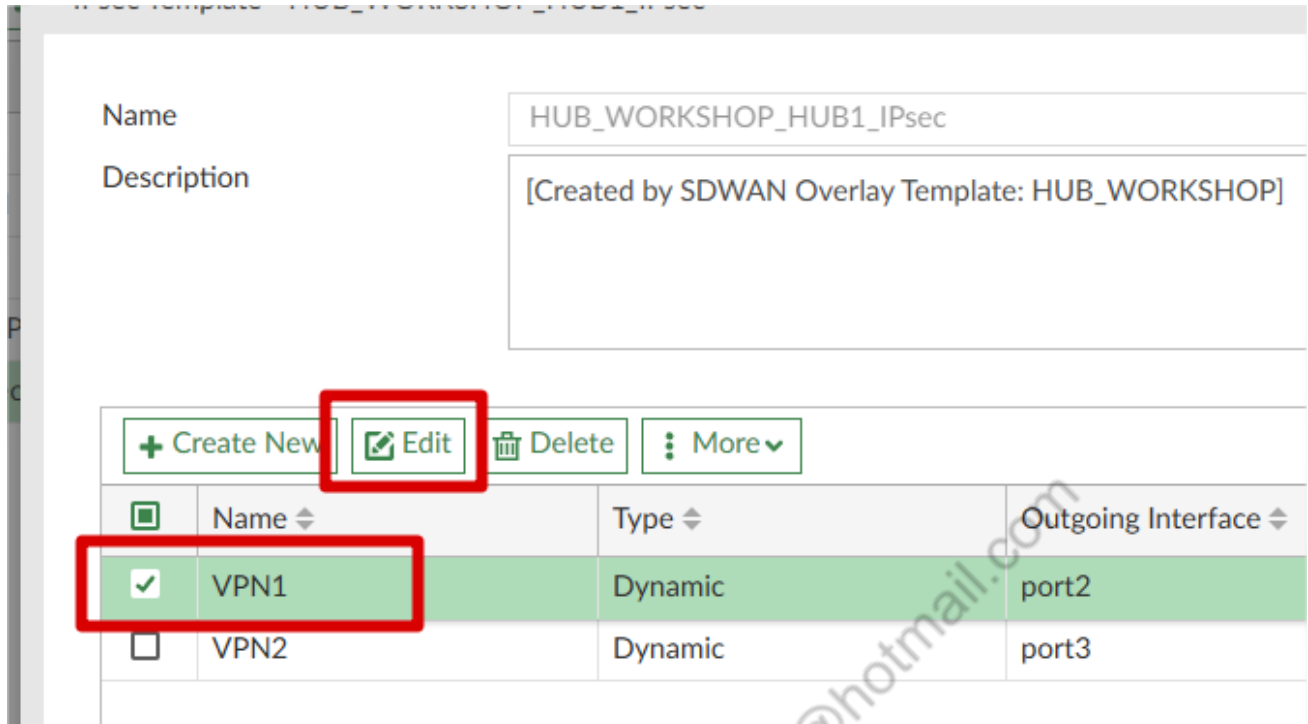


The screenshot displays the 'Install Wizard' interface for SD-WAN Overlay Templates. The left sidebar shows a list of template groups, with 'IPsec Tunnel' highlighted. The main area shows a list of templates under the 'IPsec Tunnel' group. The 'HUB_WORKSHOP_HUB1_IPsec' template is selected, indicated by a checkmark in the selection column and a red box around the template name.

Template Groups		Actions	
		+ Create New	Edit
IPsec Tunnel			
SD-WAN			
BGP			
SD-WAN Overlay			
System Templates			
Static Route			
CLI			
Feature Visibility			
Toggle Horizontal Menu			

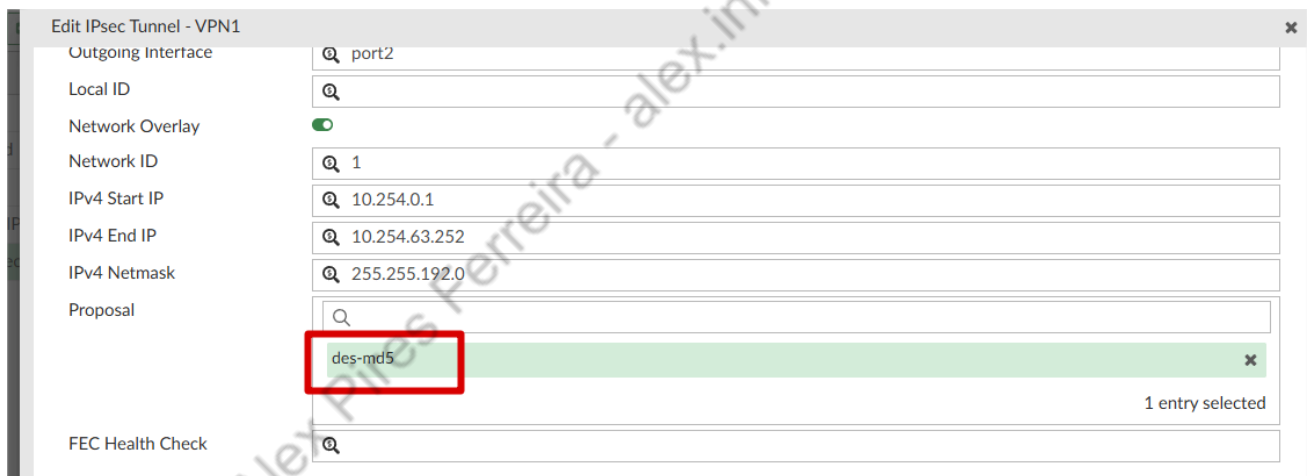
Name
HUB_IPsec_Recommended
BRANCH_IPsec_Recommended
IPsec_Fortinet_Recommended
HUB_WORKSHOP_BRANCH_IPsec
☒ HUB_WORKSHOP_HUB1_IPsec

Edite a VPN1:



Name	Type	Outgoing Interface
VPN1	Dynamic	port2
VPN2	Dynamic	port3

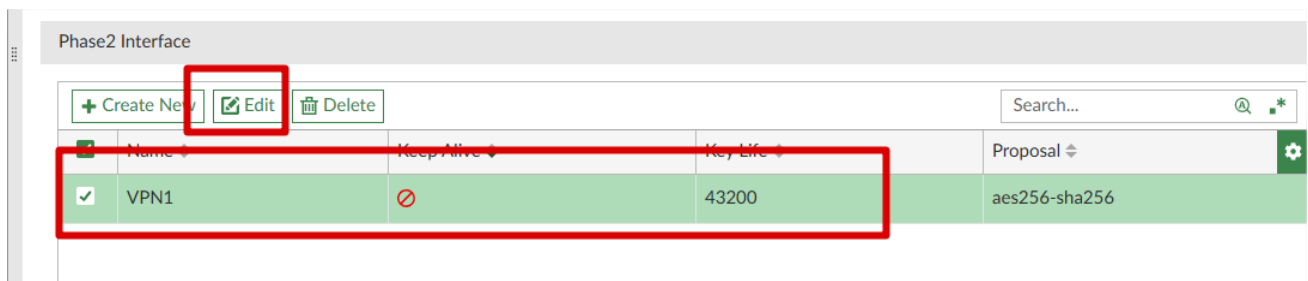
Alterar criptografia da Phase1 para des-md5:



des-md5

1 entry selected

Após isso, na Phase2 iremos alterar a criptografia e habilitar o Keep Alive:



Name	Keep Alive	Key Life	Proposal
VPN1	⊘	43200	aes256-sha256

Edit Phase 2 Interface

Name

VPN1

Keep Alive

☒

Key Life

43200

Proposal

des-md5

1 entry selected

Advanced Options >

Em advanced Options da Phase1 precisamos alterar estas 2 opções (alteramos para evitar erro no template):

exchange-ip-addr4

0.0.0.0

exchange-ip-addr6

::

fec-base

Click to select

fec-codec

Click to select

fec-egress

☐

fec-ingress

☐

fec-mapping-profile

Click to select

fec-receive-timeout

1

fec-redundant

1

fec-send-timeout

1

fgsp-sync

☐

forticlient-enforcement

☐

fragmentation

☒

fragmentation-mtu

1200

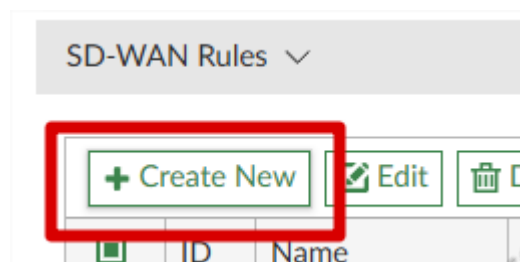
Realizar as mesmas alterações na VPN2, e após isso, também realizar as mesmas alterações no template HUB_WORKSHOP_BRANCH_IPSec.

4.3.2.AJUSTES TEMPLATE SD-WAN FILIAL

Vamos editar o Template SD-WAN Filial e começar ajustando os membros, você vai ver que o FortiManager criou 2 zonas “WAN1” e “WAN2”, não vamos utilizá-las, os membros deverão ficar assim:

SD-WAN Zones ▾							
+ Create New ▾ Edit Delete Q Where Used Search...							
☐	ID ▾	Interface ▾	Gateway ▾	Cost ▾	Priority ▾	Status ▾	Installation Target ▾ 
☐	virtual-wan-link						
☐	1	port2	\$(port2_prefix)1	0	0	✓ Enable	
☐	2	port3	\$(port3_prefix)1	0	0	✓ Enable	
☐	HUB1						
☐	3	HUB1-VPN1	0.0.0.0	0	0	✓ Enable	
☐	4	HUB1-VPN2	0.0.0.0	0	0	✓ Enable	
							6

Agora vamos criar uma SDWAN Rule do tráfego do HUB:



Edit SD-WAN Rule - 2

Name

MATRIZ_FILIAIS

Status

☒

IP Version

☒ IPv4
☐ IPv6

Installation Target

Click to select

Source

Source Address

RD_LAN
IP/Netmask: 192.168.30.0/255.255.255.0

1 entry selected

User Group

Click to select

Destination

Address

RFC1918-GRP
Group Members (3): RFC1918-10, RFC1918-172, RFC1918-192

1 entry selected

Route Tag

0

Protocol

☐ TCP
☐ UDP
☒ Any
☐ Specify
 0

Internet Service

Click to select

Outgoing Interfaces

Strategy

Lowest Cost (SLA)

Interface Preference

+

HUB1-VPN1
HUB1-VPN2

Zone Preference

+

Measured SLA

Click to select

Required SLA Target

+

HUB1_HC#1
Ping: 172.31.255.253 ; Latency: 100ms, Jitter: 15ms, Packet Loss: 1%

Quality Criteria

Latency

Forward DSCP

☐

Reverse DSCP

☐

Advanced Options >

OK

Cancel

Ajustar ordem da SDWAN Rules:

SD-WAN Rules										
+ Create New Edit Delete More Search...										
☒	ID	Name	Source	Destination	Criteria	Members	Performance SLA	Port	Protocol	Status
☒	2	MATRIZ_FILIAIS	RD_LAN	RFC1918-GRP		HUB1-VPN1 HUB1-VPN2	SLA (HUB1_HC#1)		any	Enable
☐	1	TO_INTERNET	RD_LAN	RFC1918-GRP		port2 port3	SLA (SLA_INTERNET#1)		any	Enable
☐		sd-wan	All	All	Source IP	All			any	
3										

4.3.3.AJUSTE BGP TEMPLATES

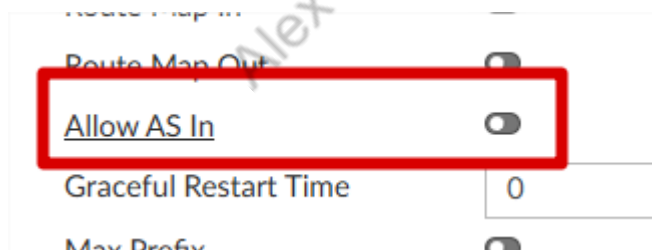
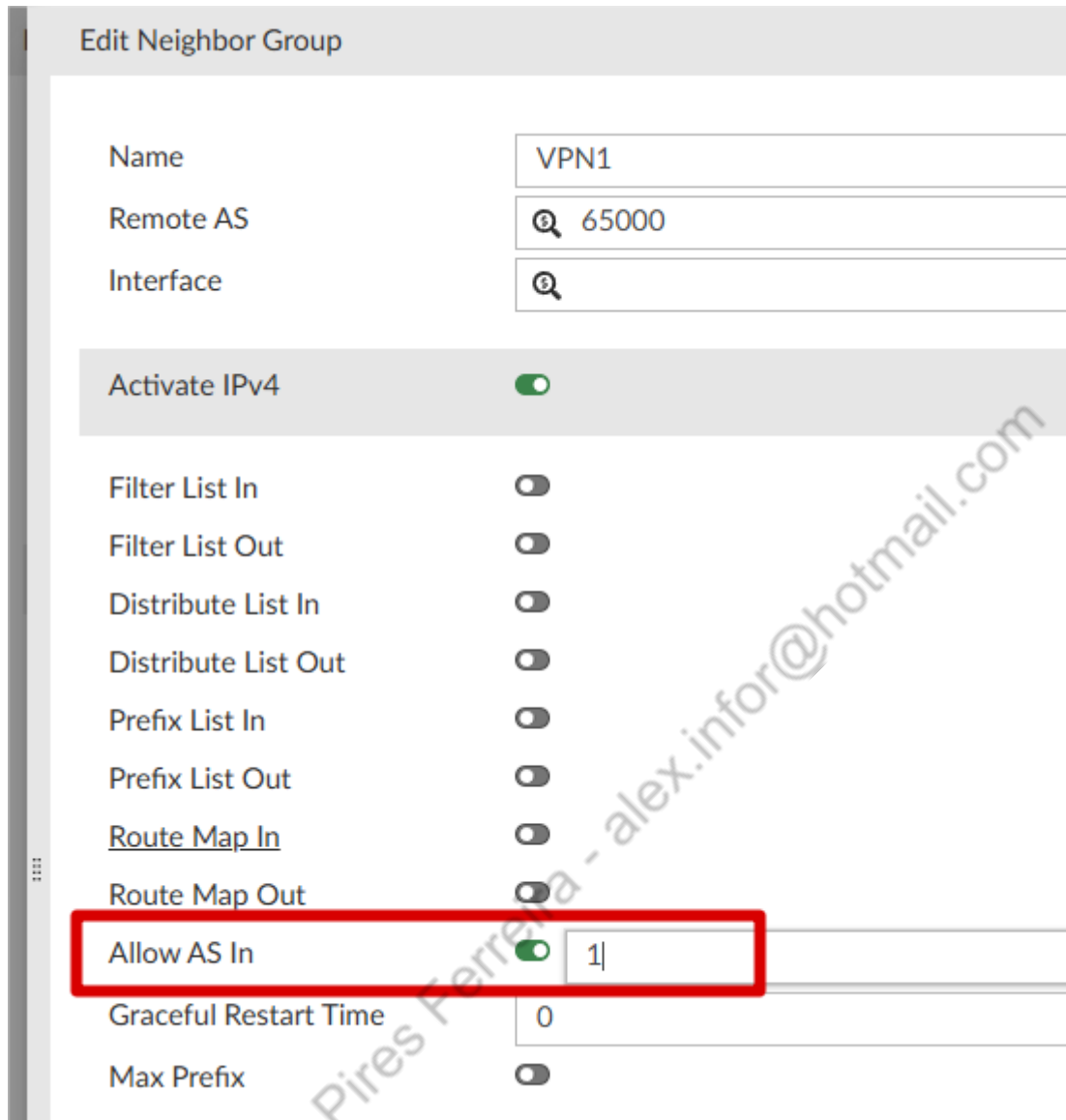
Vamos começar editando o template BGP do HUB:

Install Wizard																	
Template Groups IPsec Tunnel SD-WAN BGP SD-WAN Overlay System Templates Static Route CLI Feature Visibility	+ Create New Edit Delete Assign to Device/Group More	<table> <tr> <th>☒</th> <th>Name</th> <th>Assigned to Device/Group</th> </tr> <tr> <td>☐</td> <td>HUB_BGP_Recommended</td> <td>0 Devices in Total</td> </tr> <tr> <td>☐</td> <td>BRANCH_BGP_Recommended</td> <td>0 Devices in Total</td> </tr> <tr> <td>☐</td> <td>HUB_WORKSHOP_BRANCH_BGP</td> <td>0 Devices in Total</td> </tr> <tr> <td>☒</td> <td>HUB_WORKSHOP_HUB1_BGP</td> <td>0 Devices in Total</td> </tr> </table>	☒	Name	Assigned to Device/Group	☐	HUB_BGP_Recommended	0 Devices in Total	☐	BRANCH_BGP_Recommended	0 Devices in Total	☐	HUB_WORKSHOP_BRANCH_BGP	0 Devices in Total	☒	HUB_WORKSHOP_HUB1_BGP	0 Devices in Total
☒	Name	Assigned to Device/Group															
☐	HUB_BGP_Recommended	0 Devices in Total															
☐	BRANCH_BGP_Recommended	0 Devices in Total															
☐	HUB_WORKSHOP_BRANCH_BGP	0 Devices in Total															
☒	HUB_WORKSHOP_HUB1_BGP	0 Devices in Total															

Vamos alterar antes o Neighbor Group VPN1:

Neighbor Group		
+ Create New Edit Delete More		
☒	Name	Remote AS
☒	VPN1	65000
☐	VPN2	65000

Vamos ativar Allow AS IN em IPV4, e colocar 1, e depois desativar, isso é necessário para evitar erro no template (bug FortiManager):



Vamos fazer o mesmo procedimento na parte do IPV6 do neighbor group.

Após isso é necessário realizar no neighbour group VPN2, tanto IPV4 quanto IPV6.

Na tela principal do template BGP, no final da tela, teremos advanced Options, expandir Advanced Options e marcar o Recursive Next Hop:

Advanced Options ▾

Cluster ID	0.0.0.0
Default Local Preference	100
Distance external	20
Distance internal	200
Distance Local	200
Keepalive	60
Hold time	☒ 180
Background scan	☒ 60
Recursive Next Hop	☒
Tag Resolve Mode	Disable

Best Path Selection >

Ajustes no BGP Template BRANCH:

+ Create New Edit Delete Assign to Device/Group More ▾		
☐	Name ▾	Assigned
☐	HUB_BGP_Recommended	0 Device
☐	BRANCH_BGP_Recommended	0 Device
☒	HUB_WORKSHOP_BRANCH_BGP	0 Device
☐	HUB_WORKSHOP_HUB1_BGP	0 Device

Vamos realizar o mesmo procedimento, em ambos os Neighbor:

Ativar Allow AS IN, colocar 1, e depois desativar:

Edit Neighbor

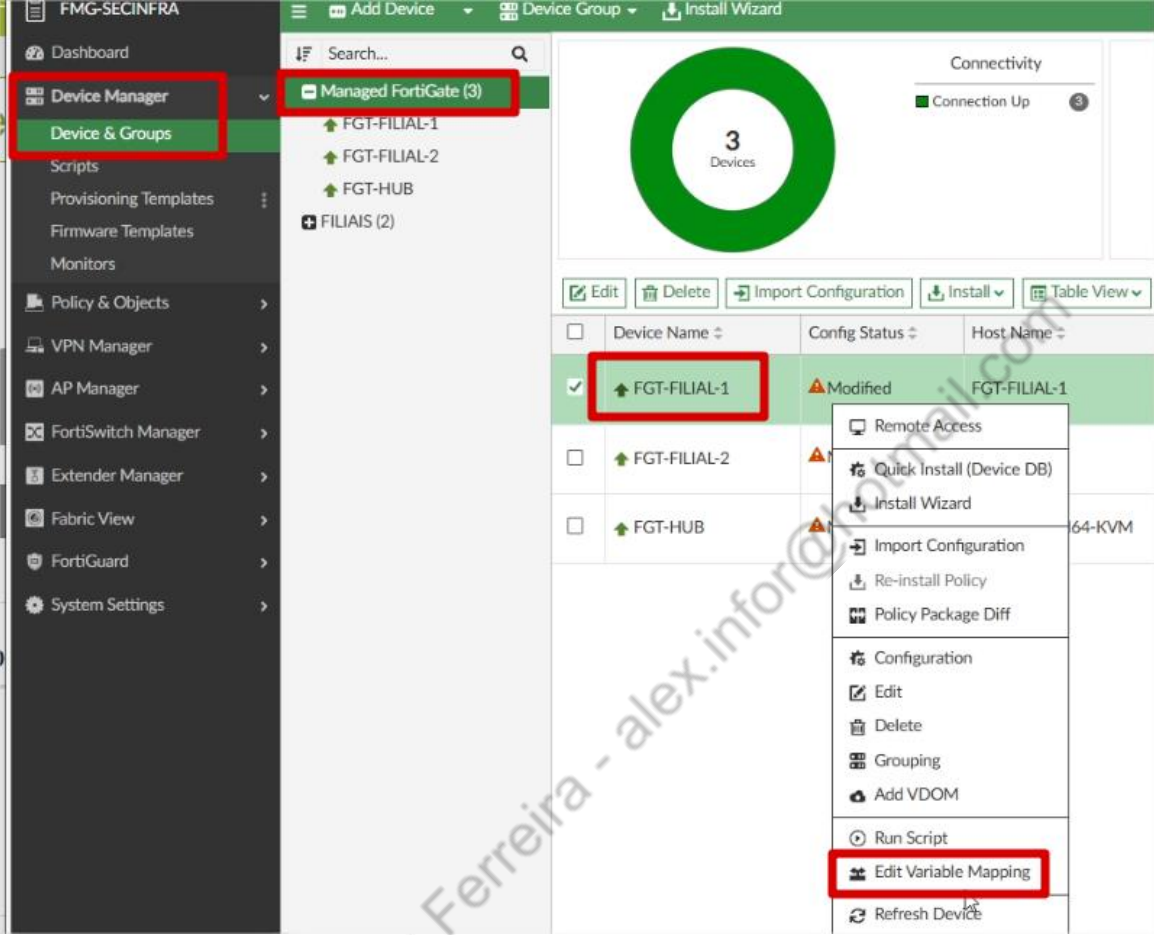
Connect Timer	10
Bidirectional Forwarding	☐
Detection	
Link Down Failover	☒
Shutdown Enable	☐
IPv4 Filtering	☒
Filter List In	☐
Filter List Out	☐
Distribute List In	☐
Distribute List Out	☐
Prefix List In	☐
Prefix List Out	☐
Route Map In	☐
Route Map Out	☐
Allow AS In	☒ 1
Max Prefix	☐

Route Map In
Route Map Out
Allow AS In
Graceful Restart Time
Max Prefix

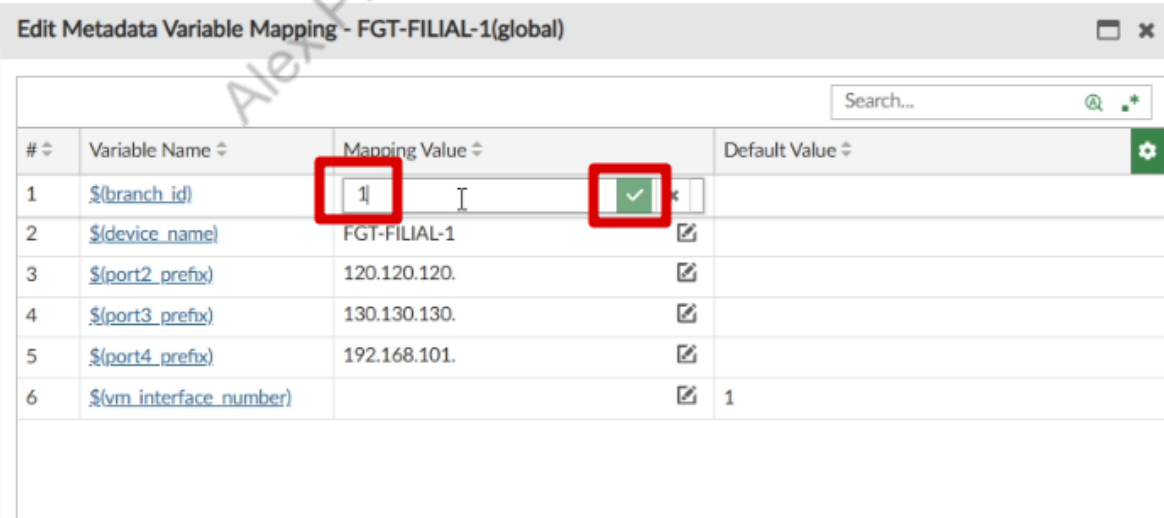
Fazer no IPV6 também.

4.3.4. INFORMAR VARIÁVEIS NOS FORTIGATES DAS FILIAIS

De volta ao Device Manager > Device Groups, vamos clicar com o botão direito no FGT-FILIAL-1 e ir na opção Edit Variable Mapping:



Informar o valor 1 na variável Branch_id, lembrar de confirmar no “V”:

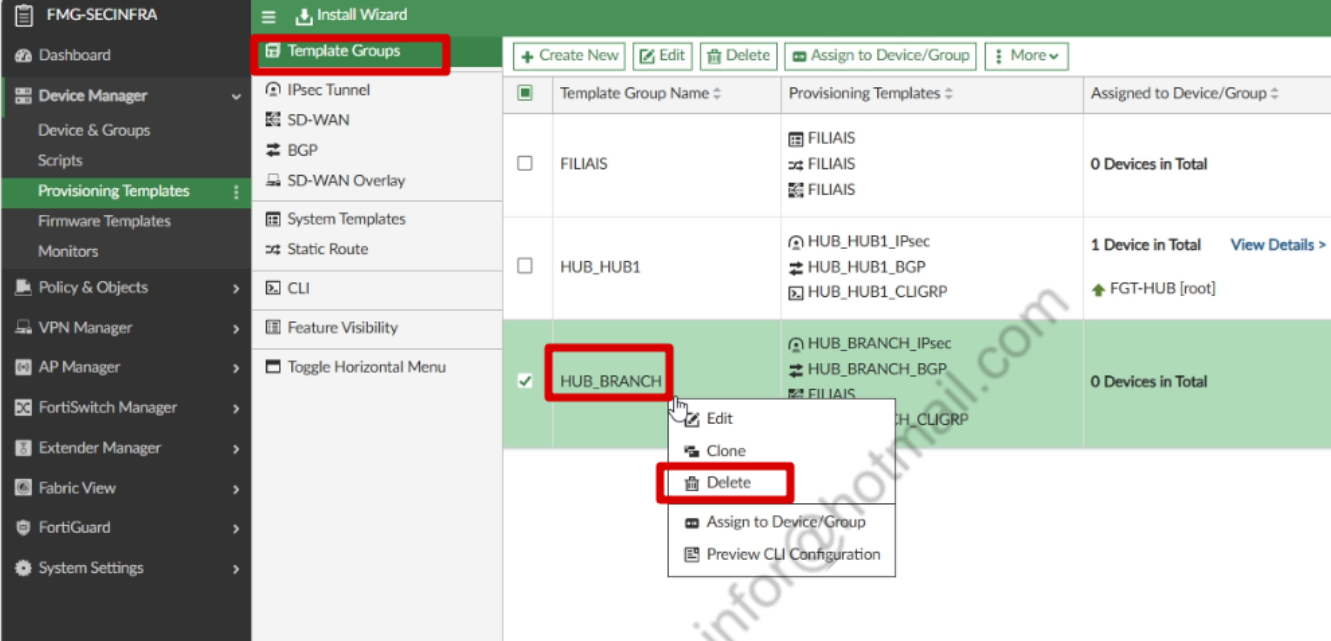


#	Variable Name	Mapping Value	Default Value
1	\$(branch_id)	1	
2	\$(device_name)	FGT-FILIAL-1	
3	\$(port2_prefix)	120.120.120.	
4	\$(port3_prefix)	130.130.130.	
5	\$(port4_prefix)	192.168.101.	
6	\$(vm interface number)		1

Repetir o mesmo processo no FGT-FILIAL-2, mudando o valor para 2.

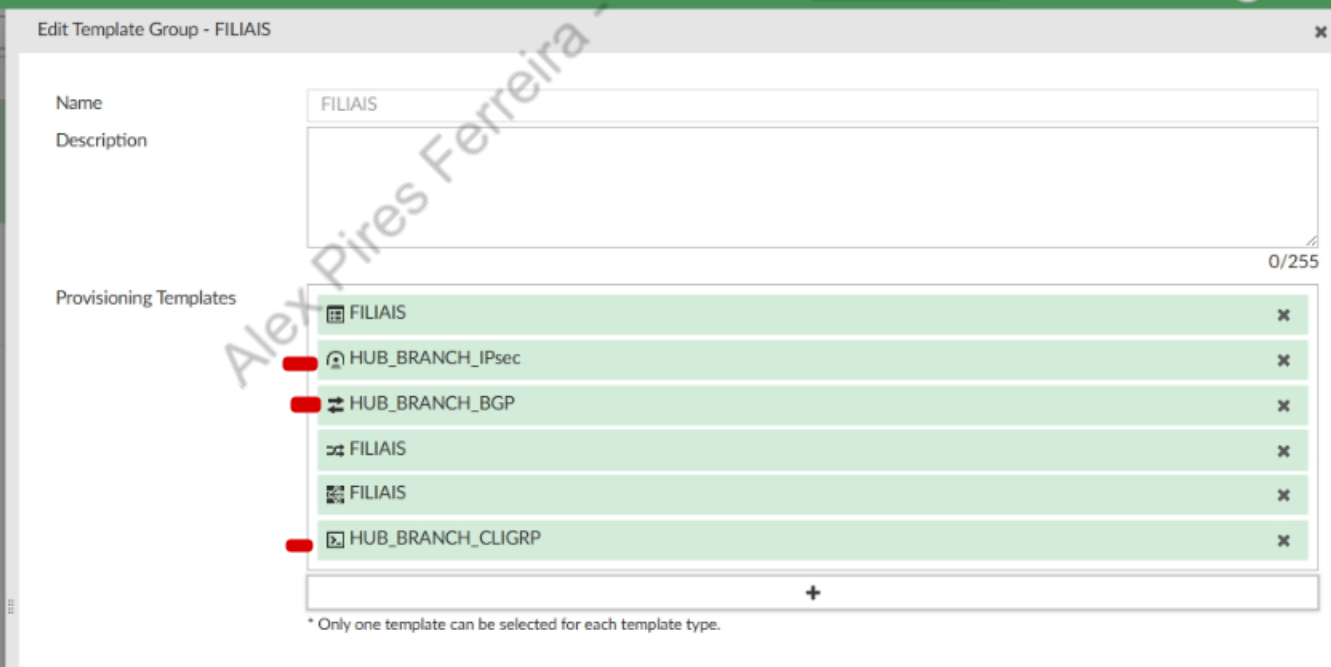
4.3.5.AJUSTE TEMPLATE GROUP FILIAIS

Vamos excluir o template group “HUB_BRANCH”, que foi criado pelo FortiManager, e ajustar o template group “FILIAIS”, que criamos anteriormente:



Template Group Name	Provisioning Templates	Assigned to Device/Group
☐ FILIAIS	- FILIAIS - FILIAIS - FILIAIS	0 Devices in Total
☐ HUB_HUB1	- HUB_HUB1_IPsec - HUB_HUB1_BGP - HUB_HUB1_CLIGRP	1 Device in Total View Details > FGT-HUB [root]
☒ HUB_BRANCH	- HUB_BRANCH_IPsec - HUB_BRANCH_BGP - FILIAIS - HUB_BRANCH_CLIGRP	0 Devices in Total

Adicione os seguintes templates no grupo:



Edit Template Group - FILIAIS

Name: FILIAIS

Description:

Provisioning Templates (0/255):

- FILIAIS
- HUB_BRANCH_IPsec
- HUB_BRANCH_BGP
- FILIAIS
- FILIAIS
- HUB_BRANCH_CLIGRP

* Only one template can be selected for each template type.

Após isso, vamos clicar com o botão direito e clicar em “Assign to Device/Group”, precisamos vincular novamente ao Device Group “FILIAIS”:

Template Group Name	Provisioning Templates	Assigned to Device/Group	Description
☒ FILIAIS	- FILIAIS - HUB_BRANCH_IPsec - HUB_BRANCH_BGP - FILIAIS - FILIAIS - HUB_BRANCH_CLIGRP	0 Devices in Edit Clone Delete Assign to Device/Group Preview CLI Configuration	
☐ HUB_HUB1	- HUB_HUB1_IPsec - HUB_HUB1_BGP - HUB_HUB1_CLIGRP	1 Device in FGT-HUB	Created by SDV

Assign to Devices/Groups

System Template: FILIAIS

Use Shift key for multiple selections and double click for moving one item.

Available Entries (3)

- ☐ FGT-FILIAL-1 [root] (IP: 120.120.120.2, Platform: FortiGate-VM64-KVM)
- ☐ FGT-FILIAL-2 [root] (IP: 140.140.140.2, Platform: FortiGate-VM64-KVM)
- ☐ FGT-HUB [root] (IP: 192.168.31.1, Platform: FortiGate-VM64-KVM)

Selected Entries (1)

- ☐ FILIAIS

4.3.6.CLI SCRIPTS ADICIONAIS

Vamos criar 2 CLI scripts adicionais, para isso, vamos em Device Manager > Provisioning Templates > CLI > Create New:

Dashboard

Device Manager

- Device & Groups
- Scripts
- Provisioning Templates**
- Firmware Templates
- Monitors

Policy & Objects

VPN Manager

Template Groups

- IPsec Tunnel
- SD-WAN
- BGP
- SD-WAN Overlay
- System Templates
- Static Route
- CLI**
- Feature Visibility

+ Create New

Name

Pre-Run CLI Template

- ☐ provision_interf
- ☐ split_hardware_s
- ☐ split_hardware_s

CLI Template (2)

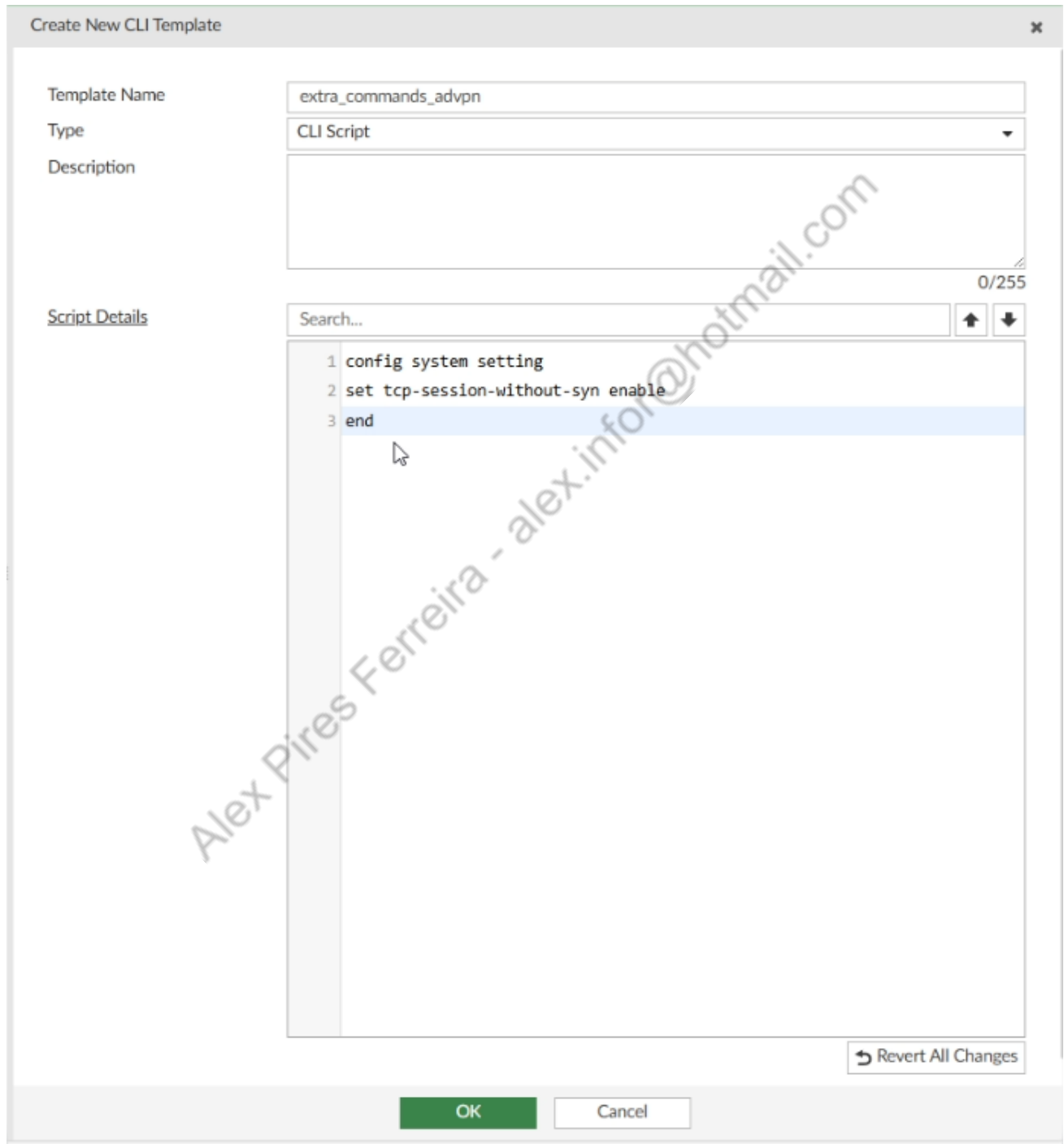
- ☒ HUB_BRANCH_
- ☐ HUB_HUB1_CL

O primeiro CLI Script, será “extra_commands_advpn”, com o código abaixo:

```
config system setting
```

```
set tcp-session-without-syn enable
```

```
end
```



Create New CLI Template

Template Name: extra_commands_advpn

Type: CLI Script

Description:

Script Details

Search...

- 1 config system setting
- 2 set tcp-session-without-syn enable
- 3 end

0/255

Revert All Changes

OK Cancel

O Segundo será "config_vpn_interfaces", com o código a seguir:

```
config system interface
```

```
edit HUB1-VPN1
```

```
set allowaccess ping
```

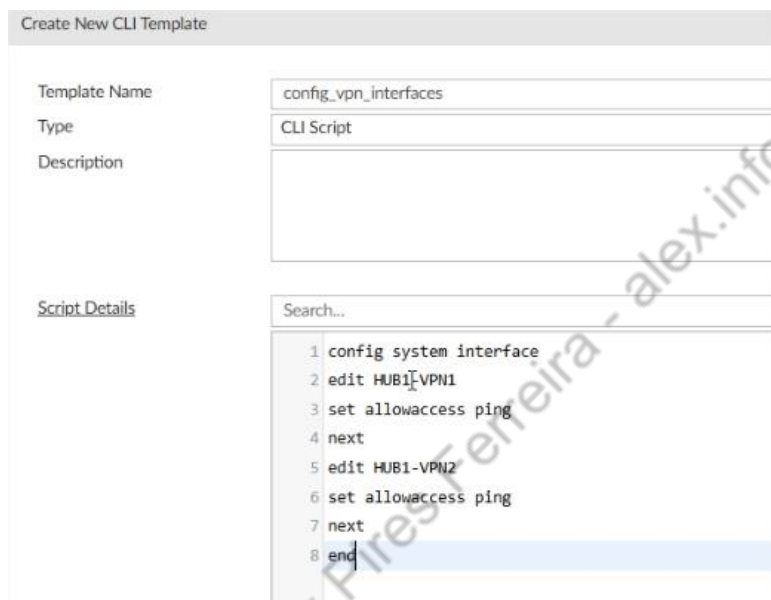
```
next
```

```
edit HUB1-VPN2
```

```
set allowaccess ping
```

```
next
```

```
end
```



Create New CLI Template

Template Name: config_vpn_interfaces

Type: CLI Script

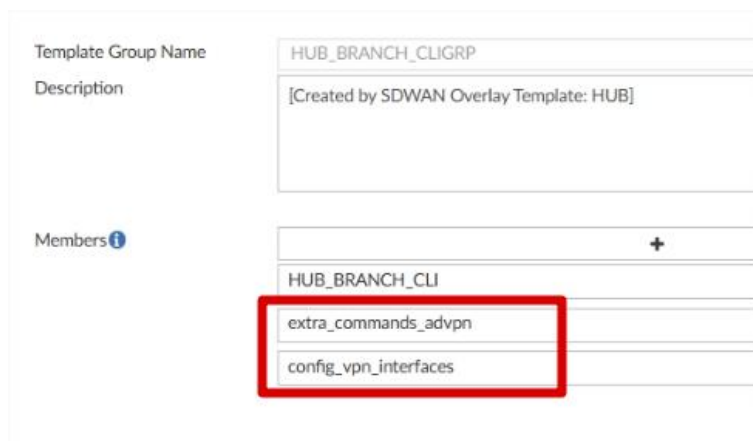
Description:

Script Details

Search...

```
1 config system interface
2 edit HUB1-VPN1
3 set allowaccess ping
4 next
5 edit HUB1-VPN2
6 set allowaccess ping
7 next
8 end
```

Agora vamos editar os CLI TEMPLATE Group, no HUB_BRANCH_CLIGRP, vamos adicionar os 2 CLI novos:



Template Group Name: HUB_BRANCH_CLIGRP

Description: [Created by SDWAN Overlay Template: HUB]

Members

- HUB_BRANCH_CLI
- extra_commands_advpn
- config_vpn_interfaces

No CLI Template Group HUB_HUB1_CLIGRP, vamos adicionar apenas o "extra_commands_advpn":

Template Group Name	HUB_HUB1_CLIGRP						
Description	[Created by SDWAN Overlay Template: HUB]						
Members 	40/255 <table><tr><td colspan="2">+</td></tr><tr><td>HUB_HUB1_CLI</td><td>×</td></tr><tr><td>extra_commands_advpn</td><td>×</td></tr></table>	+		HUB_HUB1_CLI	×	extra_commands_advpn	×
+							
HUB_HUB1_CLI	×						
extra_commands_advpn	×						

Alex Pires Ferreira - alex.infor@hotmail.com

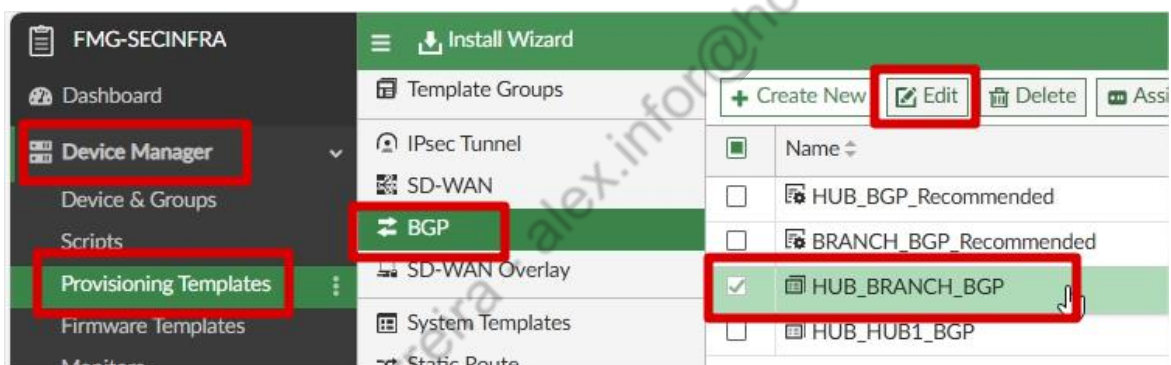
4.4. AJUSTES FINAIS E IMPLEMENTAÇÃO

Estamos chegando quase ao fim de nossa jornada, porém ainda temos os ajustes mais importantes para serem realizados. Com o que fizemos até aqui, podemos já colocar o ADVPN para rodar, de uma maneira simples. Basicamente só faltam policies para nosso ambiente funcionar.

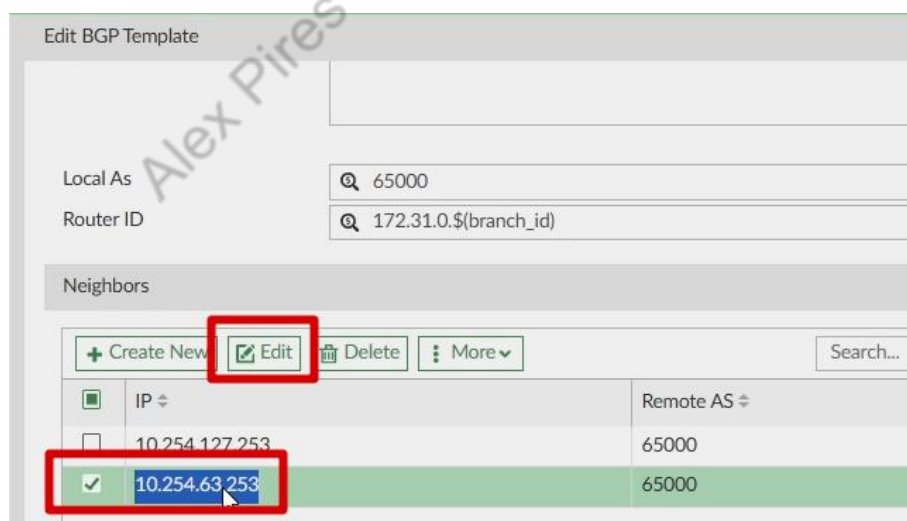
Porém para realizarmos o Self Healing com o BGP, iremos ter que realizar algumas configurações adicionais nos templates BGP, sinalizar o envio do Self healing no SD-Wan Template da FILIAL, e criar um SD-WAN Template para trabalharmos com as TAGs na Matriz.

4.4.1. Configurações BGP Template FILIAIS

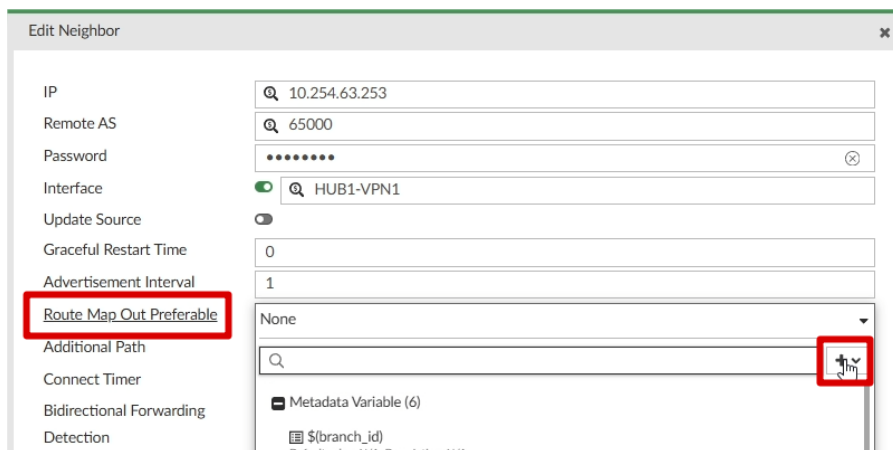
Vamos editar o template "HUB_BRANCH_BGP", para configurar os Route Maps:



Editar o Neighbor 10.254.63.253:



Em Route Map Out Preferable, vamos clicar em "+":



E após em Create New Route Map:



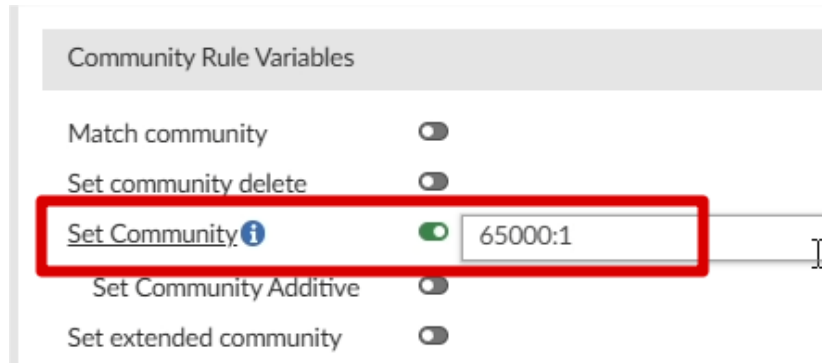
Na tela que abrir, vamos colocar o Name: RM_VPN1, e em Rules, clicar em Create New:



Colocar o valor 1, em ID:



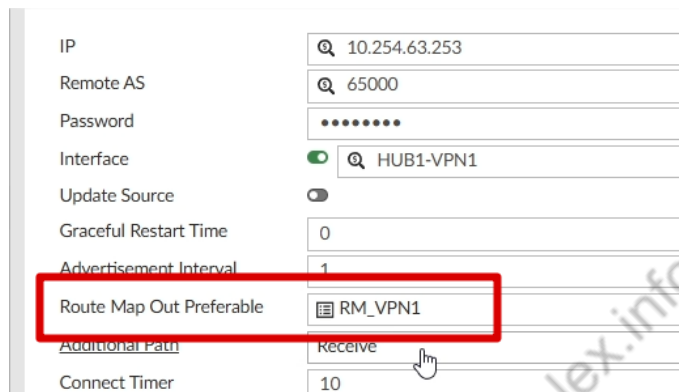
Marcar Set Community, e colocar o valor 65000:1:



Community Rule Variables

- Match community ☐
- Set community delete ☐
- Set Community** ☒ 65000:1
- Set Community Additive ☐
- Set extended community ☐

Após isso clicar em OK, na próxima tela, clicar em OK novamente e irá criar o Route Map, em Route Map Out Preferable, vamos marcar o Route Map Criado RM_VPN1):



IP: 10.254.63.253

Remote AS: 65000

Password:

Interface: ☒ HUB1-VPN1

Update Source: ☐

Graceful Restart Time: 0

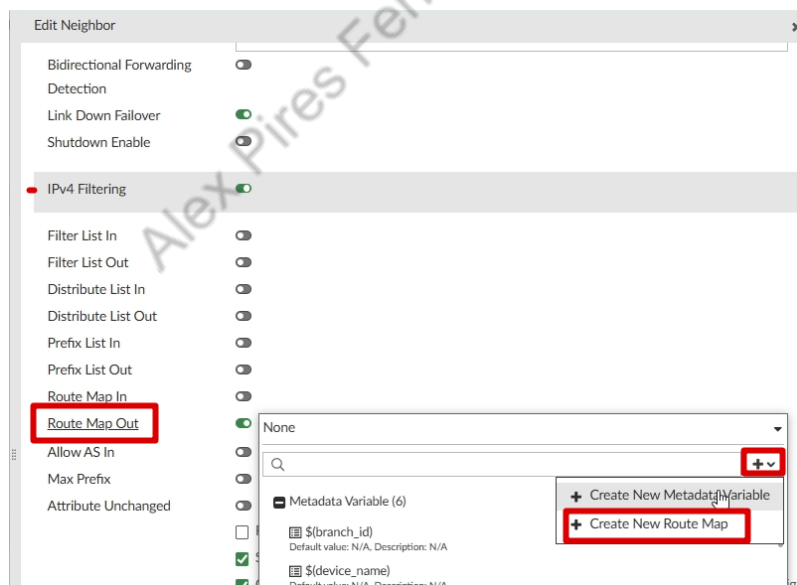
Advertisement Interval: 1

Route Map Out Preferable: RM_VPN1

Additional Path: Receive

Connect Timer: 10

Rolando mais para baixo, vamos usar a opção “Route Map Out” em IPV4 Filtering, mais uma vez clicando em + > Create New Route Map:



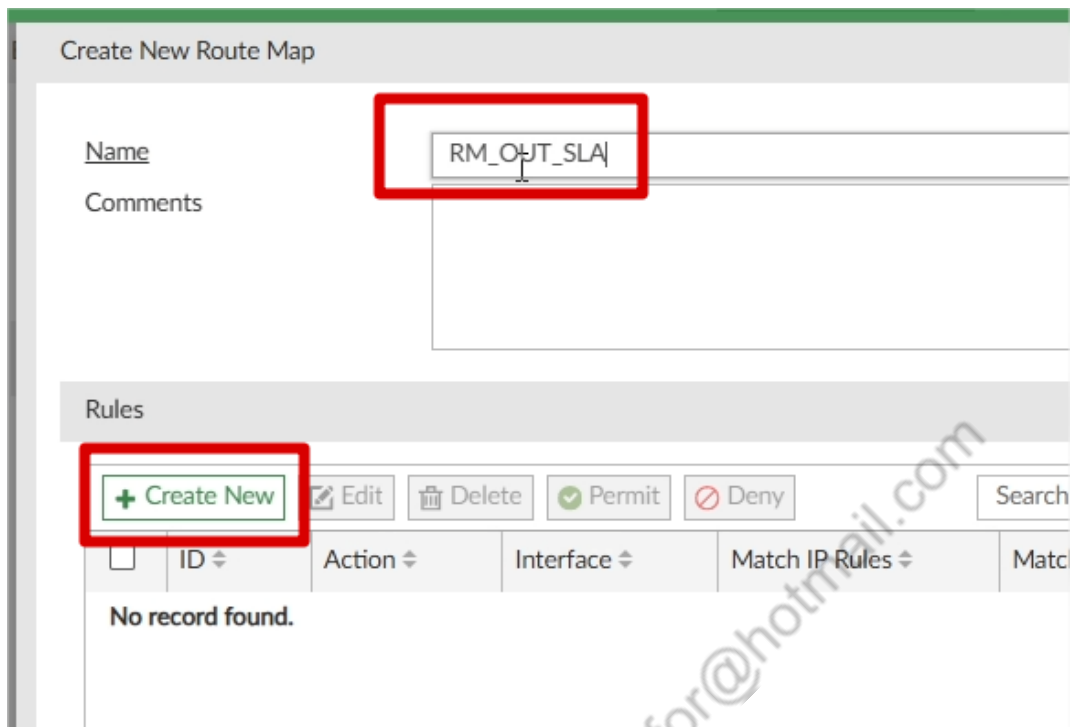
Edit Neighbor

- Bidirectional Forwarding ☐
- Detection ☐
- Link Down Failover ☒
- Shutdown Enable ☐
- IPv4 Filtering** ☒
 - Filter List In ☐
 - Filter List Out ☐
 - Distribute List In ☐
 - Distribute List Out ☐
 - Prefix List In ☐
 - Prefix List Out ☐
 - Route Map In ☐
 - Route Map Out** ☒
 - Allow AS In ☐
 - Max Prefix ☐
 - Attribute Unchanged ☐

None

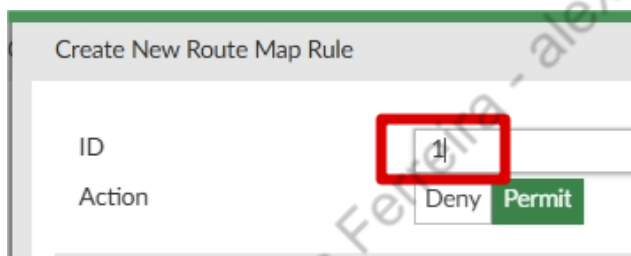
+ > Create New Route Map

Desta vez, vamos dar o nome de "RM_OUT_SLA", e após em Rules, clicar em Create New:



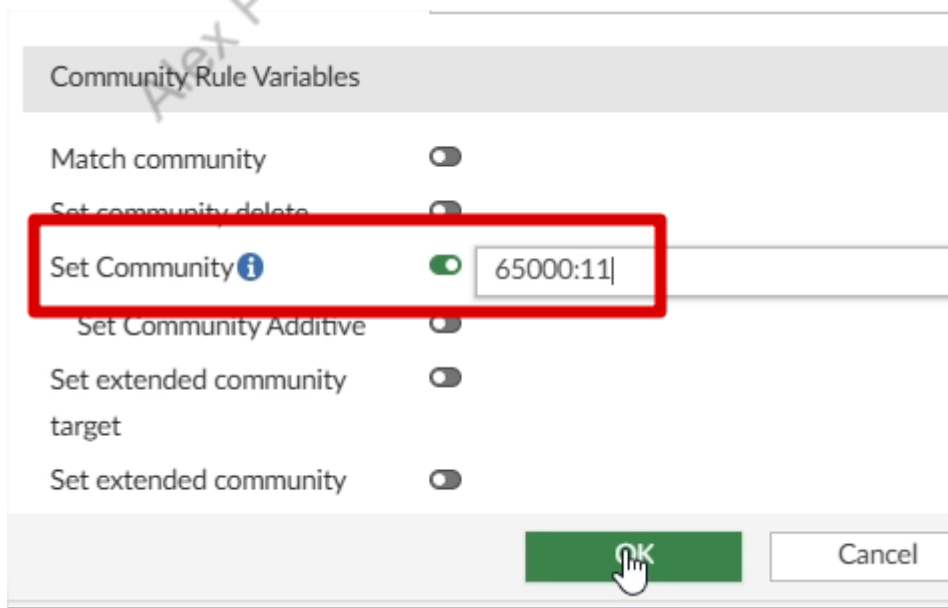
The screenshot shows the "Create New Route Map" form. The "Name" field is highlighted with a red box and contains the text "RM_OUT_SLA". Below the name field is a "Comments" text area. Under the "Rules" section, there is a table with columns: ID, Action, Interface, Match IP Rules, and Match. The "Create New" button is highlighted with a red box. Below the table, it says "No record found."

Setar o ID 1:



The screenshot shows the "Create New Route Map Rule" form. The "ID" field is highlighted with a red box and contains the value "1". Below the ID field is the "Action" section with "Deny" and "Permit" buttons. The "Permit" button is highlighted with a green box.

Em Set Community, colocar o valor 65000:11:



The screenshot shows the "Community Rule Variables" form. The "Set Community" field is highlighted with a red box and contains the value "65000:11". The "Set Community" toggle is turned on. Below the "Set Community" field are other options: "Set Community Additive", "Set extended community target", and "Set extended community". At the bottom, there are "OK" and "Cancel" buttons. The "OK" button is highlighted with a green box and a hand cursor.

Após isso, clicar em OK, OK novamente, e o Route Map será criado.

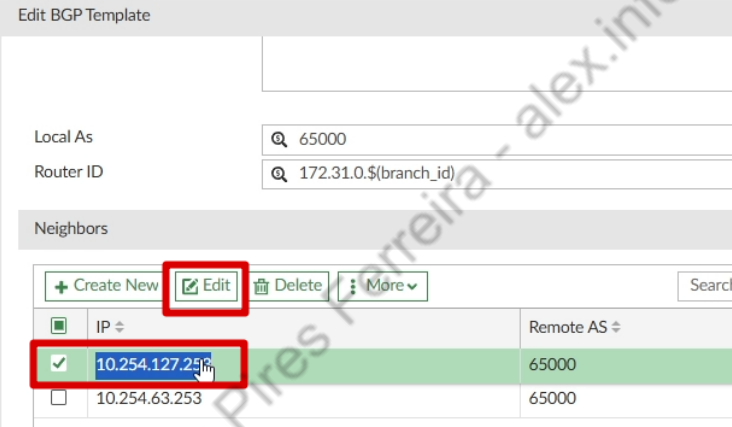
Em Route Map Out, vamos selecionar o Route Map RM_OUT_SLA, caso ele não apareça, salve o Neighbor sem setar o Route Map Out, e abra novamente a tela do Neighbor:



IPv4 Filtering

- Filter List In
- Filter List Out
- Distribute List In
- Distribute List Out
- Prefix List In
- Prefix List Out
- Route Map In
- Route Map Out** (RM_OUT_SLA)
- Allow AS In
- Max Prefix

Vamos fazer a mesma configuração no Neighbor 10.254.127.253:



Edit BGP Template

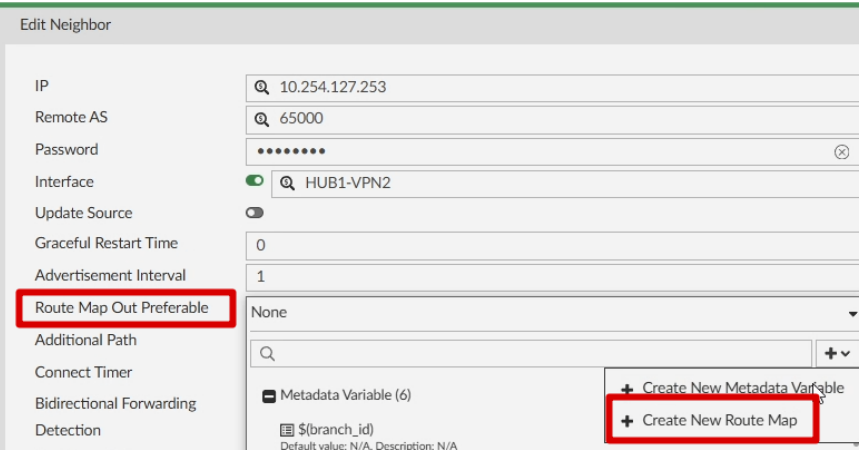
Local AS: 65000
Router ID: 172.31.0.\$(branch_id)

Neighbors

+ Create New [Edit] Delete More

IP	Remote AS
☒ 10.254.127.253	65000
☐ 10.254.63.253	65000

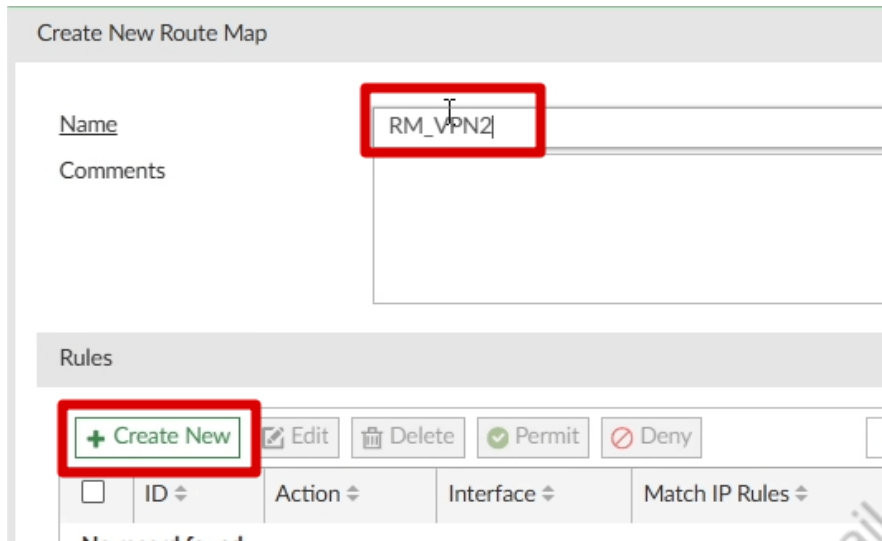
Vamos criar um novo Route Map em Route Map Out Preferable:



Edit Neighbor

IP: 10.254.127.253
Remote AS: 65000
Password:
Interface: HUB1-VPN2
Update Source:
Graceful Restart Time: 0
Advertisement Interval: 1
Route Map Out Preferable: None
Additional Path:
Connect Timer:
Bidirectional Forwarding:
Detection:
Link Down Failover:
Metadata Variable (6): \$(branch_id)
+ Create New Metadata Variable
+ Create New Route Map

Desta vez, com o nome RM_VPN2, e também iremos criar uma nova Rule clicando em Create New:



Create New Route Map

Name

Comments

Rules

☐	ID	Action	Interface	Match IP Rules
☐				

ID novamente será 1:



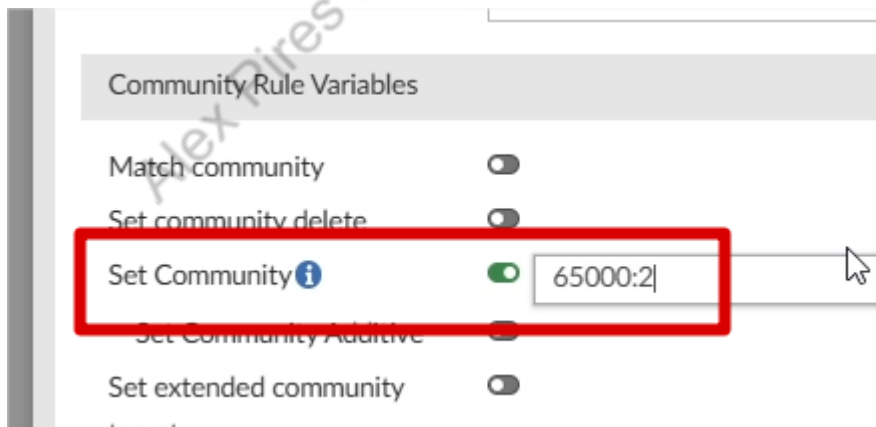
Create New Route Map Rule

ID

Action

IP Address Rule Variables

Porém em Set Community, agora será 65000:2:



Community Rule Variables

Match community ☐

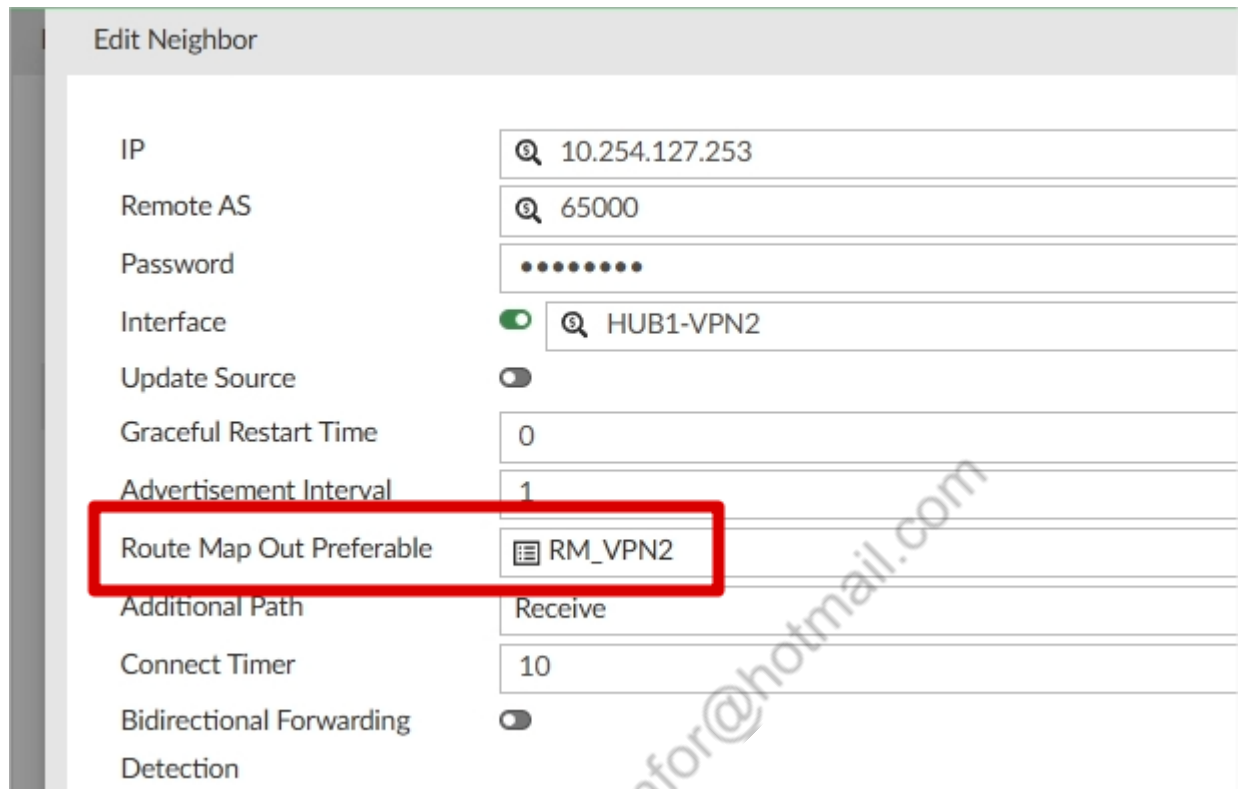
Set community delete ☐

☒ Set Community

☐ Set Community Additive

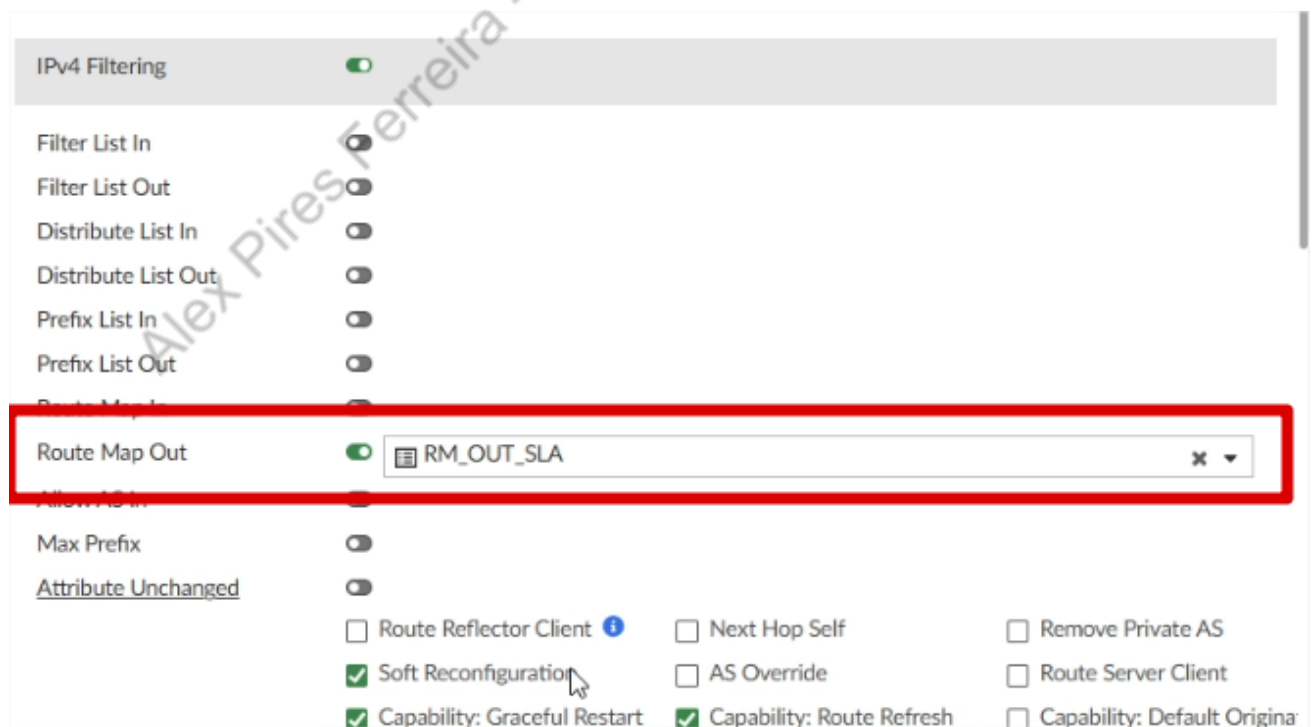
Set extended community ☐

Agora em Route Map Out Preferable, selecione RM_VPN2:



IP	10.254.127.253
Remote AS	65000
Password	
Interface	☒ HUB1-VPN2
Update Source	☐
Graceful Restart Time	0
Advertisement Interval	1
Route Map Out Preferable	RM_VPN2
Additional Path	Receive
Connect Timer	10
Bidirectional Forwarding Detection	☐

Agora vamos colocar o Route Map Out, o mesmo do outro Neighbor, RM_OUT_SLA:



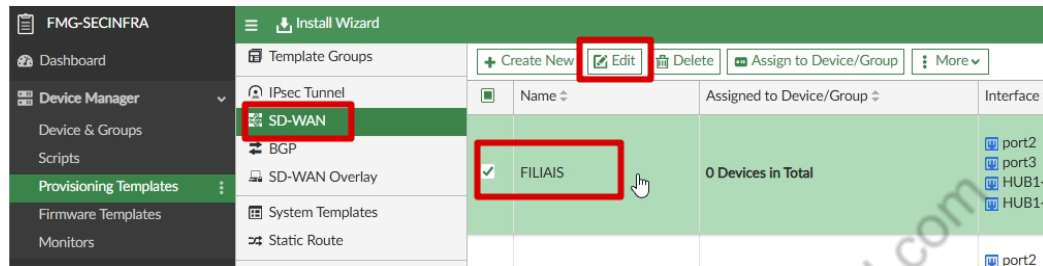
IPv4 Filtering	☒
Filter List In	☐
Filter List Out	☐
Distribute List In	☐
Distribute List Out	☐
Prefix List In	☐
Prefix List Out	☐
Route Map Out	☒ RM_OUT_SLA
Allow AS In	☐
Max Prefix	☐
Attribute Unchanged	☐
Route Reflector Client	☐
Next Hop Self	☐
Remove Private AS	☐
Soft Reconfiguration	☒
AS Override	☐
Route Server Client	☐
Capability: Graceful Restart	☒
Capability: Route Refresh	☒
Capability: Default Origina	☐

Agora basta clicar em OK, e OK novamente para salvar o template.

4.4.2. Configurações SD-WAN Template FILIAIS

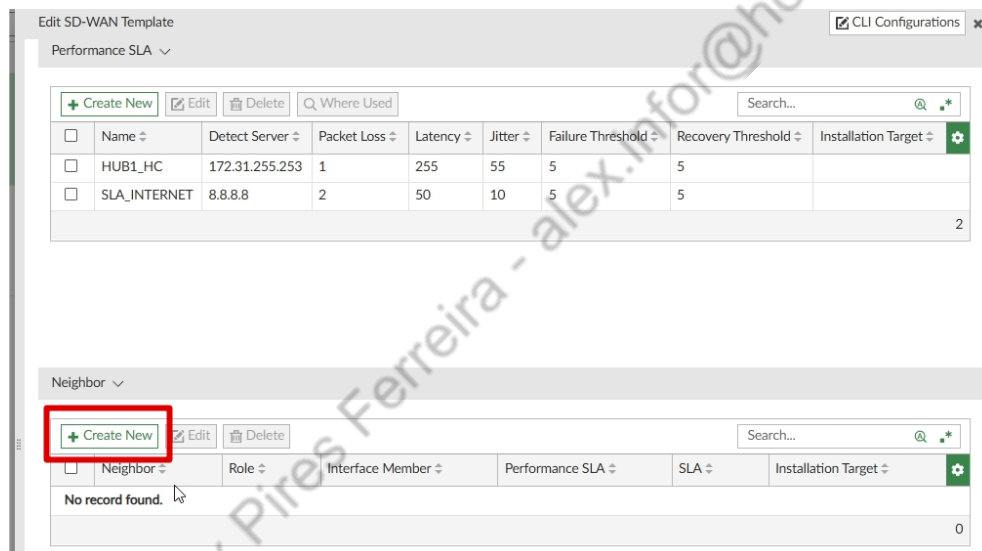
Após realizarmos as configurações do BGP, precisamos vincular o SLA do SDWAN com o Neighbor do BGP, para isso, vamos configurar SDWAN Neighbors.

Vamos editar o template SD-WAN Filiais:



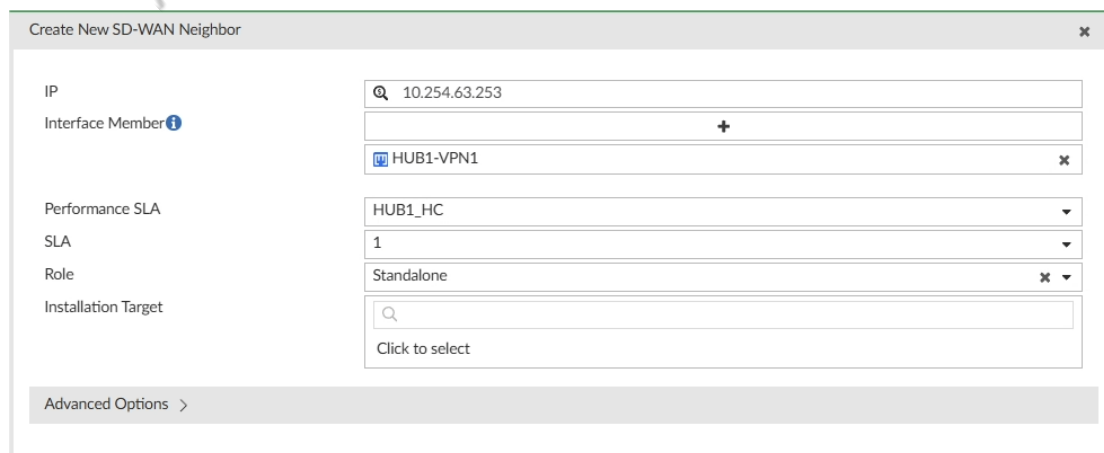
The screenshot shows the FMG-SECINFRA interface. On the left, the 'Provisioning Templates' menu is expanded, and 'SD-WAN' is selected. In the main area, the 'Template Groups' section shows a table with columns: Name, Assigned to Device/Group, and Interface. The 'FILIAIS' entry is highlighted with a red box. Above the table, the 'Edit' button is also highlighted with a red box.

Abaixo do Performance SLA, temos a área "Neighbor", vamos clicar em Create New:



The screenshot shows the 'Edit SD-WAN Template' interface. The 'Performance SLA' section is expanded, showing a table with columns: Name, Detect Server, Packet Loss, Latency, Jitter, Failure Threshold, Recovery Threshold, and Installation Target. Below this, the 'Neighbor' section is expanded, showing a table with columns: Neighbor, Role, Interface Member, Performance SLA, SLA, and Installation Target. The 'Create New' button in the 'Neighbor' section is highlighted with a red box.

Vamos preencher os dados como abaixo:



The screenshot shows the 'Create New SD-WAN Neighbor' form. The form contains the following fields:

- IP: 10.254.63.253
- Interface Member: HUB1-VPN1
- Performance SLA: HUB1_HC
- SLA: 1
- Role: Standalone
- Installation Target: Click to select

At the bottom, there is an 'Advanced Options' section with a right-pointing arrow.

Vamos adicionar o segundo neighbor, e preencher conforme abaixo:

Create New SD-WAN Neighbor

IP

10.254.127.253

Interface Member

+

HUB1-VPN2

Performance SLA

HUB1_HC

SLA

1

Role

Standalone

Installation Target

Click to select

Advanced Options >

Desta forma ficaremos como abaixo:

Neighbor

Create New

Edit

Delete

Search...

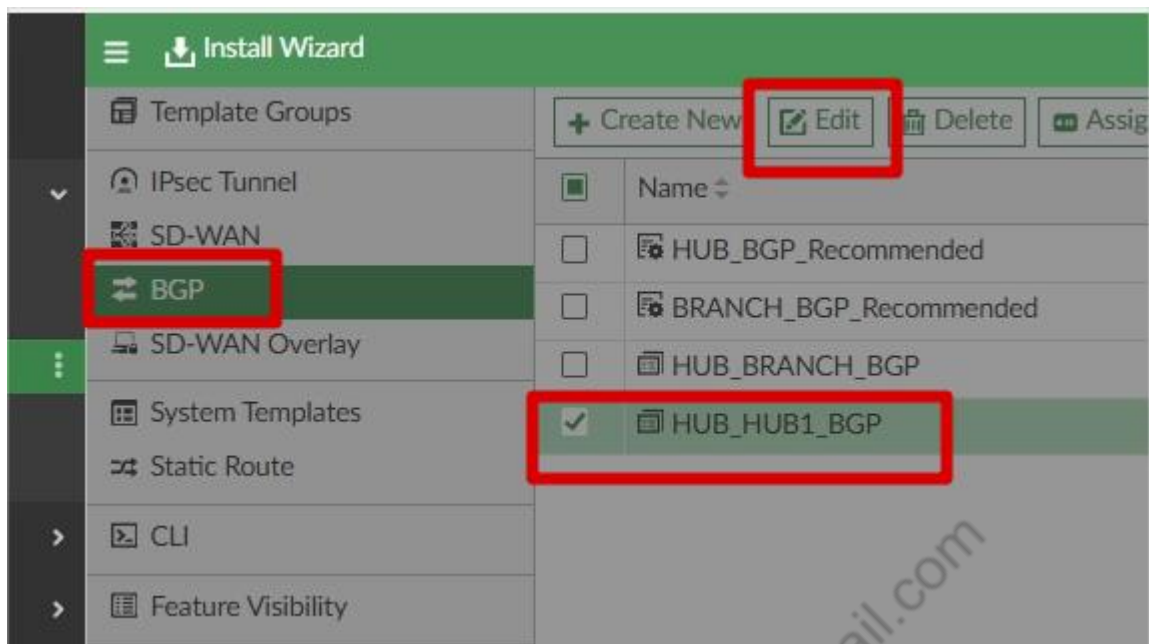
☐	Neighbor	Role	Interface Member	Performance SLA	SLA	Installation Target	
☐	10.254.63.253	Standalone	HUB1-VPN1	HUB1_HC	1		
☐	10.254.127.253	Standalone	HUB1-VPN2	HUB1_HC	1		

2

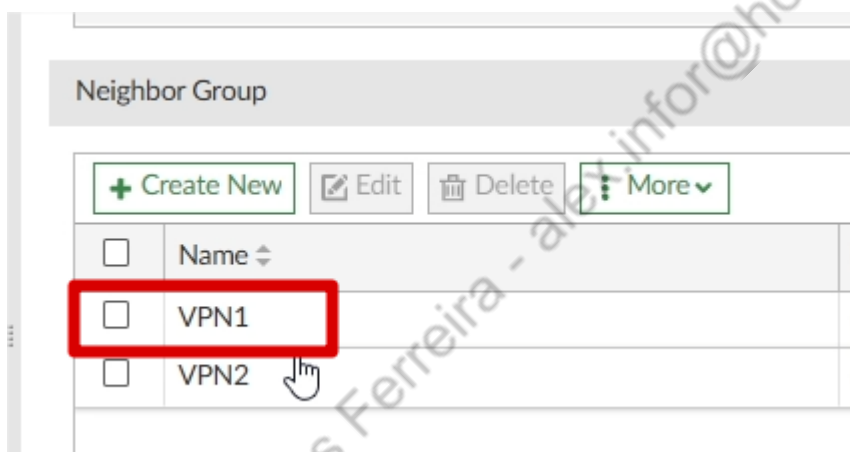
Agora basta clicar em OK para salvar o SD WAN Template.

4.4.3. Configurações BGP Template HUB

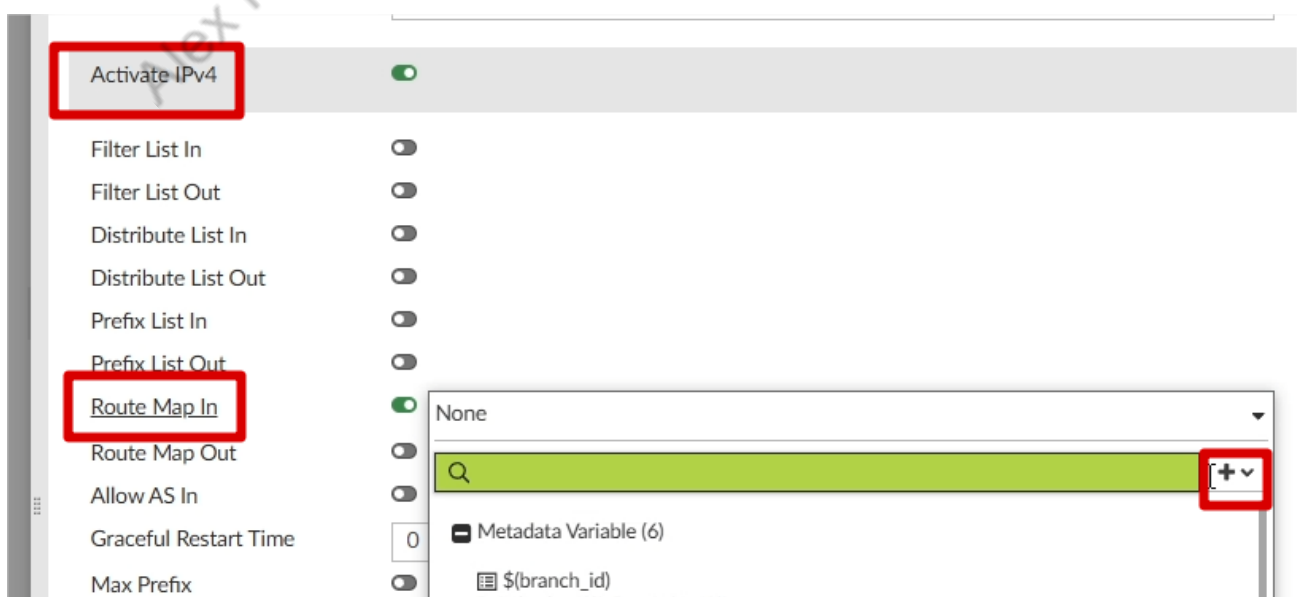
Agora iremos preparar o HUB para receber as Community Tags e setar a Route TAG correta, vamos começar editando o BGP Template HUB_HUB1_BGP:



Vamos editar o Neighbor Group VPN1:

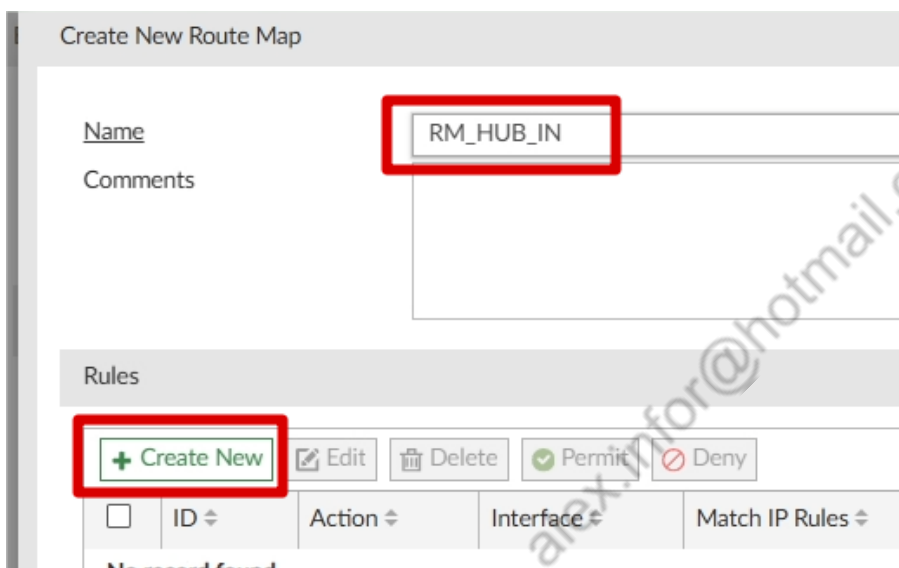


Vamos criar uma Route Map In:





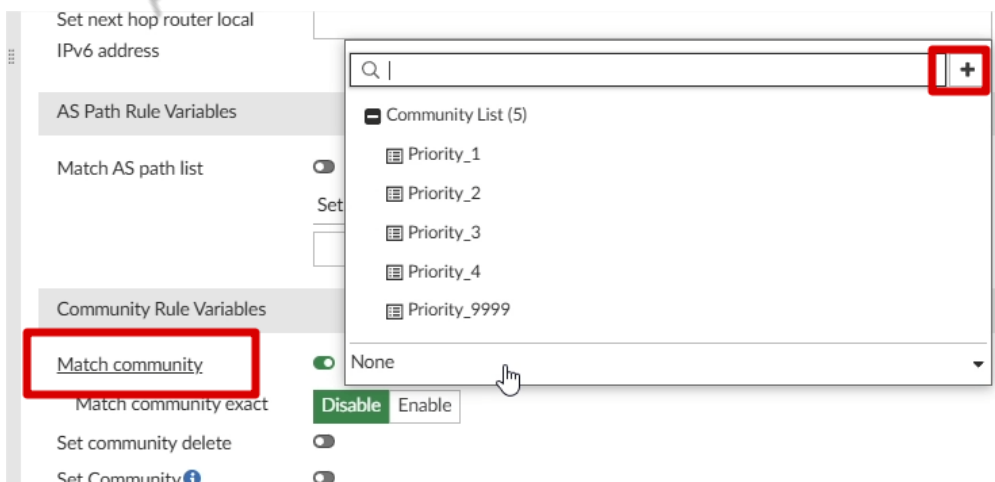
O nome será RM_HUB_IN, vamos clicar em Create New nas rules:



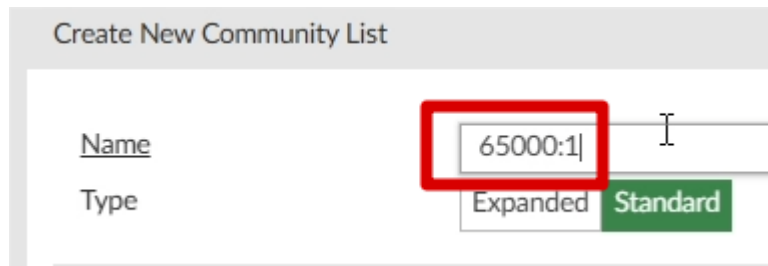
O ID será 1:



Vamos habilitar "Match Community", e criar um novo objeto:



O nome dessa Community List, será 65000:1, vamos clicar em Create New nas rules:

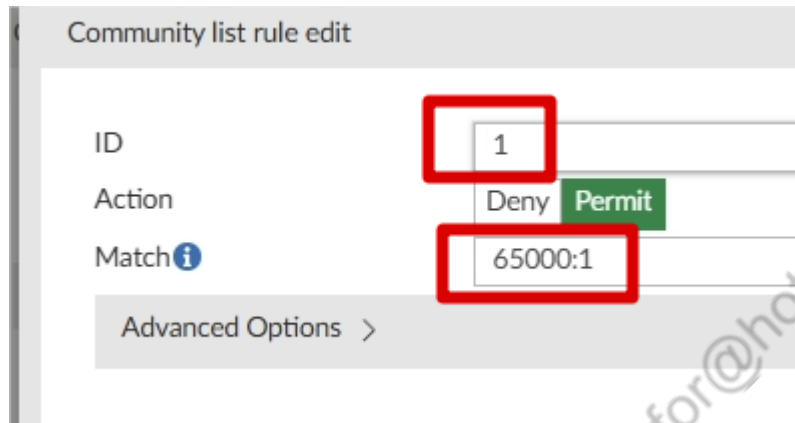


Create New Community List

Name

Type ☐ Expanded ☒ Standard

Vamos colocar o ID 1, e em Match, vamos colocar 65000:1:



Community list rule edit

ID

Action ☐ Deny ☒ Permit

Match

Advanced Options >

Clique em OK para confirmar, e agora selecione a Community List 65000:1:



Match community ☒

Match community exact ☒ Disable ☐ Enable

Set community delete ☐

Logo abaixo, ainda na mesma Route Map Rule, vamos colocar o valor 1 no campo Route Tag:

Create New Route Map Rule

Match AS path list
☐

Set AS path

+

Action

Community Rule Variables

Match community
☒

65000:1

Match community exact

Disable
Enable

Set community delete
☐

Set Community
☐

Set extended community target
☐

Set extended community site-of-origin
☐

Other Rule Variables

Match metric

Match origin

Egp
Igp
Incomplete
None

Set origin

Egp
Igp
Incomplete
None

Match route type

External-type1
External-type2
None

Set metric type

External-type1
External-type2
None

Set metric

Set tag value

Set route tag

1

Match tag

Set aggregator AS
☐

Clicamos em OK, para salvar a Rule.

Agora criaremos outra Rule, de ID 2:

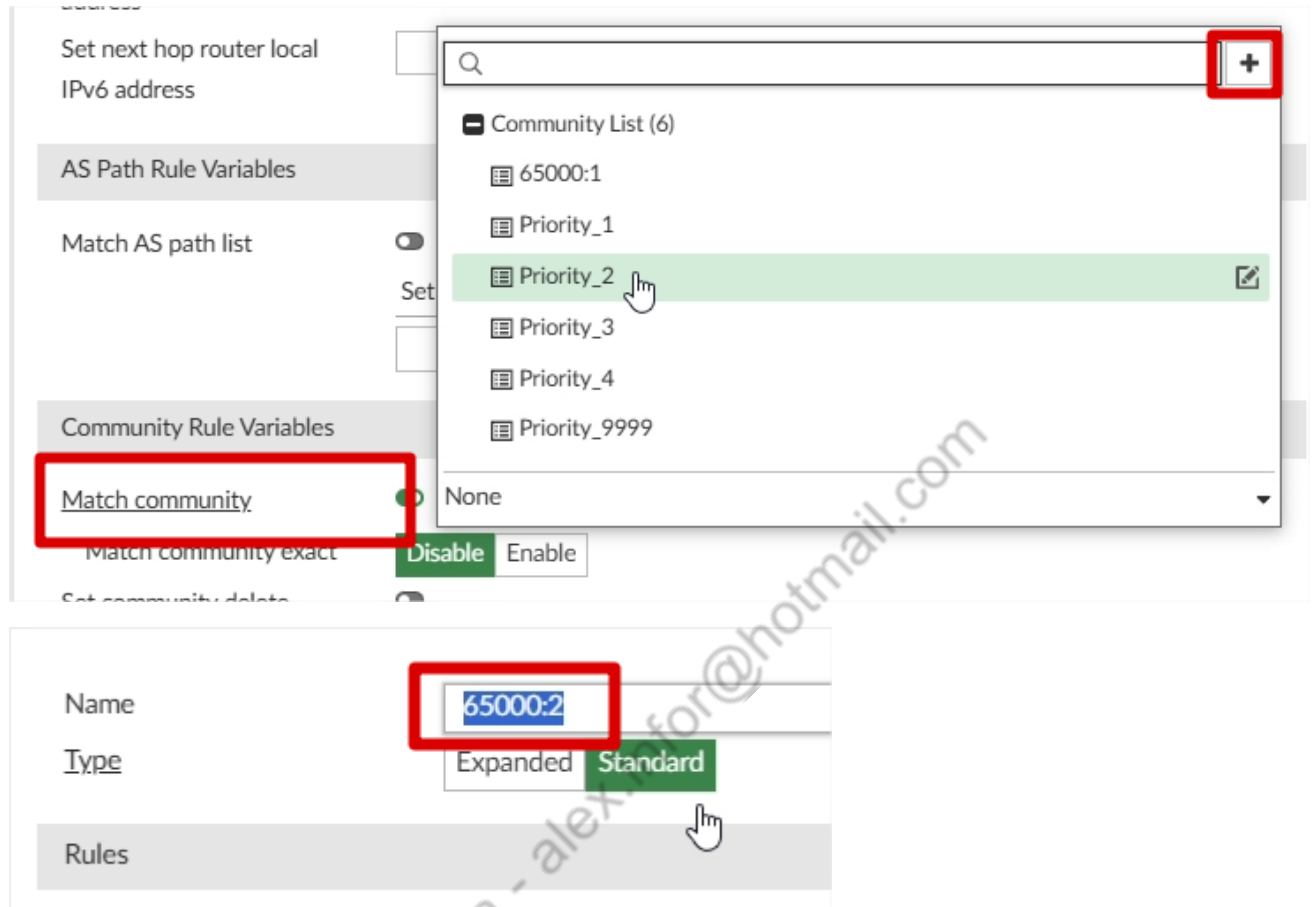
ID

2

Action

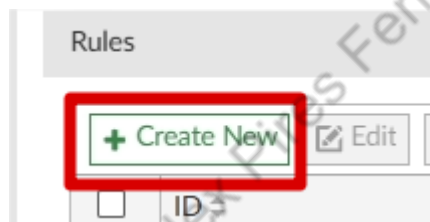
Deny
Permit

Vamos criar um Community List de nome 65000:2, para usarmos no Match Community:



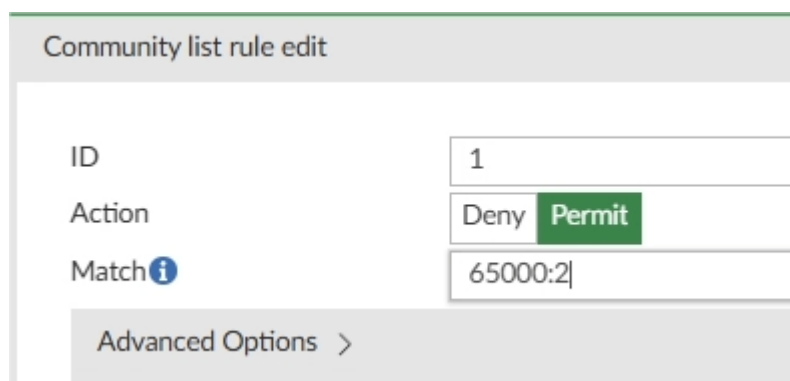
The screenshot shows the configuration page for a Juniper device. On the left, under 'Community Rule Variables', the 'Match community' option is selected and highlighted with a red box. On the right, a dropdown menu is open, showing a list of community lists: 65000:1, Priority_1, Priority_2, Priority_3, Priority_4, and Priority_9999. 'Priority_2' is highlighted with a green background and a mouse cursor. Below the dropdown, the 'Match community exact' option is selected, and the 'Enable' button is visible. In the bottom section, the 'Name' field is set to '65000:2' (highlighted with a red box), and the 'Type' is set to 'Standard' (highlighted with a green box and a mouse cursor).

Clicamos em Create New Rule na Community list:



The screenshot shows the 'Rules' section of the configuration page. A button labeled '+ Create New' is highlighted with a red box. To its right is an 'Edit' button with a pencil icon.

ID 1 e em Match, colocamos 65000:2:



The screenshot shows the 'Community list rule edit' form. The 'ID' field is set to '1'. The 'Action' field has two options: 'Deny' and 'Permit', with 'Permit' selected and highlighted in green. The 'Match' field is set to '65000:2'. At the bottom, there is an 'Advanced Options' link with a right-pointing arrow.

Ok para salvar a rule, OK novamente para salvar o Community List, e selecionamos o Community list na nossa Route Map Rule:

Match community ☒ 65000:2

Match community exact Disable Enable

Nesta Route Map Rule, vamos colocar o valor "2" em Set Route Tag:

Create New Route Map Rule

Match AS path list

Set AS path

Action

Community Rule Variables

Match community ☒ 65000:2

Match community exact Disable Enable

Set community delete ☐

Set Community ☐

Set extended community target ☐

Set extended community site-of-origin ☐

Other Rule Variables

Match metric

Match origin ☐

Set origin ☐

Match route type

Set metric type

Set metric

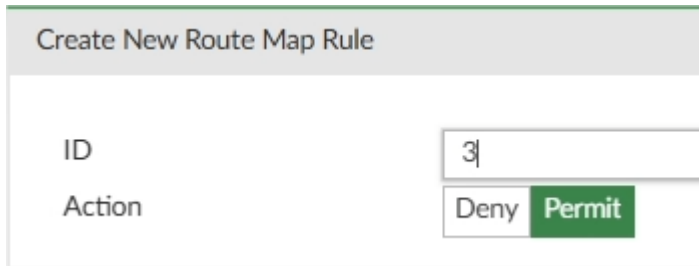
Set tag value

Set route tag 2

Match tag

Clicamos em OK para salvar a Route Map rule.

Agora vamos criar nossa última Rule, com ID número 3:

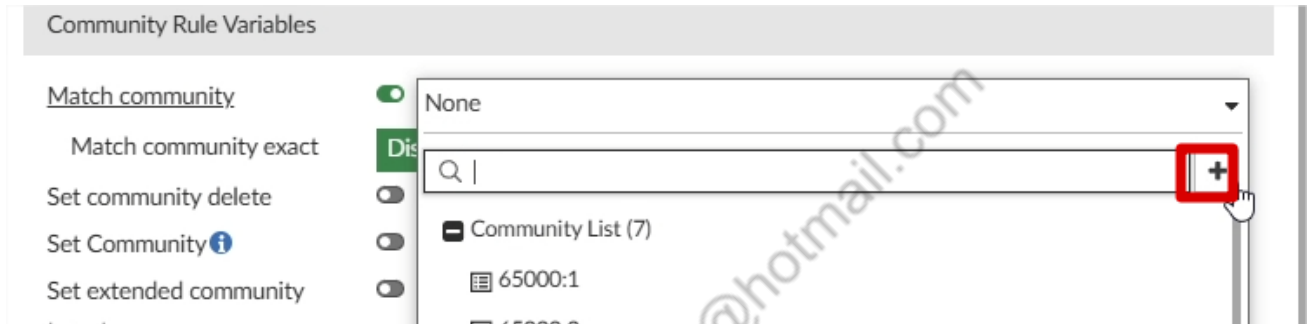


Create New Route Map Rule

ID: 3

Action: Deny Permit

Em Match Community, vamos criar uma nova Community List, com o nome de 65000:11:



Community Rule Variables

Match community: ☒ None

Match community exact: ☒ Dis

Set community delete: ☐

Set Community: ☐

Set extended community: ☐

Community List (7)

65000:1

65000:2

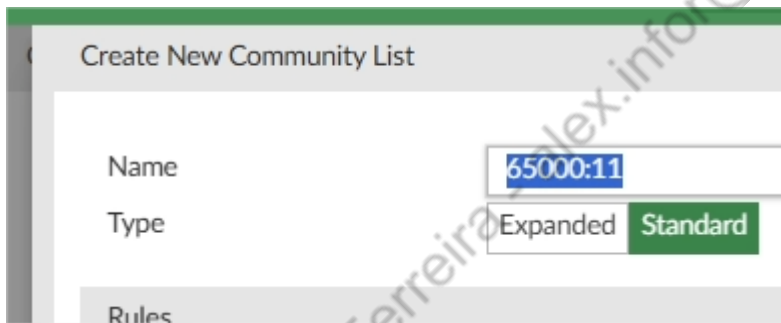
65000:3

65000:4

65000:5

65000:6

65000:7



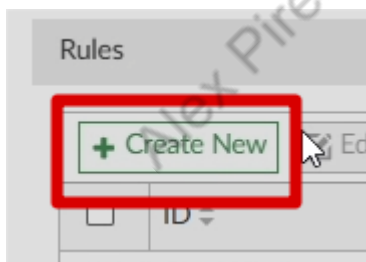
Create New Community List

Name: 65000:11

Type: Expanded Standard

Rules

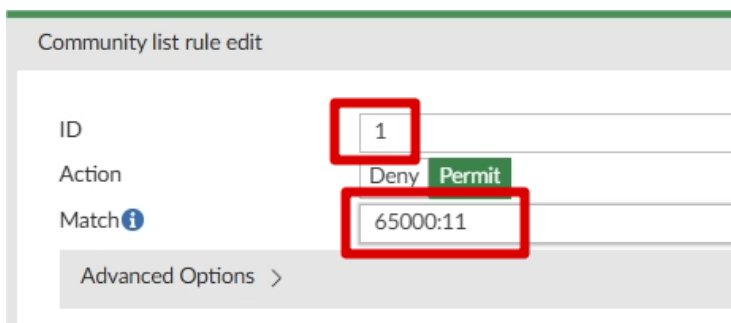
Criamos uma nova rule, com ID 1 e o Match de 65000:11:



Rules

+ Create New

ID



Community list rule edit

ID: 1

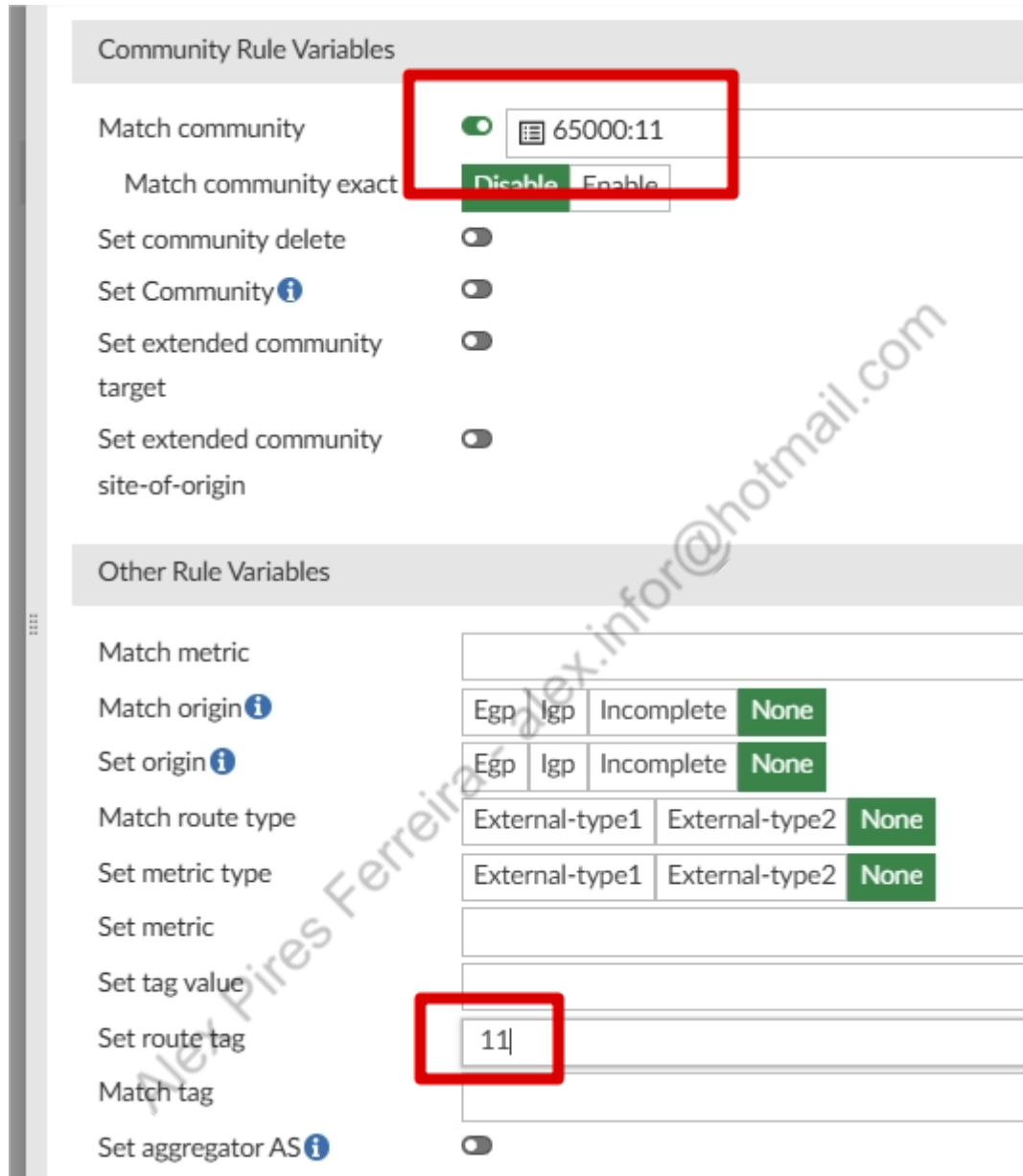
Action: Deny Permit

Match: 65000:11

Advanced Options >

Ok para Salvar a Community List Rule, e ok para salvar a Community List.

Na Route Map rule, vamos seleccionar a Community 65000:11 e setar Route Tag como 11:



Community Rule Variables

Match community ☒ 65000:11

Match community exact

Set community delete ☐

Set Community ☐

Set extended community target ☐

Set extended community site-of-origin ☐

Other Rule Variables

Match metric

Match origin

Set origin

Match route type

Set metric type

Set metric

Set tag value

Set route tag

Match tag

Set aggregator AS ☐

Clicamos em OK para salvar a Route Map Rule, e em OK novamente, para salvar a Route Map.

Agora vamos seleccionar a Route Map IN RM_HUB_IN:

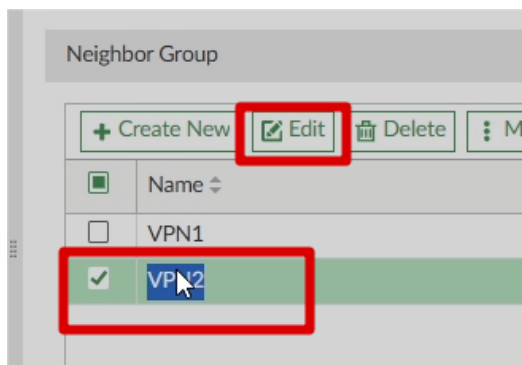


Route Map In ☒ RM_HUB_IN

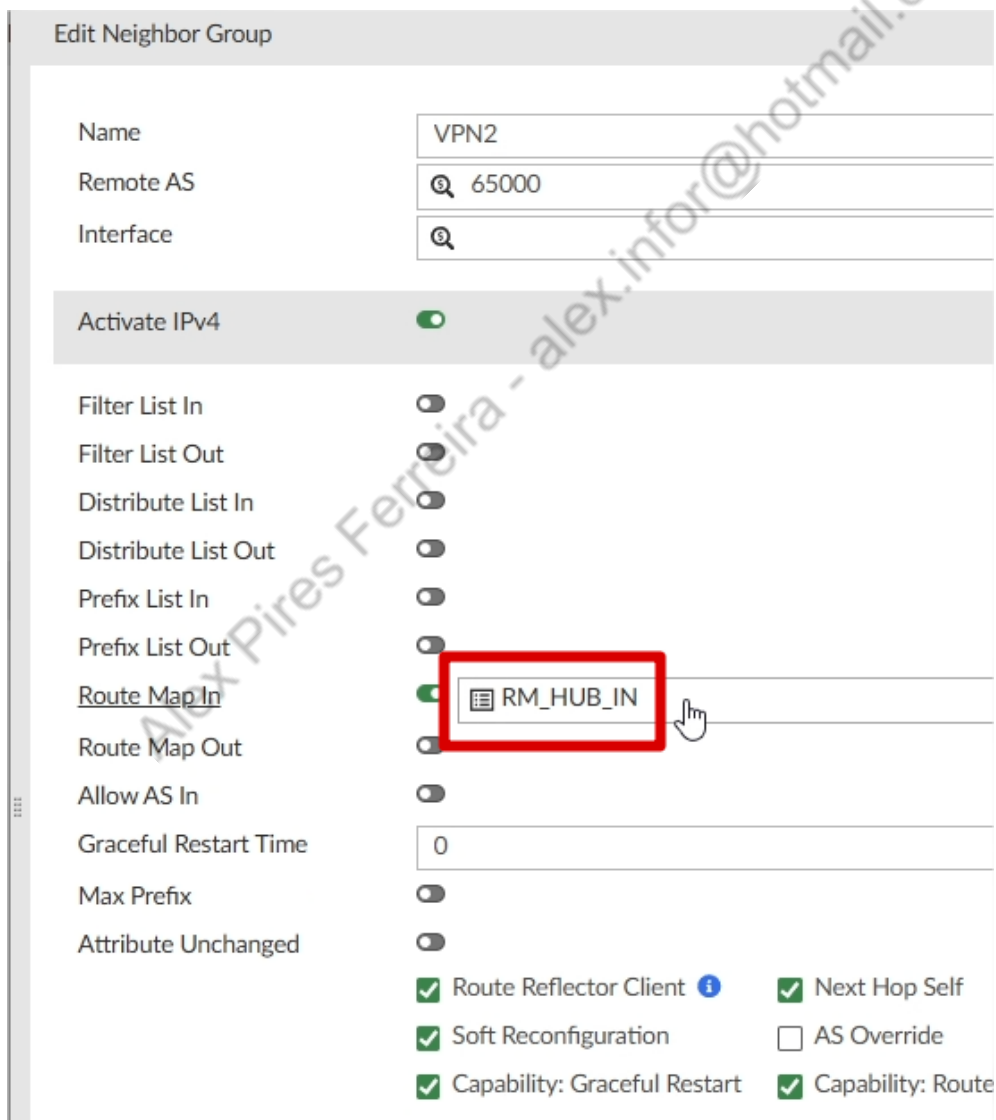
Route Map Out ☐

Clicamos em OK para salvar o Neighbor Group.

Agora vamos editar o outro Neighbor Group:



Aqui, basta selecionarmos o mesmo Route Map em Route Map In do IPV4:



Com isso finalizamos os ajustes no BGP Template do HUB.

4.4.4.Criação SD-WAN Template HUB

Agora iremos criar o SD-WAN Template do HUB, a zona virtual-wan-link, interfaces port2 e port3, assim como a SDWAN RULE para a Internet e o Health Check, serão iguais ao que já tínhamos configurado diretamente no Fortigate. Abaixo prints da configuração:

Name: HUB

Description:

SD-WAN Status: ●

0/4096

SD-WAN Zones

ID	Interface	Gateway	Cost	Priority	Status	Installation Target
virtual-wan-link						
1	port2	100.100.100.1	0	0	Enable	
2	port3	110.110.110.1	0	0	Enable	

A origem da "SAIDA_INTERNET", é os objetos RD_DMZ e RD_LAN, o destino é o RFC1918_GRP com DST negate.

SD-WAN Rules

ID	Name	Source	Destination	Criteria	Members	Performance SLA	Port	Protocol	Status	Installation Target
1	SAIDA_INTERNET	RD_DMZ RD_LAN	RFC1918		port2 port3			any	Enable	

Performance SLA simples:

Edit Performance SLA - INTERNET

Name

INTERNET

IP Version

IPv4 IPv6

Probe Mode

Active

Enable Probe Packets

☒

Protocol

Ping

Server

8.8.8.8

Participants

All SD-WAN Members Specify

port2

port3

2 entries selected

Installation Target

Click to select

SLA Target

Latency Threshold

☒ 50 ms

Jitter Threshold

☒ 10 ms

Packet Loss Threshold

☒ 2 %

Mos Threshold

☐

Action

+

Link Status

Check Interval

500 Milliseconds (20 - 3600000)

Failure Before Inactive

5 (1 - 3600)

Restore Link After

5 Check(s) (1 - 3600)

Probe Timeout

500 Milliseconds (1 - 3600000)

OK

Cancel

Vamos adicionar uma zona "VPN_FILIAIS"

SD-WAN Zones

+ Create New

Edit

Delete

SD-WAN Member

SD-WAN Zone

Name

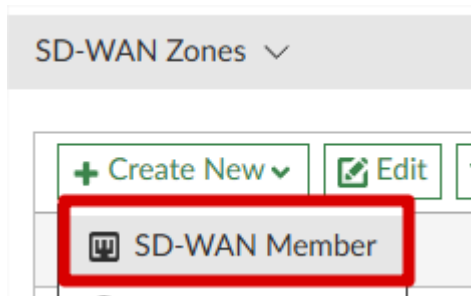
VPN_FILIAIS

Interface Members

Click to select

Clique em OK para confirmar a criação.

Vamos criar 2 membros



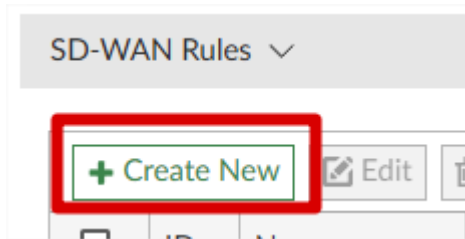
VPN1:

Sequence Number	3
Interface Member	VPN1
SD-WAN Zone	VPN_FILIAIS
Gateway IP	0.0.0.0
Cost	0
Priority	0
Status	☒
Installation Target	Click to select

Sequence Number	4
Interface Member	VPN2
SD-WAN Zone	VPN_FILIAIS
Gateway IP	0.0.0.0
Cost	0
Priority	0
Status	☒
Installation Target	Click to select

A zona VPN_FILIAIS não utilizará Health Check.

Vamos criar 3 SD_WAN Rules:



VPN1:

Edit SD-WAN Rule - 2

Name	VPN1
Status	☒
IP Version	IPv4 IPv6
Installation Target	 Click to select

Source

Source Address	 RFC1918-GRP Group Members (3): RFC1918-10, RFC1918-172, RFC1918-192
User Group	 Click to select

Destination

Address	 Click to select
Route Tag	1
Protocol	TCP UDP Any Specify 0
Internet Service	

Outgoing Interfaces

Strategy	Manual
Interface Preference ⓘ	VPN1
Zone Preference ⓘ	
Measured SLA	Click to select
Required SLA Target ⓘ	
Quality Criteria	Latency
Forward DSCP	☐
Reverse DSCP	☐

Advanced Options >

OK Cancel

VPN2:

Name VPN2

Status On

IP Version IPv4 IPv6

Installation Target Click to select

Source

Source Address RFC1918-GRP
Group Members (3): RFC1918-10, RFC1918-172, RFC1918-192

User Group Click to select

Destination

Address Click to select

Route Tag 2

Protocol TCP UDP Any Specify 0

Internet Service Click to select

Outgoing Interfaces

Strategy Manual

Interface Preference VPN2

Zone Preference

Measured SLA Click to select

Required SLA Target

Quality Criteria Latency

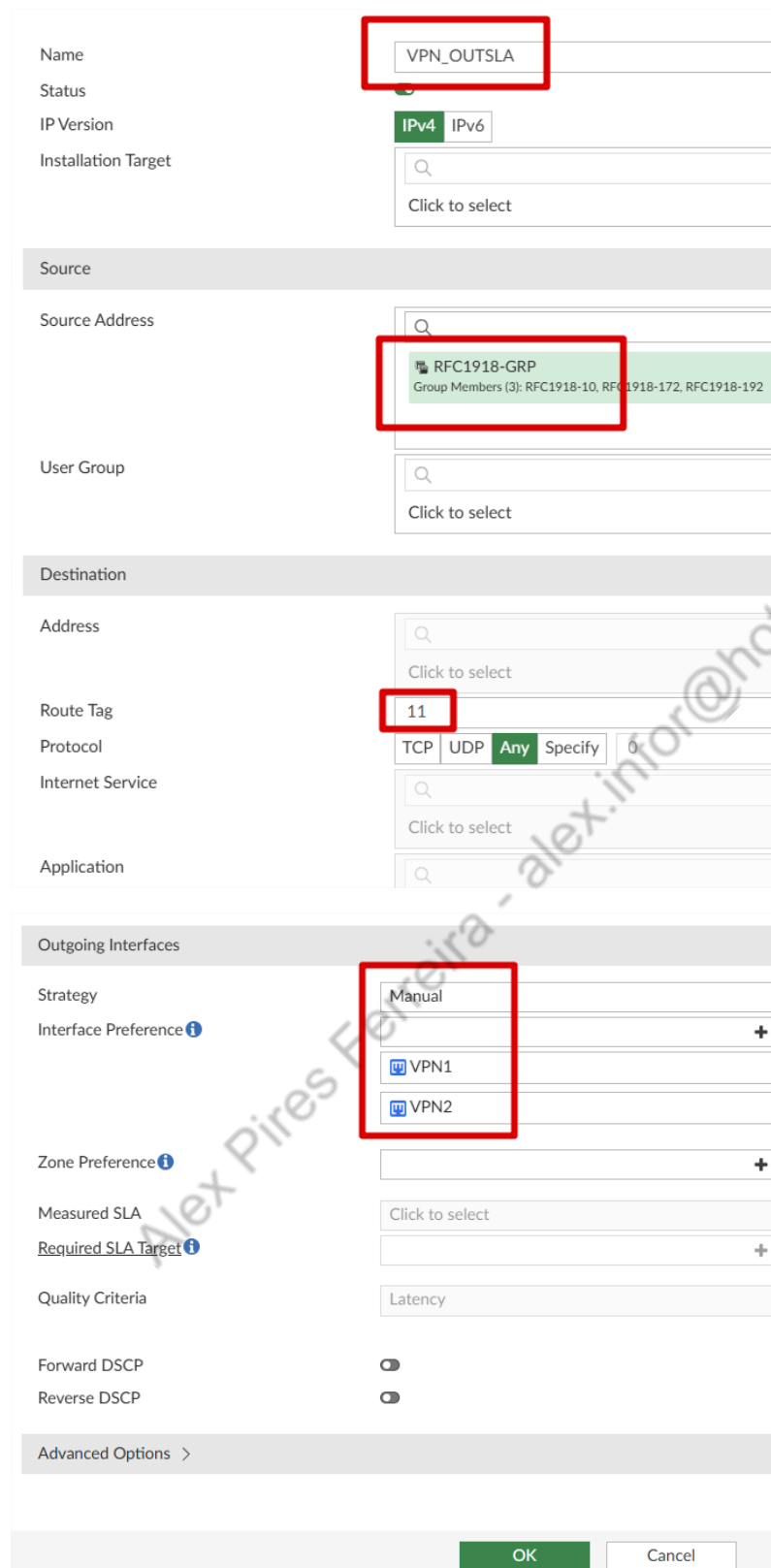
Forward DSCP Off

Reverse DSCP Off

Advanced Options >

OK **Cancel**

VPN_OUTSLA:



The screenshot shows the configuration form for a VPN_OUTSLA. Red boxes highlight the following fields:

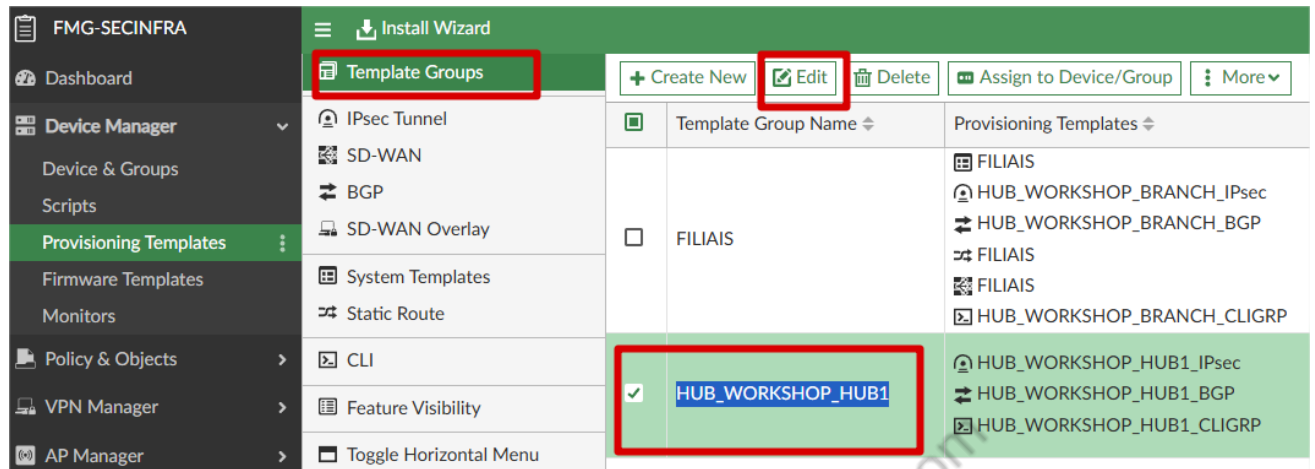
- Name:** VPN_OUTSLA
- Source Address:** RFC1918-GRP (Group Members (3): RFC1918-10, RFC1918-172, RFC1918-192)
- Route Tag:** 11
- Outgoing Interfaces:** Manual (with VPN1 and VPN2 listed below it)

The form includes sections for Source, Destination, and Outgoing Interfaces. The Destination section includes fields for Address, Route Tag, Protocol (TCP, UDP, Any, Specify), and Internet Service. The Outgoing Interfaces section includes Strategy, Interface Preference, Zone Preference, Measured SLA, Required SLA Target, Quality Criteria, Forward DSCP, and Reverse DSCP. The Advanced Options section is also visible at the bottom.

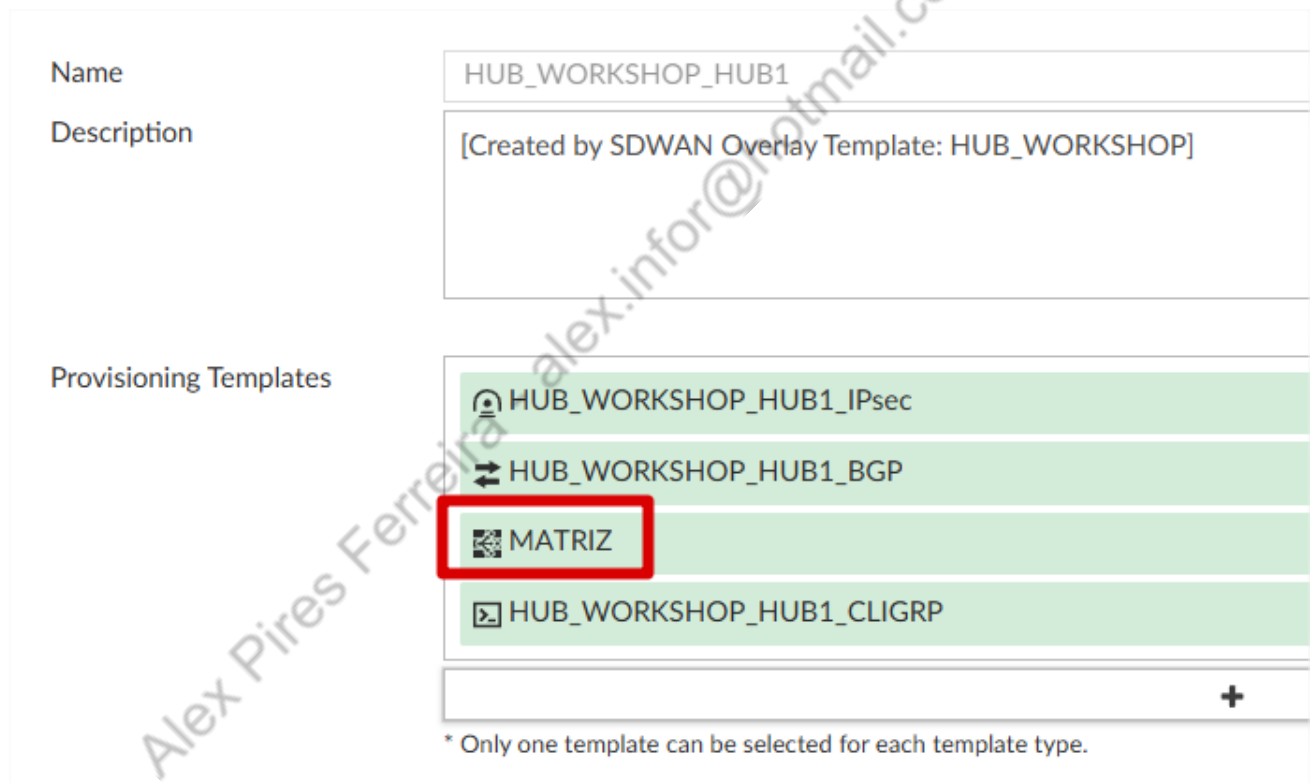
OK Cancel

Agora podemos clicar em OK para salvar nosso SD-WAN Template.

Vamos em Template Groups, e adicionaremos o SD-WAN Template ao grupo do HUB:



Template Groups		Provisioning Templates
☐	FILIAIS	- FILIAIS - HUB_WORKSHOP_BRANCH_IPsec - HUB_WORKSHOP_BRANCH_BGP - FILIAIS - FILIAIS - HUB_WORKSHOP_BRANCH_CLIGRP
☒	HUB_WORKSHOP_HUB1	- HUB_WORKSHOP_HUB1_IPsec - HUB_WORKSHOP_HUB1_BGP - HUB_WORKSHOP_HUB1_CLIGRP



Name: HUB_WORKSHOP_HUB1

Description: [Created by SDWAN Overlay Template: HUB_WORKSHOP]

Provisioning Templates:

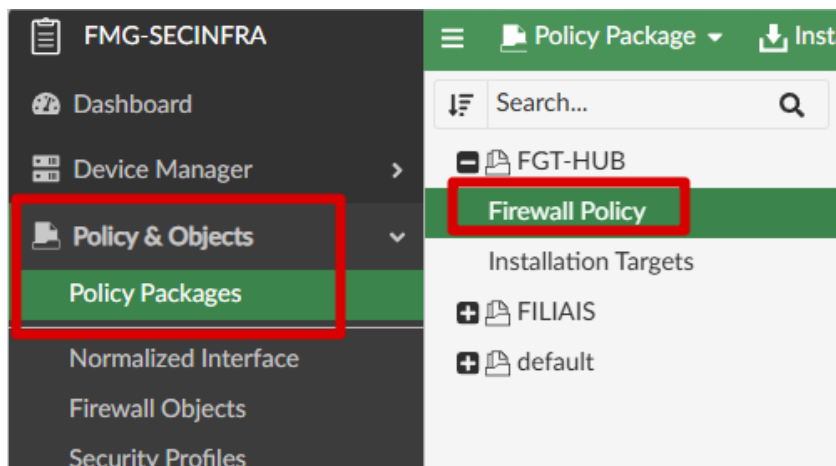
- HUB_WORKSHOP_HUB1_IPsec
- HUB_WORKSHOP_HUB1_BGP
- MATRIZ**
- HUB_WORKSHOP_HUB1_CLIGRP

* Only one template can be selected for each template type.

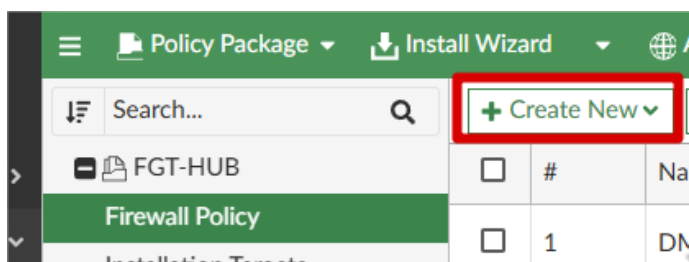
4.4.5.Criação Políticas FGT-HUB

Agora precisamos criar as políticas do FGT-HUB, nas políticas, vou utilizar os address corretos, conforme as redes que utilizamos, mas se preferir, pode criar as regras usando o RFC1918_GRP.

Vamos em Policy & Objects > Policy Packages > FGT-HUB

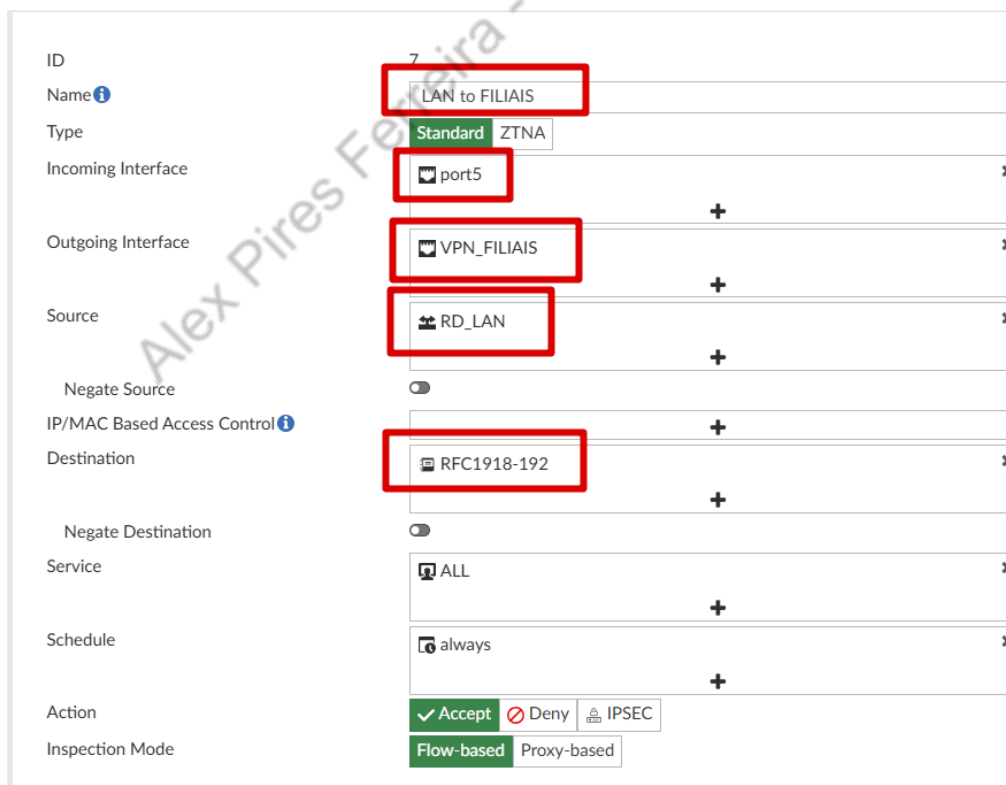


Para criar novas políticas, basta clicar em Create New:



Vamos criar as seguintes políticas:

LAN to FILIAIS, no print estamos usando a port5, caso você tenha criado uma zona para LAN da MATRIZ, será necessário normalizar essa interface.



DMZ to FILIAIS:

Name	DMZ to FILIAIS
Type	Standard ZTNA
Incoming Interface	DMZ
Outgoing Interface	VPN_FILIAIS
Source	RD_DMZ
Negate Source	☐
IP/MAC Based Access Control	☐
Destination	RFC1918-192
Negate Destination	☐
Service	ALL
Schedule	always
Action	☒ Accept ☐ Deny ☐ IPSEC
Inspection Mode	Flow-based Proxy-based
Firewall/Network Options	
NAT	☐
Protocol Options	PROT default

FILIAIS to DMZ:

Name	FILIAIS to DMZ
Type	Standard ZTNA
Incoming Interface	VPN_FILIAIS
Outgoing Interface	DMZ
Source	RFC1918-192
Negate Source	☐
IP/MAC Based Access Control	☐
Destination	RD_DMZ
Negate Destination	☐
Service	ALL
Schedule	always
Action	☒ Accept ☐ Deny ☐ IPSEC
Inspection Mode	Flow-based Proxy-based
Firewall/Network Options	
NAT	☐
Protocol Options	PROT default
OK Cancel	

FILIAIS TO LAN:

Name	FILIAIS to LAN
Type	Standard ZTNA
Incoming Interface	VPN_FILIAIS
Outgoing Interface	port5
Source	RFC1918-192
Negate Source	☐
IP/MAC Based Access Control	Destination
	RD_LAN
Negate Destination	☐
Service	ALL
Schedule	always
Action	☒ Accept ☐ Deny ☐ IPSEC
Inspection Mode	☒ Flow-based ☐ Proxy-based
Firewall/Network Options	
NAT	☐
Protocol Options	PROT default

FILIAIS to FILIAIS:

Name	FILIAIS to FILIAIS
Type	Standard ZTNA
Incoming Interface	VPN_FILIAIS
Outgoing Interface	VPN_FILIAIS
Source	RFC1918-192
Negate Source	☐
IP/MAC Based Access Control	Destination
	RFC1918-192
Negate Destination	☐
Service	ALL
Schedule	always
Action	☒ Accept ☐ Deny ☐ IPSEC
Inspection Mode	☒ Flow-based ☐ Proxy-based
Firewall/Network Options	
NAT	☐
Protocol Options	PROT default

Nesta Policy FILIAIS to FILIAIS, vamos ter que ajustar 2 opções em Advanced Options, desabilitar anti-replay e setar all em tcp-session-without-syn:

Advanced Options ▾

anti-replay ⓘ	☐
auth-cert ⓘ	Click to select
auth-path ⓘ	☐
auth-redirect-addr ⓘ	
auto-asic-offload ⓘ	☒
cgn-eif ⓘ	☐
tcp-mss-receiver ⓘ	
tcp-mss-sender ⓘ	
tcp-session-without-syn ⓘ	all
tcp-timeout-pid ⓘ	Click to select
timeout-send-rst ⓘ	☐
tos ⓘ	0x00

E finalmente a policy HEALTH CHECK FILIAIS, para essa policy, você deverá criar os objetos IP_LOOPBACK_MTZ com o valor de 172.31.255.253/32 e RD_OVERLAY com o valor de 10.254.0.0/16:

Name ⓘ	HEALTH CHECK FILIAIS
Type	Standard ZTNA
Incoming Interface	VPN_FILIAIS +
Outgoing Interface	HUB1-Lo +
Source	RD_OVERLAY +
Negate Source	☐
IP/MAC Based Access Control ⓘ	+
Destination	IP_LOOPBACK_MTZ +
Negate Destination	☐
Service	ALL +
Schedule	always +
Action	☒ Accept ☐ Deny ☐ IPSEC
Inspection Mode	Flow-based Proxy-based
Firewall/Network Options	
NAT	☐
Protocol Options	PROT default
OK Cancel	

4.4.6.Criação Policies FILIAIS

No Policy Package FILIAIS vamos criar as seguintes políticas,

LAN to MTZ e FILIAIS:

Name	LAN to MTZ e FILIAIS
Type	Standard ZTNA
Incoming Interface	LAN_FILIAL
Outgoing Interface	HUB1
Source	RD_LAN
Negate Source	☐
IP/MAC Based Access Control	
Destination	RFC1918-192
Negate Destination	☐
Service	ALL
Schedule	always
Action	Accept Deny IPSEC
Inspection Mode	Flow-based Proxy-based
Firewall/Network Options	
NAT	☐
Protocol Options	PoT default

Nessa policy, também vamos desabilitar anti-replay e setar all em tcp-session-without-syn:

Advanced Options	
anti-replay	☐
auth-cert	Click to select
auth-path	☐
auth-redirect-addr	
auto-asic-offload	☒
cgn-eif	☐
tcp-mss-receiver	
tcp-mss-sender	
tcp-session-without-syn	all
tcp-timeout-pid	Click to select
timeout-send-rst	☐
tos	0x00

MTZ e FILIAIS to LAN:

Name	MTZ e FILIAIS to LAN
Type	Standard ZTNA
Incoming Interface	HUB1
Outgoing Interface	LAN_FILIAL
Source	RFC1918-192
Negate Source	☐
IP/MAC Based Access Control	
Destination	RD_LAN
Negate Destination	☐
Service	ALL
Schedule	always
Action	Accept Deny IPSEC
Inspection Mode	Flow-based Proxy-based
Firewall/Network Options	
NAT	☐
Protocol Options	PROT default

Nessa policy, tambem vamos desabilitar anti-replay e setar all em tcp-session-without-syn:

Advanced Options	
anti-replay	☐
auth-cert	Click to select
auth-path	☐
auth-redirect-addr	
auto-asic-offload	☒
cgn-eif	☐
tcp-mss-receiver	
tcp-mss-sender	
tcp-session-without-syn	all
tcp-timeout-pid	Click to select
timeout-send-rst	☐
tos	0x00

HEALTH CHECK ADVPN:

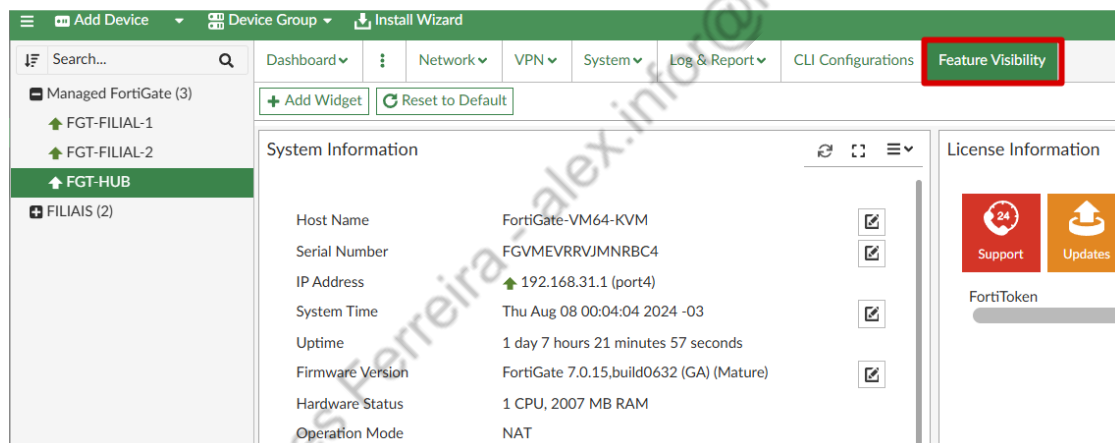
Name 	HEALTH CHECK ADVPN
Type	Standard ZTNA
Incoming Interface	 HUB1
Outgoing Interface	 HUB1
<u>Source</u>	 RD_OVERLAY
Negate Source	☐
IP/MAC Based Access Control 	
Destination	 RD_OVERLAY
Negate Destination	☐
Service	 ALL
Schedule	 always
Action	☒ Accept ☐ Deny  IPSEC
Inspection Mode	Flow-based Proxy-based
Firewall/Network Options	
NAT	☐
Protocol Options	PROT default

4.4.7.CRIAÇÃO DE POLICY ROUTE NO FGT-HUB

Em Device Manager > Device & Groups, vamos dar 2 cliques em FGT-HUB para criarmos uma configuração:

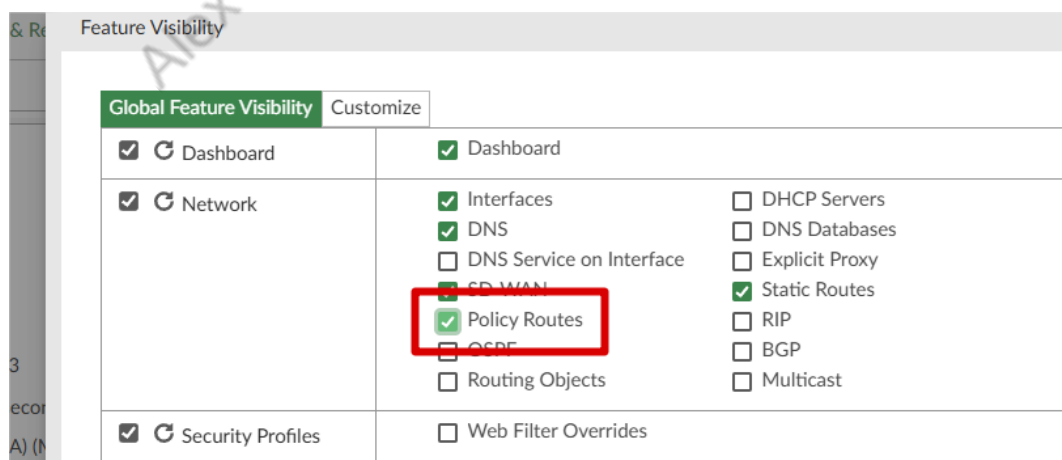
☐	Device Name	Config Status	Host Name	IP Address
☐	↑ FGT-FILIAL-1	✓ Auto-update	FGT-FILIAL-1	120.120.120.2
☐	↑ FGT-FILIAL-2	✓ Synchronized	FGT-FILIAL-2	140.140.140.2
☒	↑ FGT-HUB	✓ Synchronized	FortiGate-VM64-KVM	192.168.31.1

Clique em Feature Visibility:



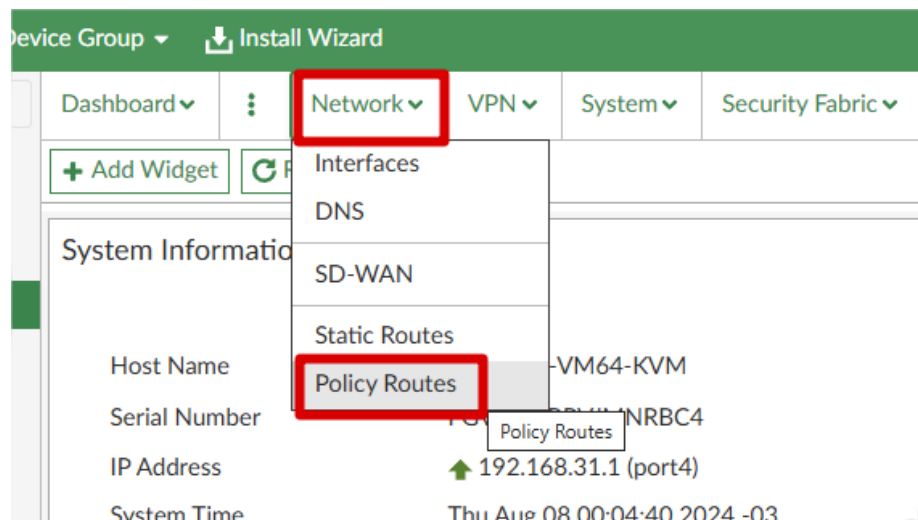
The screenshot shows the FortiGate Device Manager interface. The 'Feature Visibility' tab is selected and highlighted with a red box. The left sidebar shows a list of devices: FGT-FILIAL-1, FGT-FILIAL-2, and FGT-HUB. The main area displays system information for the selected device (FortiGate-VM64-KVM), including Host Name, Serial Number, IP Address, System Time, Uptime, Firmware Version, Hardware Status, and Operation Mode. The 'License Information' section on the right shows 'Support' and 'Updates' buttons.

Habilite Policy Routes:

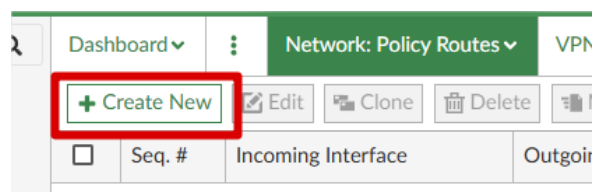


The screenshot shows the 'Feature Visibility' settings page. The 'Global Feature Visibility' tab is selected. Under the 'Network' section, 'Policy Routes' is checked and highlighted with a red box. Other features like 'Interfaces', 'DNS', 'Static Routes', and 'Routing Objects' are also visible. The 'Security Profiles' section shows 'Web Filter Overrides' is unchecked.

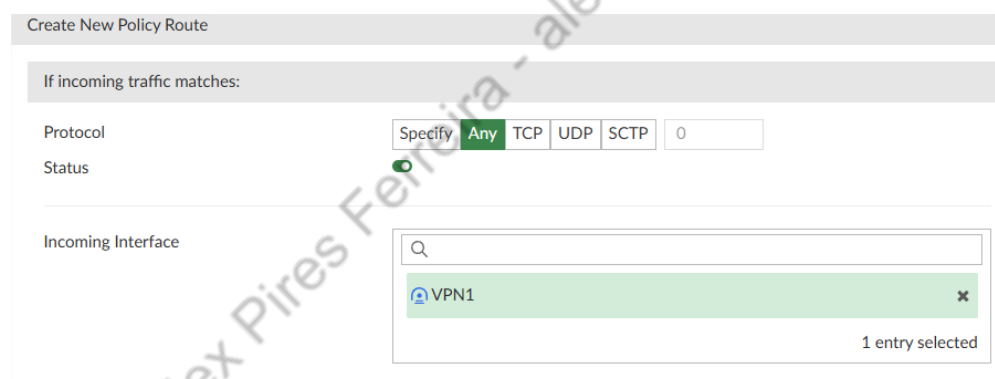
Clique em Network > Policy Routes:



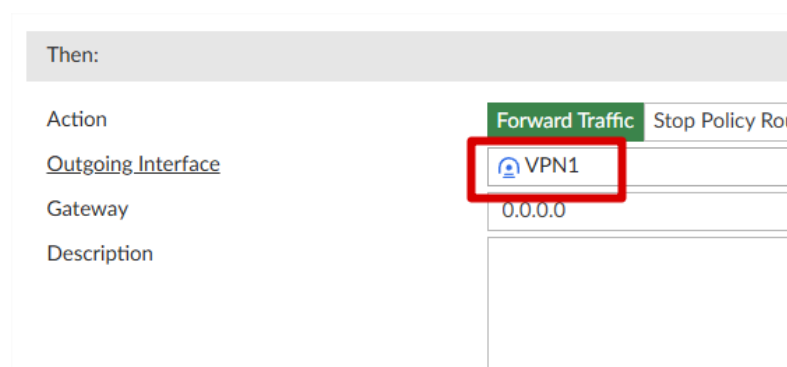
Clique em Create New:



Agora em Incoming Interface, marque VPN1:



Em Outgoing interface, marque VPN1 também:



Clique em OK, agora repita o processo e faça outra policy Route com incoming e outgoing como VPN2.

Agora suas policy routes devem estar assim:

Dashboard ▾ Network: Policy Routes ▾ VPN ▾ System ▾ Security Fabric ▾ Log & Report ▾ CLI Configurations Feature Visibility									
+ Create New Edit Clone Delete Move Copy Cut Paste Above Paste Below Search...									
	Seq. #	Incoming Interface	Outgoing Interface	Source	Destination	Source Ports	Protocol	Action	Status
☐	1	VPN1	VPN1			0-65535	ANY	Forward Traffic	Enable
☐	2	VPN2	VPN2			0-65535	ANY	Forward Traffic	Enable

Agora podemos instalar as políticas de HUB e Filiais.

Alex Pires Ferreira - alex.infor@hotmail.com

5. TESTES

Os testes serão realizados durante o Workshop ao vivo, e você terá eles disponíveis na gravação.

Durante o Workshop, poderemos realizar diversos testes, inclusive com ideias dos participantes, para que dessa forma, todos entendam o processo, e também saibam das capacidades da implementação que realizamos.

6. CONCLUSÃO

Novamente, gostaria de agradecer a confiança depositada na minha pessoa e no canal Sec Infra! Espero que o conteúdo tenha sido explicado da forma mais clara possível, e que lhe ajude na sua caminhada profissional. Qualquer dúvida, reclamação ou sugestão sobre o Workshop, você poderá enviar para o WhatsApp (54) 99357-8866.

Obrigado!

Matheus de Castro Azzi.